AIRPORTS AUTHORITY OF INDIA
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SCOPE OF WORK FOR DESIGN, DEVELOPMENT, IMPLEMENTATION AND MAINTENANCE OF AIRSEWA AND PACR.

1. Project Background

Ministry of Civil Aviation had launched AirSewa web portal with mobile Apps on Android & IOS on 25th November 2016 as proof of concept. AirSewa provides a common platform to air travellers to submit grievances and seek flight/Airport information covering all stakeholders. The selected bidder should develop the AirSewa which shall be an advanced version of the existing portal and mobile application. AAI desires to improve upon the existing web portal &App and enhance the services. MoCA/AAI wants to develop AirSewa with addition of new features to current AirSewa application (AirSewa 1.0 features) as outlined in the below scope of work requirements. The focus of AirSewa would be to provide an upgraded platform to all stakeholders of civil aviation including air travellers to address various requirements and concerns of civil aviation passengers. AirSewa platform shall make good use of the latest SMAC (Social, Mobile, Analytics and Cloud) technologies, CRM technologies, location technologies, UI/UX technologies and other related technologies to provide a comprehensive digital platform to provide various desired features and functionalities needed for a delightful digital air travel experience.

2. Scope of Work

The purpose of this section is to define the scope of work of the bidder for design, implementation, and maintenance of the AirSewa Portal, Mobile Application and Passenger Assistance Control Room (PACR).

The scope of the bidder primarily consists of the following four components:

a) **Design and Development of AirSewa Grievance Redressal Platform:** The bidder shall design, develop, operate, and maintain the AirSewa Platform, which will function as the core system for grievance redressal platform. The platform shall interact with users through a web portal, mobile application, and shall provide APIs to support integration with airlines, airport operators, and other required stakeholders.

Operations and Maintenance of the AirSewa Platform: The bidder shall be responsible for the operation and maintenance of the AirSewa for a period of 3 years after the commissioning of the system which shall be extendable to 2 more years (total 5 years) on mutually agreeable terms. It would include application support and maintenance, database management, web-based SLA monitoring tool, backup and archival services etc. It shall also include administration, monitoring and management of system, storage and portal.

b) Provisioning, Setup, Implementation – PACR

Provisioning and deployment of complete call centre hardware and backend infrastructure in compliance with the prescribed specifications, including but not limited to desktops, headsets, telephony systems, and network components. Implementation and configuration of omni-channel contact centre solution with IVRS, including call flows, routing logic, dashboards, and reporting modules. API-based integration of all communication channels with AirSewa Agent Module for seamless ticketing and interaction handling. End-to-end setup covering testing, security compliance, go-live, and stabilization to ensure PACR is fully operational on a 24x7 basis and provide adequate manpower.

S. No.	Component	Scope Description
1	Centralized 24×7 Operations	SP shall establish and operate PACR as a centralized, single point of contact for stakeholder assistance and grievance redressal on a 24×7 basis, ensuring seamless and uninterrupted service delivery.
2	Manpower Deployment & Multilingual Support	SP shall deploy trained agents, supervisors, and support staff with mandatory support in Hindi and English, scalable to additional Indian languages as required. Resources shall be aligned with SLAs, response timelines, and quality benchmarks.
3	Contact Centre Infrastructure (Hardware & Operations)	SP shall provision, install, and maintain complete call centre hardware including desktops, headsets, printers, telephony devices, and networking components as specified in the attached document , but not limited to the same. Vendor shall also provide backend infrastructure such as CTI/telephony servers, UCM, storage, SIP/PRI systems, routers, firewalls, load balancers, LAN, switches, and redundancy for high availability and BCP. Any additional hardware/software required to make PACR fully functional shall be included. Annexure-IV Compliance "Minimum specification for Desktop & with IVRS"
4	Multi-channel Support via AirSewa Agent Module	SP shall provide support Across channels including email, social media, WhatsApp, IVRS, CPGRAMS, INGRAMS, and other stakeholder systems through integration with the AirSewa Agent Module.
5	Case Management & Escalation	SP shall handle end-to-end ticket lifecycle including case creation, acknowledgment, categorization, routing, and coordinated escalation with airlines, airport operators, and stakeholders, ensuring timely resolution and structured feedback.
6	Integration with AirSewa	SP shall perform API-based integration of contact centre systems (CTI, IVR, email, social media, etc.) with the AirSewa platform for seamless ticketing and interaction handling. CRM development, customization, licensing, or maintenance is included in vendor scope.
7	Technology & Monitoring	SP shall implement and manage integrated telephony and IVRS systems aligned with AirSewa for real-time monitoring, dashboards, MIS reporting, and adherence to defined SLAs/KPIs.
8	Service Delivery & Operations	SP shall ensure SLA/TAT adherence, quality of service, escalation management, workflow compliance, response and resolution timelines, and continuous service improvement.
9	Reporting & Performance Management	SP shall provide operational MIS, dashboards, and analytics based on integrated data, without ownership or modification of the CRM system.

10	Compliance & Scalability	SP shall ensure data security, privacy, and regulatory compliance. The operating model shall be scalable to handle fluctuations in demand, with flexibility in manpower and infrastructure deployment.
11	Regulatory Compliance (TRAI/OSP/TCCCPR)	SP shall ensure compliance with all applicable telecom regulations. OSP (Other Service Provider) registration as per TRAI/DOT guidelines shall be mandatory prior to Go-Live for operating SIP/VoIP/PRI-based call centre. SP shall also ensure compliance with Telecom Commercial Communications Customer Preference Regulations, 2018 including mandatory DLT (Distributed Ledger Technology) registration of headers/templates for outbound calls and SMS. Proof of compliance shall be submitted before commencement of operations.
12	Telephony Security (SIP/VoIP Controls)	Vendor shall ensure secure deployment of SIP/VoIP infrastructure with adequate controls to prevent toll fraud, spoofing, and vishing attacks. This shall include implementation of secure session management at network boundaries (such as SBC or equivalent), encryption of signalling and media (SIP TLS/SRTP) wherever supported, access controls, configuration hardening, and continuous monitoring/logging of telephony systems. The solution shall comply with industry best practices and applicable government security guidelines.

1. **Capacity building:** The bidder shall organize training sessions for Ministry of Civil Aviation, a sample set of passengers, Airport operators, Airlines, DGCA, CISF, BCAS & other aviation ecosystem stakeholders to enable them to understand the features in the portal, the resolution mechanism, the escalation mechanism, their roles and responsibilities, troubleshooting, feedback, etc. The responsibilities of the bidder will include the creation of training manuals, digital training content, conducting the trainings, hosting of training content and manuals on AirSewa portal and app

2.1. Project Objective

The main objectives of the AirSewa platform and PACR are as follows:

2.1.1. Enhance efficiency of grievance redressal mechanism

The primary purpose of the AirSewa platform is to provide passengers with a platform to escalate, submit, track, and resolve grievances related to air travel — including escalation regarding issues involving flights, refunds, baggage, airport facilities, and service quality in an easy and convenient way. Passengers can raise complaints digitally through the AirSewa web portal or mobile application, and these grievances would be routed to the concerned stakeholder for timely resolution.

2.1.2. Improve Accountability of stakeholders

AirSewa strengthens accountability by ensuring that MOCA/AAI has visibility into grievances of passengers or citizens against various aviation ecosystem stakeholders such as airlines, airport operators, and other stakeholders to ensure CAR compliance, response timelines, and quality of service delivery. The platform enables structured categorization of complaints (e.g., baggage, delays, ticketing, refunds), supports detailed reporting, and allows passengers to provide feedback on the resolution experience.

Some of the complaints that can be registered on this portal are given below. The list is indicative and not exhaustive.

S. No	Complainant	Type of complaints
1	Passenger	Flight Delay/ Cancellation
		Special assistance
		Ticket refund
		Baggage Complaints
		Issues related to lost items
		Check-in Complaints long queues
		Security Check Complaints
		Meals-Related Complaints
		Staff behavior & Customer Service Issues
		About Airport facilities
		Airline responsiveness
		Issue in Airport due to VIP movement
		Delay in baggage offloading
		Accessibility Complaints
		Unfair demands, malpractice
		Customs
		Immigration
		Safety and Security issues at airport
		Medical Emergency at Airport
		Critical Emergency: such as Emergency Medicine left at baggage Student Admit Card left at baggage & Baggage not found
		Parking & Transportation Complaints
		Cleanliness & Facility Complaints
		Staff Behaviour & Customer Service Issues
		Accessibility Complaints for Persons with Disabilities
		Air Fare

2.1.3. Enhance Aviation Governance Through Centralized Monitoring

AirSewa will work as a central system that keeps track of:

i. Complaints by passengers towards stakeholders in aviation sector

ii. Quality of services provided

iii. Issues that need escalation and prioritization

This will enable MOCA/AAI to maintain an oversight on the responses and the passengers/citizens will be able to directly reach out to MOCA/AAI for any grievance.

2.1.4. Make the grievance redressal system more comprehensive

AirSewa aims to add options to address grievances of passengers through multiple modes of input.

2.1.5. Assure Superior Service Delivery & Performance Management:

i. SLA-Driven Performance: Embed Service Level Agreements (SLAs) directly into operational workflows, with automated monitoring, alerts, and escalations to ensure timely service delivery.

ii. Performance Analytics: Provide granular performance data for stakeholders/ SPOCs/, departments, and specific service categories to identify rectification measures which could be related to policy, SOP, or training, etc.

iii. Root Cause Analysis: Utilize collected data to identify systemic issues and implement long-term corrective and preventive actions, moving beyond reactive problem-solving.

2.1.6. Maximize Operational Efficiency & Resource Optimization:

i. Automation of Routine Tasks: Automate manual processes like coordination, tagging of complaints, etc to reduce the chances of human error, reduce response time and for optimum utilisation of human resources.

ii. Real-time Resource Visibility: Provide real-time insights into complaints through dashboards, MIS system and reports.

2.1.7. Cultivate Deep Citizen Engagement & Satisfaction:

i. Simplified Feedback Mechanisms: Offer diverse, user-friendly channels for citizens/ passengers to provide feedback, suggestions, and report issues with minimal friction.

ii. Transparent Accountability: Ensure citizens have full visibility into the status and progress of their requests, fostering trust and demonstrating MoCA/AAI's commitment to resolution.

iii. Reduce TAT of resolution with regular real-time updates on status of resolution, options for re-opening complaints, options to escalate, options to give feedback, etc.

2.1.8. Setup and operate Call Center for using Airsewa Agent module

i. To establish a centralized 24x7 Passenger Assistance and Grievance Redressal Centre (PACR) as a single point of contact for all stakeholder interactions.

ii. To enable seamless, omni-channel support integrated with AirSewa for efficient case management, tracking, and resolution.

iii. To ensure timely, transparent, and accountable grievance handling through defined SLAs, escalation mechanisms, and real-time monitoring.

iv. To create a scalable, secure, and technology-driven support ecosystem capable of handling growing volumes and diverse stakeholder needs.

2.2. Stakeholders Ecosystem

This section details the key stakeholders for AirSewa and PACR along with their role. However, more stakeholders may be integrated as per the requirement of MoCA/AAI.



AirSewa+PACR

Figure-1: AirSewa and PACR Stakeholders

Ecosystem Table 1 Key Functions of Stakeholders

S No.	Stakeholders	Key Functions
1	MoCA	MoCA is the nodal ministry responsible for formulation of policies, planning, development, and regulation of civil aviation in India. There are multiple authorities/ organizations that report to MoCA such as Directorate General of Civil Aviation (DGCA), Bureau of Civil Aviation Security (BCAS) and Aircraft Accident Investigation Bureau (AAIB).
2	Airports Authority of India (AAI)	AAI manages airport infrastructure, air navigation services, and CNS/ATM systems Across the country. AIMS supports integrated airport information management by consolidating operational data to enhance situational awareness and decision-making.
3	Directorate General of Civil Aviation (DGCA)	DGCA functions as the regulatory authority responsible for civil aviation safety oversight, airworthiness certification, and licensing. It ensures compliance with national regulations and ICAO standards through inspections, audits, and enforcement actions.
4	Bureau of Civil Aviation Security (BCAS)	BCAS formulates and enforces aviation security standards. It conducts security audits, oversees training, and monitors compliance across airports and aviation operators.
5	CISF / State Police/ Private Security Agency	CISF and State Police are responsible for airport security, including passenger screening, perimeter protection, and safeguarding of critical aviation assets. They also manage law-and-order functions and coordinate emergency response activities.
6	Airport Operator	The airport operator oversees all aspects of airport management, including operations, infrastructure maintenance, safety, and passenger services. It ensures coordination among airlines, ATC, security agencies, and ground service providers for efficient airport performance.
7	Airline Operator	Airline operators manage flight planning, crew scheduling, aircraft operations, and passenger handling. They coordinate with airports, ATC, and regulatory authorities to maintain safe, timely, and compliant flight operations.
8	Passenger	Passenger is the end user of the aviation sector; they usually raise their grievances through social media platforms or AirSewa web portal or mobile application. This portal / application was launched by Ministry of Civil Aviation (MoCA/AAI) in 2016 to monitor and track passenger grievances related to air travel and airports.

2.3. Driving / Key Success factors for AirSewa

Following are some of the key driving factors for the successful implementation and operation of AirSewa:

2.3.1. Strong backend mechanism

One of the main objectives of the AirSewa is to enhance efficiency of grievance redressal mechanism in aviation landscape. To achieve this, the platform should have be easy to use, have quick TAT for resolution and strong backend mechanism for grievance redressal.

2.3.2. Ease of Integration with multiple stakeholders

AirSewa is a grievance redressal portal in aviation sector which has multiple stakeholders. Thus, integration of the portal with other systems with these stakeholders is key success factor for quick resolution of grievances.

2.3.3. Protection of Information and Data Privacy

AirSewa will have the data of users that raise ticket in the portal. Thus, the system should be compliant to Government of India's policy on Data Privacy and Information security.

2.3.4. User Experience (Front end)

To achieve high adoption and low bounce rate, AirSewa should provide superior user experience. Passenger/ CRO at PACR should find the process of raising and tracking ticket, creating a login, etc. very easy to use. There should not be any issues like filling long forms, login issues, OTP related issues, etc.

2.3.5. Design Principles for AirSewa

The following indicative design principles shall be adopted by the selected System Integrator (SI) for successful design of the solution.

Design Principle	Description
Accessibility	The platform should support multiple access channels such as web, mobile app, IVRS, WhatsApp, walk-in assisted filing. Forms should be designed in easy-to-interpret language, with suitable font types and sizes to ensure readability. Design should be inclusive to ensure accessibility for users with low technical literacy. Suitable accessibility tools and mechanisms should be provided as per latest GIGW and as amended from time to time (**Guidelines for Indian Government Websites)** norms to ensure accessibility for Specially-abled users, including WCAG 2.2 guidelines or latest, IS 17802.
Performance metrics	The system should support a minimum of 600 Concurrent users including complainants, redressor departments and admin users. Transaction processing timelines and execution should be within the defined SLAs. The bidder shall size the infrastructure considering daily average and peak API transaction loads for all integrations, including future scalability requirements
Security by design	• Ensure the developed software is compliance with industry cyber-security frameworks (e.g., CERT-In, ISO 27001, OWASP). • Enforce encryption in transit, at rest and during processing. • Use secure coding practices and conduct regular code scans, vulnerability assessments, and penetration testing to identify buds.
Privacy & Data Protection	• Adhere strictly to applicable data-protection laws (DPDP, IT Act 2000 and amendments) and government policies. • Implement privacy-first data handling, with data minimization and audit controls. • Enable granular consent management and role-based access controls (RBAC). • Separate Development, Testing, Staging/UAT and Production environments shall be maintained. Use of real passenger PII or production data in non-production environments shall be prohibited unless expressly approved by MoCA/AAI
Modular, Open and Interoperable Architectures	Modularization, standard interfaces, and interoperability to enable flexible upgrades and horizontal and vertical scalability of the platform. Microservices approach should be followed to ensure the same.

Design Principle	Description
User friendly interfaces & inclusive UX	Web portal and mobile apps must conform to GIGW 3.0, DBIM, UX4G Solution should provide simple and standardized user experience
Multilingual	Content must support major Indian languages and ensure responsive layouts for all language scripts
Availability, reliability and serviceability	Solutions must achieve a level of availability, reliability and serviceability in line with the business criticality and requirements.
Scalability, flexibility	Architecture must ensure that system can deliver consistently high performance and handle a large volume of transactions as defined in the Performance requirements. The system should permit vertically and horizontal scalability to handle the scales upward and must anticipate order-of-magnitude growth beyond today's state-of-the-art standard (user scale, data scale, and application scale). The architecture should allow for scalability as and when the number of registered resources have increased. Provide a flexible platform to support ongoing changes in the business and effective integration of new business initiatives. A microservices based architecture should be followed to ensure a scalable. Modular platform with circuit break mechanisms for troubleshooting and platform resilience
Openness and Extensibility	o Adoption of open API, open standards and wherever prudent open- source products are of paramount importance for the system. This will ensure the system is lightweight, scalable and secure. Openness comes from use of open standards and creating vendor neutral APIs and interfaces for all components.
Innovative, sustainable and modern solutions	o Technology solutions should be innovative and apply modern approaches as per current trends. It should enable more effective ways to meet aviation related business demand and requirements.
Resilience and Robustness	o The platform should promote resilience between its platform components and act as a robust solution. It should have ability to provide and maintain an acceptable level of service in case of faults and challenges to the normal platform operation.

Design Principle	Description
Responsive Change Management	o Changes to the platform information environment are implemented in a timely manner.
Control Technical Diversity	o Technological diversity should be controlled to minimize the non-trivial cost of maintaining expertise in and connectivity between multiple processing environments.
Security risks are considered	o Technology environment should maintain a level of security that is commensurate with the risk and magnitude of the harm that could result from the loss, misuse, disclosure or modification of information.
Simplify and standardize	o Standardize the processes and services. Introduction of new solutions should simplify the operations.
Technology Refresh	o System should have provision to adopt technology refresh in every 1-2 years or whenever there are significant updates and upgrades available as decided by competent authority

2.3.6. Application

Table 2 Guiding Principles - Application

Application Principles	
Supreme User Interface / Experience	o Solution should provide simple and standardised user experience as per latest GIGW, DBIM, UX4G norms. o Applications are easy to use. The underlying technology is transparent to users.
Technology Agnostic	The solution should be developed using open-source libraries and software that are compatible with multiple operating systems and infrastructures, ensuring that it remains vendor- and platform-agnostic and can be easily ported accross different environments

2.3.7. Data

Table 3 Guiding Principles - Data

Data Principles	
Data security and privacy	Data should be protected from unauthorized use and disclosure
Data is accessible and available	○ Quality data should be available and accessible to enable users to perform their functions
Common Vocabulary and Data Definitions	○ Data is defined consistently throughout, and the definitions are understandable and available to all users. ○ Information, data and its metadata are consistently defined and structured through a Data Model. Data definitions and vocabulary are easy to comprehend and published to all users. Calculations, terms, definitions and business rules defined and approved by the business for the business and to be used by the entire business.

2.4. Process Involved in AirSewa and PACR for Grievance Redressal

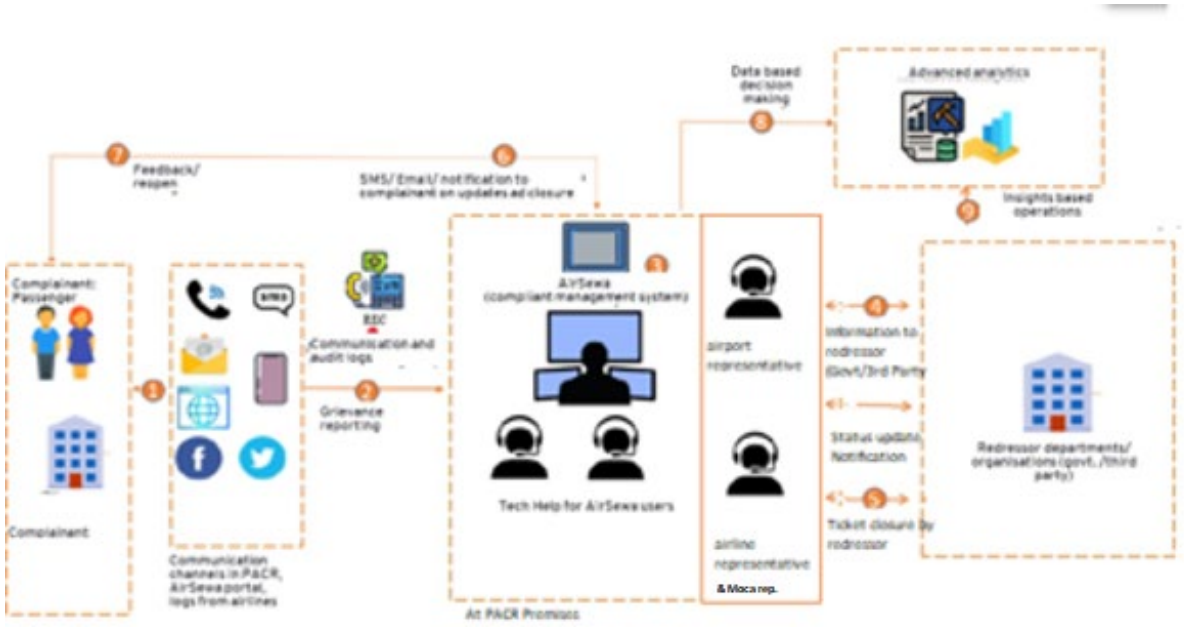


Figure - 2: AirSewa process for Grievance Redressal (Indicative)

The AirSewa portal is one of the grievance redressal channels amongst many. Other channels can be social media, IVR, phone calls, voice mails, e-mails, etc. But AirSewa is also a case management platform which shall monitor each ticket from start till complete closure. Even the grievances received on other portals should be logged in the AirSewa portal for a unique id generation or ticket number generation. To enable this platform, it will have separate landing pages for the following type of users:

1. **Complainant:** The complainants can be bonafide air passengers that might want to issue a complaint or share grievance against the concerned stakeholders using this app or through other channels.

2. **Redressor:** The departments or organisations like the airlines, airport operators, government agencies like DGCA, etc are the ones that can redress the grievances mapped to them. They will need to access the AirSewa portal to address the complaints mapped to them and act as per the government guidelines.

   The representatives of Airline/Airport operators will also be seated at PACR premises to co-ordinate between the Call Center team and respective operators. They should also get ticket dashboards and can also comment on individual tickets if required.

3. **Dashboard:** MoCA/AAI and any other overarching agencies like DGCA, etc (with approval of MoCA) will monitor the status of overall complaints. They should be able to see a dashboard based on multiple parameters described in this RFP, extract reports, run queries, etc. to have an oversight over the grievance redressal process and intervene as and when required.

The process of using AirSewa portal shall include the following stages:

2.4.1. User raising complaint:

To raise a complaint the user shall:

i. Register on the portal

ii. Login after registration

iii. Click on option to raise a new ticket

iv. Enter their details about:

   a. PNR

   b. Name

   c. Phone number

   d. Email id

   e. Complaint type

   f. Brief description of issue

   g. Upload evidence/ supporting information in photo/ video/ pdf/ word formats

v. Once the user submit the complaint, a case number/ ticket number is generated using which the user can track the following:

   a. The agency it is assigned to,

   b. The agency acknowledging it and working on it,

   c. The agency diverting to another department/ stakeholder (in some cases if needed)

   d. The action taken by the agency/ stakeholder,

   e. The expected TAT of resolution (based on the usual TAT of the officer/ department looking into it),

   f. The ideal TAT for resolution for the type of complaint based on government guidelines

g. Cause of delay in resolution if any

h. The agency/ stakeholder shall message the user for any further information required.

vi. Closure of complaint

a. The user receives the information from the stakeholder about the action taken

b. The user may re-open the complaint ticket if not satisfied

c. The user may accept the resolution and give feedback

d. The user may initiate escalation in case of delay, urgency, etc.

2.4.2. Redressor stakeholder addressing complaint

i. The complaints raised by the user/ passenger will be allocated to the stakeholder concerned depending on the type of issue.

ii. The stakeholder shall get an alert/ notification for every new complaint they are tagged to.

iii. The stakeholder shall verify within 1 hour if the complaint allocated to them is relevant or needs to be diverted to another stakeholder in case they are wrongly mapped or in case multiple stakeholder involvement is required.

iv. The stakeholder if verifies that the complaint is rightfully allocated, shall ask the concerned team to work on resolving the complaint immediately.

v. In case the complaint is wrongly tagged, they divert it to the concerned stakeholder/ agency

vi. To resolve the complaint the stakeholder shall:

a. Identify the problem

b. Ask for additional information from the complainant if needed

c. Verify if the complaint is genuine

d. Check for the ideal resolution as per SOP and government guidelines

e. Act as per guideline / SOP

f. Keep the complainant posted/ informed through AirSewa about the action being taken even when in process

2.4.3. Verification:

i. The right closure of the complaint must be verified after the concerned agency claims to have resolved the complaint.

ii. The verification shall be AI based which will compare the resolution/ action taken to government rules/ guidelines:

iii. Feedback from complainant can help in ensuring the resolution was satisfactory

iv. In case the resolution was as per guidelines and the complainant still re-opens complaint or mentions that they are not satisfied with the resolution, manual intervention to check the issue with complainant will be required and suggested by SOP / AirSewa portal to the concerned stakeholder/ agency.

2.4.4. Monitoring

i. The AirSewa will show the following information in the form of interactive dashboard and reports that can be extracted from the backend. Some of the commonly used information is given below and there may be more such analysis needed by MoCA/AAI.

   a. The complaints raised category wise like related to flight delays, flight cancellation, baggage lost, baggage exchanged, food in flight, safety issue, security issues, staff behavioural issues, etc

   b. The stakeholder wise complaints

   c. Number of complaints per day/ month/ year, this can be further drilled down to category wise, stakeholder wise

   d. Complaints resolved, pending, re-opened

   e. Complaints as per duration of resolution time (resolved early, resolved within time as per guidelines, delayed resolution cases)

   f. Complaint wise causes of delays

   g. Officer wise pendency

   h. Department wise/ agency wise pendency (comparison of average response time for different airlines)

   i. Growth rate of complaints (trend analysis)

   j. Escalations received and their resolution time

ii. Other privilege is required for Admin access for nodal officer:

   a. To manage User Hierarchy, Accounts.

   b. Filter Report Generation.

c. User Login Activity Report.

d. Provision to auto-assign and manually re-assign grievances.

e. Provision to add comments.

f. Messaging between Admins.

g. Report of all Master Data.

2.5. Envisaged Solution landscape (indicative)

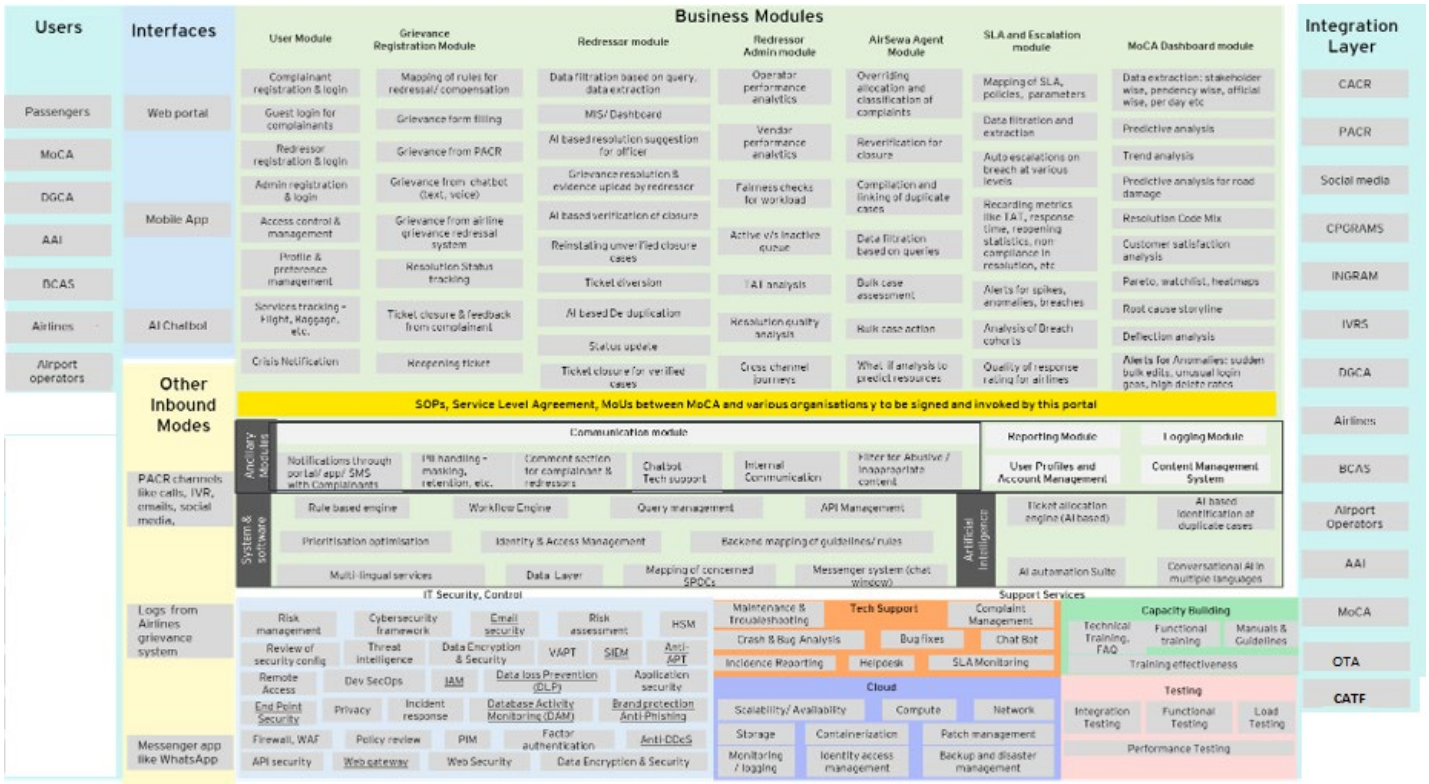


Figure – 3: High Level Solution

Above is an indicative solution landscape for AirSewa. The bidder is expected to design the detailed solution landscape as per the requirements gathered during project execution. Integrations should be done where ever feasible and required

2.5.1. User Personas

The following section describes the personas of the various users that will be accessing the AirSewa platform. This is indicative and final user personas would be finalized with the bidder in discussion with MoCA/AAI

1. Complainants

   Passengers are the users logging complaints on the AirSewa portal from the complainant side.

2. Redressor

   These are staff / officers from the departments that will address the complaints logged on the AirSewa portal. Each department would be responsible for reviewing, owning, actioning and closing the complaints relevant to them. Indicative list of departments is as follows –

   i. Airline entities
   ii. Airport operator (AAI / Private agency)
   iii. OTA platforms for ticket booking like Make My Trip, Yatra, Goibibo, etc
   iv. BCAS
   v. DGCA
   vi. CISF/ State police

3. Administration & Operations

   This persona includes users from MoCA/AAI and other overarching agencies like DGCA who would be responsible for monitoring the status of complaints, generate and view reports to evaluate the performance of the platform have an oversight over the grievance redressal process and intervene as and when required. These users should have access to dashboard, report generation system, run system queries, etc. for data-based decision making, pro-active interventions, predictive analysis and granular monitoring. The process flow explained in this document may sometimes be modified as per the suitability as decided by competent authority.

2.5.2. User Interfaces

This section elaborates the interfaces that the various user personas will interact with, and a high-level set of features provided by each interface.

2.5.2.1. Complainant interface:

This section elaborates the interfaces that should be available to the complainants for accessing and interacting with the AirSewa portal. The list of interfaces to be made available to complainants is elaborated below –

a) **AirSewa mobile application in Android & iOS –** Downloadable app from app stores that can be leveraged for raising and submitting a ticket, tracking its status and providing feedback.

b) **AirSewa web portal** – Web portal accessible through all major internet browsers for accessing AirSewa

c) **AI Text Chatbot with Multilingual support** – An AI text Chatbot that is integrated with the mobile app, web portal, WhatsApp and other social media channels. It should support 12 major Indian languages, excluding English and Hindi

d) **AI Voice Chatbot with Multilingual support** – An AI voice Chatbot that is integrated and accessible Across the mobile app, web portal, and messenger app like WhatsApp. It should support 12 major Indian languages, excluding English

e) **Other channels** – Apart from the above, complainants also register complains / express grievances Across various other channels. These complaints need to be documented and registered in the AirSewa portal in an automated manner, for further processing. These complaints would be reviewed and approved by the PACR team for further processing. An indicative list of other channels is below –

   i. WhatsApp/ Telegram
   ii. Phone call
   iii. Voice mail
   iv. E-mail
   v. IVR
   vi. Complaints registered through Social Media platforms of airlines, airports, regulators and the Ministry (Twitter, FB, Instagram, YouTube)
   vii. Existing government public grievance platforms such as CPGRAMS, INGRAMS and others
   viii. Physical letter
   ix. Other communication channels as prescribed by MoCA/AAI in future

These modes should be digitized for input to the AirSewa portal using AI and other technologies, as applicable. The final ticket to be logged to the AirSewa portal would be reviewed by PACR. The tracking and case management of complaints raised on these modes shall be done using AirSewa which will generate a unique id for each and identify redundant complaints, help in tracking the cases end-to-end.

The workflows to be enabled by these interfaces is elaborated below -

2.5.2.2. User Registration:

i. The Passenger shall register on the portal/ app by giving details like name, phone number and e-mail id and username, etc. It should be enabled by factor authentication method and should be convenient for users at the same time. The information required to fill should be minimal to keep it short and convenient for User.

ii. Once authentication is successfully verified, the user is registered and can proceed to log in.

2.5.2.3. Passenger login:

i. The User login to be enabled by factor authentication method. Once verified the User is logged in.

2.5.2.4. Ticket Creation:

i. The user enters minimal details like PNR (if applicable), grievance type, grievance description, Upload evidence/ supporting information in photo/ video/ pdf/ word formats. The form to be filled should be short, convenient and easy to understand. It should be multi-lingual and follow accessibility standards to make it inclusive platform for users.

ii. Tickets should be auto assigned to the concerned department on creation, based on the subject matter of the complaint / grievance

2.5.2.5. Tracking of ticket till closure:

i. Once the complaint is registered, the ticket number notification should be generated and sent to the User through SMS/ WhatsApp/ notification in app and web portal, etc.

ii. The User should be able to track the complaints through app/ web portal and should keep getting updates on the status frequently through messages on phone like text message or message through messenger apps such as WhatsApp/ Telegram.

Closure of Complaint: shall have the option to close the ticket on his/her own

i. The User shall have the option to escalate the complaint at start or while it is in process.

ii. The details of the escalated tickets will come to the PACR monitoring team. The team will follow up with the concerned stakeholders based on the ticket details.

2.5.2.6. Re-opening ticket:

Once the ticket is closed by the concerned stakeholder like airlines, airport operator, etc, the User shall have the option to:

i. Re-open the ticket in case not satisfied

ii. Escalate the complaint after re-opening

2.5.2.7. Feedback:

The User shall be asked for feedback at the time of closure of ticket/ complaint. The mode of asking for feedback can be voice call, message, form in the AirSewa portal, etc as decided in discussion with MoCA/AAI. The feedback form/ format should be short, easy to understand and convenient to fill.

2.5.3. Redressor's interface:

This section elaborates the interface using which the Redressor like airlines, airport operators, government agencies like DGCA, etc can view, monitor and address the complaints.

Redressor should have access to the below interfaces to action their requests if they do not have native CRM else should have integration with their native CRM –

- AirSewa Mobile App for Redressor
- AirSewa Web portal login for Redressor

The departments should have access to the below features for handling the grievances:

1. **View & Action pending complaints**

i. User should be able to login to their account (individually or designation wise). The bidder shall be responsible for creating adequate number of logins and the platform should permit addition / removal of new departments users in case of change in organizational structure of the stakeholder.

ii. The User (Actioning department) should be able to see the cases/ complaints / tickets assigned to them in their CRM (if exist) else should follow the workflow for "no native CRM" given in this document. They should be able to see separate cases that are marked as 'priority cases' and 'escalated cases'.

iii. They should be able to divert the complaint cases to other department, in case a case is wrongly tagged to them

iv. Passanger should be able to communicate through the portal via comments/ messages/ chat with the complainant, PACR CROs via AirSewa portal. Users should receive a notification over email / WhatsApp when such a message has been sent to them.

2. **Stakeholder with no Native CRM**

i. The stakeholders should be able to add other stakeholders in case the ticket/ complaint required attention/ intervention from multiple stakeholders and departments.

ii. They should be able to escalate the complaint to their seniors in case a decision is beyond their jurisdiction

iii. They should be able to close the complaint from their end once they resolve by uploading / adding evidence of relevant resolution such as the transaction id of the compensation of ticket transferred to the passenger (complainant)

iv. They should be notified if the ticket is re-opened by the complainant

v. They should be able to sort the complaints as per date and time, priority, as per type of complaint, region, sector, etc.

vi. They should be able to see their own performance in terms of their response time for various types of complaints, pendency, average resolution time in past months, improvement in their response time, etc

vii. The bidder should configure business logic regarding policies, procedures and SOPs into the platform. The recommended solution as per policy / SOP should be recommended to a user when they are responding to a complaint. Additional, in case of an exception to policy / SOP, same maybe flagged to the user when closing the request and to their supervising officer during the approval workflow.

viii. The platform should permit the departments to implement a maker / checker workflow for responding to certain category of complaints

ix. The platform should permit the department to automate response to certain category of complaints to reduce the turnaround time.

3. User Management

i. Department users should have access to a user management portal for adding / deleting / modifying new users.

ii. The portal should allow multiple user roles to determine who will action a request, who will approve the resolution and other business activities.

4. Knowledge Repository

i. Department users should have access to a knowledge repository with relevant guidelines/ SOPs / policies for appropriate resolution

ii. Users should be able to access the manuals / videos related to training content to use the portal

5. Technical support and feedback

i. Users should be able to ask queries to a tech-support team over email, phone and chat to help them troubleshoot and resolve issues related to the portal

ii. Users should be able to give feedback to improve the portal further

2.5.4. Admin interface:

It is the interface using which MoCA/AAI and other overarching agencies nominated by MoCA/AAI can monitor the status of complaints, can see a dashboard, extract reports, run queries, etc. to have an oversight over the grievance redressal process and intervene as and when required.

The admin interface should be available to users over **the mobile app** and **web portal**

The AirSewa will show the information in the form of interactive dashboard and reports that can be extracted from the backend in the concerned section. Some of the commonly used information is given below and there may be more such analysis needed by MoCA/AAI/ AAI.

1. The complaints raised category wise like related to flight delays, flight cancellation, baggage lost, baggage exchanged, food in flight, safety issue, security issues, staff behavioural issues, etc
2. The stakeholder wise complaints

3. Number of complaints per day/ month/ year, this can be further drilled down to category wise, stakeholder wise
4. Complaints resolved, pending, re-opened
5. Complaints as per duration of resolution time (resolved early, resolved within time as per guidelines, delayed resolution cases)
6. Complaint wise causes of delays
7. Officer wise pendency
8. Department wise/ agency wise pendency (comparison of average response time for different airlines)
9. Growth rate of complaints (trend analysis)
10. Escalations received and their resolution time
11. Open v/s closed complaints

Apart from the analytics mentioned above it should also have the following capabilities:

1. **Handling complaints from other channels**

2. Provide an interface to the PACR team to reviewing, edit and approve complaints from other channels
3. **User Management**

   i. Admin users should have access to a user management portal for adding / deleting / modifying new departments and admin users.

   ii. The portal should allow multiple user roles to determine administrators, who will action a request, who will approve the resolution and other business activities.

4. **Knowledge Repository**

   i. Admin users should have access to a knowledge repository with relevant guidelines/ SOPs/ policies for appropriate resolution

   ii. Users should be able to access the manuals / videos related to training content to use the portal

5. **Technical support and feedback**

   i. Users should be able to ask queries to a tech-support team over email, phone and chat to help them troubleshoot and resolve issues related to the portal

   ii. Users should be able to give feedback to improve the portal further

2.5.5. Business modules

This section describes the backend business modules and application services that would be enabled by the AirSewa portal to meet the objectives outlined in this document.

2.5.6. User Registration Module

1. Registration of user:

   i. The user shall register on the portal/ app by giving details like name, phone number and e-mail id and username, etc. It should be enabled by factor authentication method and should be convenient for users at the same time. The information required to fill should be minimal to keep it short and convenient for User.

   ii. Once the authentication is verified, User gets registered and can login.

2. User login:

   i. The User login to be enabled by factor authentication method. Once verified, the User is logged in.

3. Guest login: the user should be enabled to login as a guest through factor authentication. The guest login shall give the option to user to give consent to save their data and create an account.

4. Access management: The nodal officers of organizations that are users should be able to control the access of staff and manage changes in the access to their staff in the portal.

5. Profile and preference management: the user can decide and opt for notification through various channels like email, SMS, notifications on app, etc.

6. Services tracking – Flight, Baggage, etc

   i. The citizen should be able to put the flight number on the home page of portal without having to login the portal/ app

   ii. The portal/ app should give the status and estimated time of arrival and landing at the correct airport to the user. For this the portal/ app should be able to fetch real-time data of flight status from the concerned stakeholder like airline, CATFM, etc, as decided by MoCA/AAI

   iii. The user should be able to track a flight from AirSewa App/ portal by adding the flight number either on home page without logging in or after logging in.

   iv. The system should allow the portal/ app to send regular updates through notifications, SMS, messenger app (Arratai/ WhatsApp/ Telegram, etc) message, email, etc to give real-time status of the flight in case the user opts for the same through option given in the portal

   v. The portal shall allow tracking of components that are useful for passengers like flights, baggage, airport, etc as directed by MoCA/AAI

7. Crisis Notification: The portal should have the option of adding any notifications on home page as per directions by MoCA/AAI especially in case of major crises to caution the passengers/ users. For instance:

   i. The flights are expected to delay in northern region due to fog for next 24 hours

   ii. There is a crisis by 'X' Airlines due to technical glitch, likely to be resolved in next 2 days, the

   users shall receive refunds for cancelled flights in next 5 business days.

2.5.7. Grievance Registration Module

The AirSewa app/ portal will allow the users to raise and track the complaints through their login. This will involve the following steps after logging in (after registration or as a guest):

1. **Creation of ticket**:

   i. Grievance form filling: The user enters minimal details like PNR (if applicable), grievance type, grievance description, Upload evidence/ supporting information in photo/ video/ pdf/ word formats. The form to be filled should be short, convenient and easy to understand. It should be multi-lingual and follow accessibility standards to make it inclusive platform for Users.

   ii. The tickets will get auto generated for complaints fetched from PACR

   iii. The tickets will be generated for cases that MoCA/AAI shall take up from logs fetched from grievance redressal system of airlines. These tickets shall be based on parameters defined in rule-based engine, it can be based on level of severity, cases unresolved within said TAT, priority cases, etc

   iv. The portal should be able to parse data/ complaints received from the voice notes/ chat of AirSewa portal.

2. **Tracking of ticket till closure**:

   i. Once the complaint is registered, the ticket number notification should be generated and sent to the User through SMS/ WhatsApp/ notification in app and web portal, etc.

   ii. The User should be able to track the complaints through app/ web portal and should keep getting updates on the status frequently through messages on phone like text message or message through messenger apps such as WhatsApp/ Telegram.

3. **Closure of Complaint**:

   i. The User shall have the option to close the ticket on his/her own

   ii. The User shall have the option to escalate the complaint at start or while it is in process.

4. **Re-opening ticket:**

   Once the ticket is closed by the concerned stakeholder like airlines, airport operator, etc, the User shall have the option to:

   i. Re-open the ticket in case not satisfied

   ii. Escalate the complaint after re-opening

5. **Feedback**

   The User shall be asked for feedback at the time of closure of ticket/ complaint. The mode of asking for feedback can be voice call, message, form in the AirSewa portal, etc as decided in discussion with MoCA/AAI. The feedback form/ format should be short, easy to understand and convenient to fill.

2.5.8. Redressor Module

1. The redressor like airlines, airport operators, DGCA, etc which shall be tagged to the complaints through AI based ticket allocation engine.

2. The ticket classification and allocation shall be based on parameters like key words, mapping of roles and responsibilities in the backend, and any other rules that are needed for correct auto-allocation in the system.

3. The redressor tagged shall also be allowed to divert to complaint to the other department/ organisation or add more redressors in case needed.

4. In case of wrong tagging of the redressor, the AirSewa agents should be able to override the tagging and assign it to correct redressor.

5. The redressor should be able to communicate through the portal via comments/ messages/ chat with the complainant, other officials and stakeholders mapped to the AirSewa portal

6. The tagged redressor should be able to add more stakeholders in case the ticket/ complaint required attention/ intervention from multiple stakeholders. The overall SLA for complaint resolution shall not be impacted if the additional redressor is added late, the timeline in which an additional redressor can be added shall be defined in the SOP.

7. The redressor should be able to compare the demand/ the action required to relevant guidelines/ SOPs to decide for the appropriate resolution. The AI tool shall assist in summarizing long complaint descriptions and in identifying the resolution suggestion for officer as per the DGCA or other relevant guidelines mapped in the system.

8. The redressor should be able to escalate the complaint to their seniors in case a decision is beyond their jurisdiction

9. The redressor should be supported by the system through AI based identification of duplicate grievances, however, this needs to be verified by the officials / AirSewa agents if the duplicate identified by the AI tool is confirmed.

10. The redressor should be able to close the complaint from their end once they resolve by uploading

    / adding evidence of relevant resolution such as the transaction id of the compensation of ticket transferred to the passenger (complainant)

11. The redressor should be notified if the ticket is re-opened by the complainant

12. Query management and dashboard for Redressor:

i. The Redressor should be able to see the cases/ complaints / tickets assigned to them. They should be able to see separate cases that are marked as 'priority cases' and 'escalated cases'.

ii. They should be able to sort the complaints as per date and time, priority, as per type of complaint, region, sector, etc

iii. They should be able to see their own performance in terms of their response time for various types of complaints, pendency, average resolution time in past months, improvement in their response time, etc

13. Tech help and feedback by redressor for portal

i. They should be able to access the manuals / videos related to training content to use the portal

ii. They should be able to ask queries to a chatbot/ tech help team to help resolve their issues, help them in troubleshooting the issues related to the portal

iii. They should be able to give feedback to improve the portal further

14. Once the complaint is escalated to a senior nodal officer at redressor's end, the nodal officers should be able to login and should be able to do the following:

i. compare the demand/ the action required to relevant guidelines/ SOPs to decide for the appropriate resolution

ii. divert the cases to other stakeholders in case they are wrongly tagged to them

iii. communicate through the portal via comments/ messages/ chat with the complainant, other officials and stakeholders mapped to the AirSewa portal

iv. add more stakeholders in case the ticket/ complaint required attention/ intervention from multiple stakeholders

v. to close the complaint from their end once they resolve by uploading / adding evidence of relevant resolution such as the transaction id of the compensation of ticket transferred to the passenger (complainant)

vi. run queries, extract reports, monitor:

a. They should be able to see the cases/ complaints / tickets assigned to them.

b. They should be able to see separate cases that are marked as 'priority cases' and 'escalated cases'.

c. They should be able to sort the complaints as per date and time, priority, as per type of complaint, region, sector, officers/ SPOCs they were assigned to, etc.

d. They should be able to see performance of their organisation/ department for grievance redressal as per the Key Performance Indicators/ benchmarks defined by MoCA/AAI/ government agencies in terms of their response time for various types of complaints, pendency, average resolution time in past months, improvement in their response time, etc

vii. Make decisions related to training and capacity building:

a. They should be able to monitor the attendees of past trainings conducted

b. They should be able to nominate officers for trainings

c. They should be able to request for additional trainings to the bidder in case needed.

viii. Get Tech help and give feedback for portal

a. They should be able to access the manuals / videos related to training content to use the portal

b. They should be able to ask queries to a chatbot/ tech help team to help resolve their issues, help them in troubleshooting the issues related to the portal

c. They should be able to give feedback to improve the portal further

2.5.9. Defined closure process

1. Stakeholders can close ticket, but passengers can reopen up to 2 times

2. Escalation triggers like escalation 1 resolution within 24 hours, escalation 2 after 72 hrs, etc. based on the benchmarks defined by MoCA/AAI and should be modified if the benchmarks are amended by MoCA/AAI from time to time. The escalation triggering shall be automatic in case:

    i. Of delays as per the guidelines set by MoCA/AAI/ DGCA.

    ii. Grievance case is urgent (like health emergency, medicines involved in the baggage, admit card of a student in a baggage, etc.)

Note: bidder shall define such priority cases to allow for prioritisation and escalation triggers for the same in discussion with MoCA/AAI and its stakeholders. There should be a provision to assign critical/ priority cases to PACR in the portal in case the SOPs suggest so.

3. Passenger feedback & verification post report closure:

   Ticket closure shall be permitted to the stakeholder responsible only once they share some evidence such as the transaction id of compensation paid, etc. The closure will be considered complete only if the passenger confirms the same in feedback

4. AI based closure verification based on CAR/ DGCA guidelines

   The resolution given/ action taken shall be verified by AI based mechanism at backend to check if the resolution is complaint to the government regulations/ guidelines which will have to be updated as per the amendments in the regulations/ guidelines from time to time.

5. Verification of resolution before closure:

   The following shall be the means to verify if the case can be considered closed:

   i. Tagged airline/stakeholder must provide evidence before closure

   ii. AI-based auto-verification aligned with CAR/DGCA rules

   iii. Passenger feedback collected for every case

The wrongfully closed or unverified closure cases shall be reinstated at any stage mentioned above. In case the passenger is not happy with the resolution while the AI based mechanism indicates the resolution was compliant to the guidelines, a manual intervention may be required to cross check the issue with the complainant. Such cases shall go to PACR for quick coordination and verification with various stakeholders, to take a call on the genuineness of the ask/ expectation by the Complainant.

2.5.10. Redressor Admin Module

The admin portal for each user organisation shall have a dashboard to monitor the workload, the TAT, quality of resolution, performance, etc. The portal should enable filtering and extraction of data in various reports to help monitor and act where needed based on data. The analytics that should be enabled in the portal shall include, but may not be limited to the following:

1. Operator performance analytics: the portal should allow the performance analysis of AirSewa Agents, operators in PACR, of officials handling the complaints for each user organisation, etc. The parameters may include Agent/queue load, average handling time, transfer rate, Quality scores from QA audits, coaching flags, etc.

2. Fairness checks for workload: the portal should enable Fairness checks for workload distribution/ allocation of tickets to operators: normalize by case complexity/severity, and suggested rebalancing of load.

3. Active v/s inactive queue: the portal should allow real-time monitoring of the movement of complaints from active to inactive queue and vice-a-versa.it shall include analysis of inflow vs. outflow, backlog, aging buckets, etc

4. Cross channel journeys: the portal should allow the monitoring of complainants moving from 1 channel to another to raise complaint such as a complainant may switch from web app to call for same complaint due to poor response time of dissatisfaction, etc.

5. TAT analysis: the portal should allow analysis of TAT of resolution category wise, region wise, stakeholder, wise, etc

6. Resolution quality analysis: the system should allow monitoring of quality of resolution

7. The portal should allow governance check such as:

    a. Audit logs: to track configuration changes, role-based access events.

    b. Data retention checks

    c. Anomalies: sudden bulk edits, unusual login geos, high delete rates.

2.5.11. AirSewa Agent Module

AirSewa Agent Module shall be a part of the portal that enables the manual overriding of erroneous activities like ticket tagging, closure, verification of ticket closure, allocation, etc. The portal should allow the agents to do the following:

1. Overriding allocation and classification of complaints, in case the complaints are wrongly classified or wrongly tagged to a redressor by AI engine.

2. Reverification for closure: in case the ticket gets closed erroneously by the system

3. Compilation to avoid duplication of cases: in case there are multiple follow ups from complainant marked as a new complaint or any other cases causing duplication of complaints. There shall be provision to merge or link the follow up complaints to the main ticket id as per the SOP for such cases or as directed by MoCA/AAI.

4. Ticket allocation engine: the portal should allow allocation of tickets based on workload, expertise and other relevant parameters of the agents but if needed, the supervisor/ nodal officer may override that method of allocation and assign cases manually.

5. Query management: the agents should be able to filter complaints based on various parameters, extract reports, etc on need basis.

6. Communication channel for additional information: the portal should allow the agents to communicate with each other, with complainant or redressor to seek or provide additional information.

7. Bulk case assessment and action: the portal should enable the agents to assess cases in bulk and take bulk action of Grievances such as override severity level, etc.

8. What-if analysis to predict resources: the portal should allow predictive analysis for increasing or decreasing of resources in case when the number of expected grievances is going to rise during certain periods. This will help in deciding the additional headcount/shift adjustments effect on cost, contract and Service Level Agreement with the agency providing operators if any.

2.5.12. SLA and escalation Module

1. Mapping of SLA:

    a. The portal should allow running of multiple SLA policies based on various parameters, priorities, escalations, severity, compensations, penalties, type of non-compliance cases, etc

    b. SLAs to be reconfigured based on PACR's SLA and changing CAR rules and terms of MoU with Redressors

    c. Resolution provided is as per rules defined for each grievance type and sub-category.

2. Query management: the portal should allow filtering of cases based on type of non-compliances, age of grievance, TAT of complaint, etc

3. Auto escalations on breach at various levels: the portal should automatically escalate the complaints-based rules defined at various stages.

4. Recording metrics: the portal should maintain logs like TAT, response time, reopening statistics, non-compliance in resolution, etc

5. Alerts: the system should be able to detect and send alerts in case of spikes, anomalies, breaches

6. Analysis of Breach cohorts: the system should be able to identify which combinations (category × location × priority) cause most misses and which combinations (category × location × priority) cause most complaints

7. Quality of response rating for airlines: The portal/ app should allow mapping of quality of resolution by redressor such as airport operators, airlines, etc and give them token of appreciation, ratings, etc as per MoCA/AAI's discretion.

8. During any crisis or emergent situation that may affect large no. of passengers, the SLA to be updated will be followed and stakeholders has to provide reply in real time basis. MoCA/AAI may ask to increase the manpower and over all capacity which selected bidder has comply on prorated rates.

9. If a guideline or advisory is issued by MeitY or any other related agencies, the Service provider need to act accordingly and patch the system. However, this will not be considered under the change request.

2.5.13. Dashboard Module

The dashboard of the portal for MoCA/AAI should allow/ enable the monitoring of various parameters such as given below:

1. The complaints raised category wise like related to flight delays, flight cancellation, baggage lost, baggage exchanged, food in flight, safety issue, security issues, staff behavioral issues, etc
2. The stakeholder wise complaints
3. Number of complaints per day/ month/ year, this can be further drilled down to category wise, stakeholder wise
4. Complaints resolved, pending, re-opened
5. Complaints as per duration of resolution time (resolved early, resolved within time as per guidelines, delayed resolution cases)
6. Complaint wise causes of delays
7. Officer wise pendency
8. Department wise/ agency wise pendency (comparison of average response time for different airlines)
9. Growth rate of complaints (trend analysis)
10. Escalations received and their resolution time
11. Open v/s closed complaints
12. Comparison of complaints in active queue v/s inactive queue
13. Queue time by priority of complaints
14. First Contact Resolution (FCR): % resolved in first agent interaction.
15. Reopen Rate: % closed tickets reopened within X days.
16. Escalation Rate: % tickets escalated beyond L1.
17. Transfer Rate: average hops between queues/agents.
18. Resolution Code Mix: root-cause or disposition categories.
19. Customer satisfaction analysis
20. Sentiment score (from NLP) & change over time.
21. SLA breach analysis for stakeholders such as airlines in grievance redressal.
22. Trend lines: volume vs. SLA attainment; CSAT vs. sentiment.
23. Pareto: top 10 categories contributing to x% (say 80%) of volume/breaches.
24. Map: grievance rate & SLA by region.

25. Watchlist: chronic categories/regions/ stakeholder with persistent breaches/ complaints sector wise/ category wise

26. Live queue: inflow vs. outflow, backlog, aging buckets.

27. SLA breach risk (predictive) by queue/category. The predictive analysis could be based on multiple factors like seasonal due to weather conditions, due to special events, change in regulation, depreciating performance/ assets of an airline, etc.

28. Agent availability & workload; suggested rebalancing, performance unit (PU).

29. Alerts: spikes, anomalies, outage-like patterns (by category/channel).

30. Verification of resolution by complaints as per priority, channel, category, location, department/ stakeholder, etc.

31. Breach cohorts: which combinations (category × location × priority) cause most misses.

32. Complaint cohorts: which combinations (category × location × priority) cause most complaints

33. Root-cause storyline: breach reasons (workload, dependencies, missing info, change in regulation without sufficient notice period, etc).

34. Category/subcategory trend (last 12–24 months).

35. Pareto of categories by volume and by negative outcomes (breaches, escalations).

36. Channel mix & shifts duration, including over-time.

37. Channel-specific conversion & abandonment (e.g., chat dropped sessions, missed calls, etc).

38. Cross-channel journeys: how many start in app then call?

39. Deflection analysis for mapping of complaints to wrong stakeholder

40. Success and popularity analysis of various complaint channels like app, chatbot, social media with time (trend analysis)

41. Operator performance: Agent/queue load, average handling time, transfer rate, Quality scores from QA audits; coaching flags,

42. Fairness checks for workload distribution/ allocation of tickets to operators: normalize by case complexity/severity to ensure apples-to-apples.

43. What-if: additional headcount/shift adjustments effect on cost, contract and Service Level Agreement with the agency providing operators for call centre if any.

44. Heatmaps:

    a. Heatmaps by district/state; per-capita grievance rates.

b. Urban vs. rural passenger trends; seasonality effects, Major v/s smaller airport trends, trend for various city type, etc

45. Language-wise performance

46. Quality Assurance & Resolution Quality

    a. Audit outcomes by agent/queue/category.

    b. Reopen reasons, deviations from SOPs.

    c. Resolution text similarity (duplicate handling), macros usage analysis.

47. Governance check:

    a. Audit logs: configuration changes, role-based access events.

    b. Data retention checks

    c. Anomalies: sudden bulk edits, unusual login geos, high delete rates

48. Filters & Drilldowns (make everything sliceable)

    a. Time (relative & custom), Location, Channel, Category/Subcategory, Priority/Severity, Status, Agent/Queue/stakeholder, Language, Source app, etc.

49. Advanced analytics:

    a. Forecasting: seasonal volume & SLA breach forecasts (SARIMA/Prophet-style).

    b. Anomaly Detection: control charts (p-charts for breach rate)

    c. Cohort & Survival Analysis: time-to-resolution curves by category/priority.

    d. Topic Modelling & Clustering: auto-discover new themes; drift detection.

    e. Root Cause Discovery: association rules (e.g., {Category X, Region Y} ⇒ Breach).

    f. Queueing Simulations: simulate SLAs/ complaints under different staffing patterns/ different situations such as crises and accidents, etc.

    g. Prioritization Optimization: optimize assignment order under SLA and fairness constraints for workforce.

50. Recommended Visuals (map KPIs to charts)

a. Run charts / sparklines: inflow, backlog, SLA attainment, resolution verification and
ticket closure, complainant's satisfaction/ feedback trend.

b. Pareto bars: categories/regions driving most volume or breaches.

c. Box & whisker: TAT distribution by category/priority.

d. Sankey: channel → queue → resolution paths; transfer chains.

e. Heatmaps: SLA attainment by (category × region), agent × hour-of-day load.

f. Aging waterfalls & cohort heatmaps: backlog health and reopen cohorts.

g. Geo choropleths: grievance rates, SLA, CSAT by district/state.

h. Word clouds / key phrase bars: top topics from Neuro-linguistic programming (NLP) (backed by drillable tables).

2.5.14. AI based analytics

The portal should be able to conduct various types of analysis that can help in pro-active decision making. Some of the examples of AI based analytics have been listed below, the bidder shall identify more such features and add based on their as-is assessment or based on request from MoCA/AAI to improve the efficiency of the system:

1. Identify trends across similar grievance types
2. Learn from provided resolutions & feedback
3. Track issue patterns, including conflicts or disagreements
4. Predict required resources based on recurring trends & skill needs for airlines, control rooms like PACR and their own team handling AirSewa portal.
5. Highlight frequently asked queries
6. The data logs of the present version of AirSewa Portal will be used to train the AirSewa AI agent.

2.5.15. Ancillary modules

Communication Module

1. Notifications through portal/ app/ SMS:

    a. The portal should allow communication through notifications to mobile number through app, SMS, messenger app like WhatsApp, e-mail, etc (as per the preferred modes defined by MoCA/AAI), for various status updates (ticket id, ticket status, etc), OTP for authentication, etc.

2. PII masking: Personal Identifier Information (PII) should be masked in all communications

3. Data retention policy and implementation to maintain data privacy especially of personal data.

4. Setting user preference for mode for notifications: the user should be able to choose one or more modes of receiving the communication like app notifications, SMS, e-mail, etc.

5. Daily report sharing: Daily summary report of grievances added, resolved, escalated and on hold should be compiled and uploaded and accessible to concerned officials.

6. Chat for tech support: the portal should allow chat for tech support by the app users especially the redressors

7. Comment section for complainant & redressors: the portal should enable communication between the complainant and redressor, through messenger/ chat / comment section to seek and share additional information to support quick resolution of the compliant.

8. Filtration/ tagging of abusive/ inappropriate content as part of compliant/ grievances coming from various channels.

Reporting Module

This module shall support in extraction of reports from various dashboards/ MIS for users like MoCA/AAI, redressor organisations, etc. The portal shall ensure that:

1. There is a provision to generate digital reports from the portal

2. There is a provision to generate printable reports that can be physically printed

3. There is a provision for dynamic reporting, allowing to run queries on reports for granular level data and analysis

Logging Module

This module shall support in maintaining records of activities and events with time stamp and marking of login through which the activity took place. The system shall ensure the following:

1. System logs

2. Grievance logs

3. Security logs

User Profiles & Account Management

Access management provides the right for users to be able to use a service or group of services. Access management is the process of granting authorised users the right to use a service, while restricting access to non-authorised users. The system should be able to identify and authorize the users and would allow access to AirSewa portal and database. The user access management system would be able to identify the rights available with the user in terms of viewing, addition, deletion, modification of the data and generation of various reports through MIS.

Content management system

Content management system (CMS) is required to maintain the content on the AirSewa web-portal/ Mobile App. It should be possible to add content in form of Text, HTML, videos or Images on the department website. Along with the facility to easily publish content on the website, CMS would also have facility to maintain access logs for security and accountability purpose.

a. Configuration management and Version Control

- As the application undergoes enhancements and modifications due to problem requests, defect fixes and change requests, it becomes increasingly important to keep the source code under version control and the system under configuration management.
- With the objective of keeping track of the version updates in the application, Bidder shall adhere to the configuration management process defined in conjunction with MoCA/AAI.
- Bidder shall ensure that a copy of the production environment is backed up and stored in the repository before the new / modified components are copied to Production.

b. Release management

- Release management procedure shall be defined in conjunction with MoCA/AAI to ensure smooth transition of the application changes from release environment to production environment.
- As part of the release management Bidder shall perform the following activities:
    - Bidders shall group the related change requests, assess their development progress and accordingly prepare a schedule for their release.
    - Bidder shall in consultation with MoCA/AAI prepare a detailed release plan for every release. This plan shall include the release number and date of release. It shall also contain details about the change request to be released.

2.5.16. System and Software Services

The portal should support with the following services:

2.5.17. Rule based engine

The rule-based engine will support in defining rules and parameters in the backend for automation. This includes parameters for selection of escalation cases, priority cases, urgencies, etc. It also involves mapping of key words to classification and allocation of complaints, etc.

This is crucial in making the backend robust as verification of closure will rely on rule-based engine as well.

2.5.18. Backend mapping of guidelines/ rules

The AirSewa portal should have rules / compliances for multiple type of grievances mapped to CAR & other relevant DGCA rules / guidelines. The bidder shall be responsible for keeping them updated as per the amendments in regulations from time to time. The selection of relevant rules and guidelines that need to be mapped to comprehensively address all types of complaints and grievances shall be undertaken by the bidder in consultation with MoCA/AAI and stakeholders. The final list must be signed off by MoCA/AAI before mapping/ updating in the portal.

2.5.19. Workflow Engine

The workflow engine will be crucial in defining the process of the software application. This has been detailed in vol 1 of the RFP.

2.5.20. Query management

Query management service shall allow users to filter out and extract subsets of data from master date as described in this document.

2.5.21. API Management

AirSewa portal shall require multiple interactions with multiple other stakeholders from government as well as private sector such as:

i. Government sector entities:

   a. PACR

   b. CACR

   c. DGCA

   d. ASG (CISF/ State Police)

   e. AAI

   f. BCAS

ii. Private sector entities:

   a. Airlines

   b. Airport Operators

c. Other third parties (like OTA's)

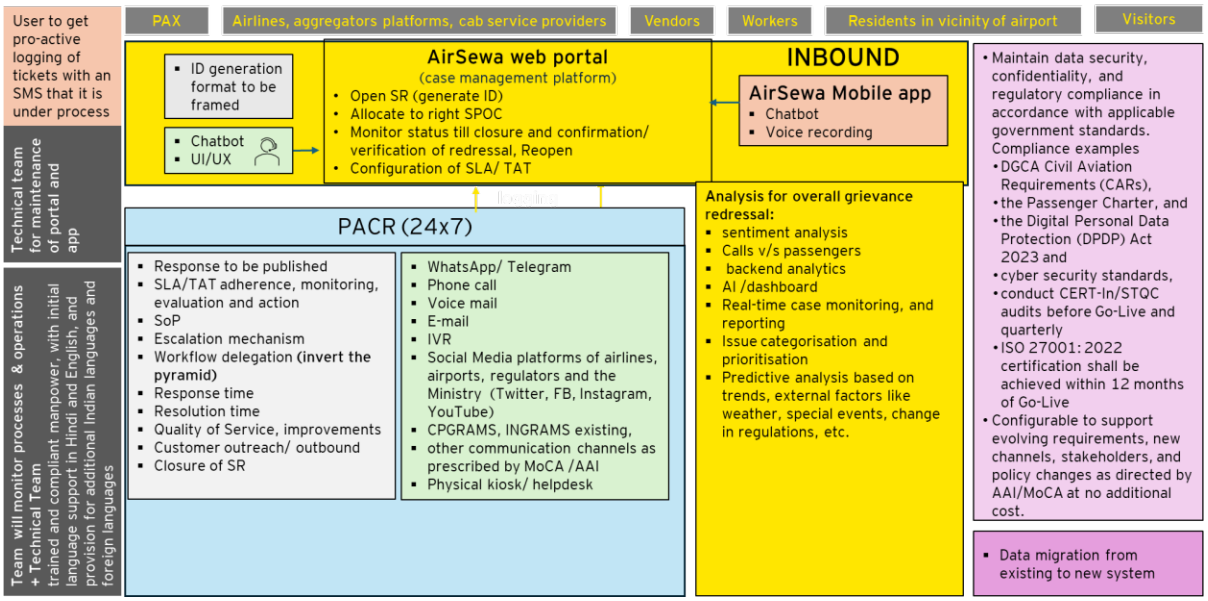


Figure 4: the MoCA/AAI's vision of AirSewa with PACR.

As can be seen in the image above, the Passenger Assistance Control Room (PACR), it is the main source of accepting grievances but the case management of the grievances in the PACR shall also be done in AirSewa portal. The input modes for grievances in PACR include:

a. Messenger apps like WhatsApp/ Telegram

b. Phone call

c. Voice mail

d. E-mail

e. IVR

f. Social Media platforms of airlines, airports, regulators and the Ministry (Twitter, FB, Instagram, YouTube)

g. CPGRAMS, INGRAMS existing,

h. other communication channels as prescribed by MoCA/AAI /AAI

i. Physical letter

1. AIRSEWA will provide an API ecosystem for integration with various stakeholder systems such as Airline, airport operators Online Travel Agents (OTAs) and other stakeholders mentioned in this document and figures above etc. to build innovative and meaningful commercial services. This must be built similarly to other Indian Digital Infrastructure projects like Digital Identity (Aadhaar), UPI etc. with the vision to create a vibrant eco-system around it.
2. MoCA/AAI intends to encourage independent application developers to develop tools (apps/portals) and other services for passengers relating to mentioned functions of AirSewa. To support this functionality, bidder shall publish and maintain the APIs, which will be consumed by these service providers. The information or query received through such apps/portals shall be received and processed by AIRSEWA.

3. Following are some of the key principles for API framework:

   i. API layer would not be exposed to untrusted connection.

   ii. All external redressors who do not have their own CRM (e.g. All AAI airports) will connect to AIRSEWA through SSL Layer of authentication along with user id, single sign authentication and other important security features. Those who have CRM like Airlines the Call to action will be open in their CRM.

   iii. All APIs level access to **AIRSEWA** (for users accessing the system) should be through **HTTPS**

   iv. AIRSEWA should have provision to support issuance of license key / sub licenses key including validation of the same in the system.

   v. All data transfer from / to AIRSEWA to happen through APIs

   vi. App signature authentication will be through the license key + time stamp + app version and other meta data

   vii. All the APIs would be stateless in nature, thus easy to load balance, even if hit through portal is very high and this requires high end processing.

viii. MoCA/AAI would prescribe the mechanism for empanelment of third-party systems who will use the AIRSEWA APIs and build apps using the same

ix. Bidder would deploy a developer sandbox to test the APIs with dummy data.

x. An API design document with the specification to start developing the interfaces. The APIs may be RESTful services with XML payload and may have the following minimum information in the design document but not limited to

a. Purpose of API
b. Author & Owner of API (controlling entity)
c. Input parameters
d. Output
e. Error codes

4. AIRSEWA is envisaged to be an API based solution where external agencies / eco system partners will also build & manage APIs as well as will set up secured networks to access the AIRSEWA. Hence, it is important that the bidder sets up, manages and monitors the API services for AIRSEWA. This entire framework needs to be set up and operationalised as part of the scope of work. The broad scope of the bidder relating to this function is detailed below:

i. Enrolment and operations

MoCA/AAI will be the overall regulator and overseer of the API based system. As part of this project, bidder will set up the requisite process as well as system to build, operate & manage and sustain APIs for third party service provider in a secured and controlled environment. This will establish secure connectivity compliant with AIRSEWA standards and specifications. Third party service provider will offer their AIRSEWA-compliant network connectivity as a service and transmit authentication requests to AIRSEWA.

Third Party Systems will use AIRSEWA authentication to enable its services and connects to the AIRSEWA through an internet connectivity after validation of license key.

ii. Authorization and License Key Management

Bidder should create an administrative portal to enable third party systems to have a user account to manage their services. System shall ensure only the authorized persons of third-party system have access to this admin portal.

iii. Standardizing API and specification

Based on the solution developed API needs to be standardized and specification must be defined so that it can be exposed in the stakeholder ecosystem and third-party systems can create their own services and expose them to the outer world for stakeholder use. Bidder should have the ability to define, document, edit and publish API specifications.

iv. Environment Management

Bidder should provide a sandbox environment where third party systems can publish a mock version of APIs. The admin portal shall be used to create third party systems Dev IDs that can be accessed by the developers of potential third-party systems for development, test and integration. User can perform testing in a sandbox environment which is distinct from production. Sandbox shall provide the same catalogue as the production framework. All the APIs shall be hosted in sand box environment to ensure at-least couple of third-party systems integrate/test before the API is moved to production.

v. User Authentication

Bidder to provide authentication services for allowing users to access the above-mentioned environments and to do the following operations

a) To authenticate user into the Sandbox.

b) To configure authorisation policy for new APIs as they are introduced to the framework.

c) Allow user to access the available APIs and associated properties in accordance with his/her entitlements.

d) Allow third party app exposed to the API, resources data in accordance with the configuration for that app.

e) Blacklist/Block Access

vi. Publishing and Management of API

There should be a mechanism that allows authorized users to publish new APIs as they are created to sandbox, test and production environments as required. Once the API are developed and deployed in the sandbox environment bidder to do a proper functional, security and performance test and certify the API before being published for production usages. An API catalogue should be maintained by the system.

vii. Version Control

Bidder should provide a controlled mechanism for API versioning control for any change. The version & release management process must cover this aspect to ensure any change is made or rolled out in a controlled & informed manner.

viii. API Retirement

Bidder should provide a mechanism to retire/archive APIs. The solution should provide full support of managing, retiring and archiving APIs as part of the life cycle and associated version control.

ix. API Governance & SLA enforcement

Bidder should provide a mechanism to define and enforce SLAs/quotas for consuming entities of the API framework. The solution should provide "Plan" such as to control how much traffic can be sent by each user through the interface. A Plan to make available a collection of resources from one or more APIs, or a plan defines a rate limiting policy that specifies how many requests an application is allowed to make during a specified time interval, and what action should be taken when the threshold is exceeded. The solution should support both a hard limit which throttles the traffic and a soft limit which notifies the administrator about the policy violation. For soft limit the solution should be further customised to buffer the extra messages to shape the traffic. Application developers gain access to APIs by registering their applications to access plans by using the developer portal.

Solution should also provide real time SLA monitoring capability. Solution should enable configuration of system SLAs. The APIs load shall be continuously and pro-actively monitored for suitable & prompt actions in case of excessive loads, failures or performance bottlenecks. Administrator should perform appropriate actions as SLA thresholds are approached/ reached
- including notifications, throttling, rejection of requests, raising alarms.

API framework and AIRSEWA must work together to synchronize API and back-end service policies. Additionally, API lifecycle stages should be synchronized with system service implementation stages. An API Governance experience may provide a straightforward set of lifecycle stages (e.g., created, published, deprecated, retired, blocked) that may be customized by the development team.

x. <u>API Updates, Notification and tech support</u>

Bidder should provide a mechanism to provide consuming entities with appropriate notifications with respect to APIs. The solution should ensure that application developers who wish to access and use the APIs are self-sufficient. Documentation about an API, such as URL used to call the API and the security mechanism used by the API to authenticate application user, is automatically generated when defining the API and exposed through the developer portal. Additional supporting documentation that can further help application developer to use the API, such as samples and/or tutorials and other supporting documentation shall be available through the developer portal.

xi. <u>API Security Governance</u>

Solution should provide appropriate & adequate security mechanisms governing access to API framework- i.e. Authentication strength. The system should inspect the headers for APIs genuineness before acceptance. It should apply all security checks e.g. DDoS Attacks, XML Denial of Service (xDoS), Slow down or disable an XML based System, Message Snooping, XML Document Size Attacks, XML Document Width Attacks, XML Document Depth Attacks, Jumbo Payloads, Recursive Elements, Public Key DoS, XML Flood, Resource Hijack etc. to ensure rightful and secured access to API consumers.

xii. Audit Trail

Solution should allow the audit tracking of dev/client apps consuming APIs and portal API. Audit log should have capability to track (not an exhaustive list)-

a) User ID/Credential
b) Transaction data
c) IP addresses of source and destination
d) Date & time stamp
e) Session duration
f) Session ID/key (Session key must be dynamically generated for every transaction session key must not be reused)
g) License Key

xiii. API Usage Activity

Solution should provide with reporting data on the usage/transactions performed by users/consuming entities on exposed API. The solution should be capable of generating reports/Online view that shows API usage information such as most/least accessed APIs, highest/lowest/average response time, data volumes transferred, transfer rates, the maximum and minimum response time over a period, rejected API requests. Custom queries can be created to further refine the API usage information such as identifying API calls that falls within a window of response time.

2.5.22. Prioritisation optimisation

The prioritisation optimisation refers to the service that supports in managing workload and queue of the AirSewa agents for optimisation of workload.

2.5.23. Identity & Access Management

The system should allow nodal officers to create more logins, manage access to various officers in their organisation in case of new hires, transfers, addition of responsibilities, etc.

2.5.24. Multi-lingual services

i. The portal should support multi-lingual interaction as the complainant can be an Indian who knows only their regional language or an international traveller who does not speak English.
ii. The portal should be meticulously engineered for accessibility to enable its use by Persons with Disabilities (PwD).
iii. Verification process of complaint closure

The following shall be the means to verify if the case can be considered closed:

a. Tagged airline/stakeholder must provide evidence before closure

b. AI-based auto-verification aligned with CAR/DGCA rules

c. Passenger feedback collected for every case

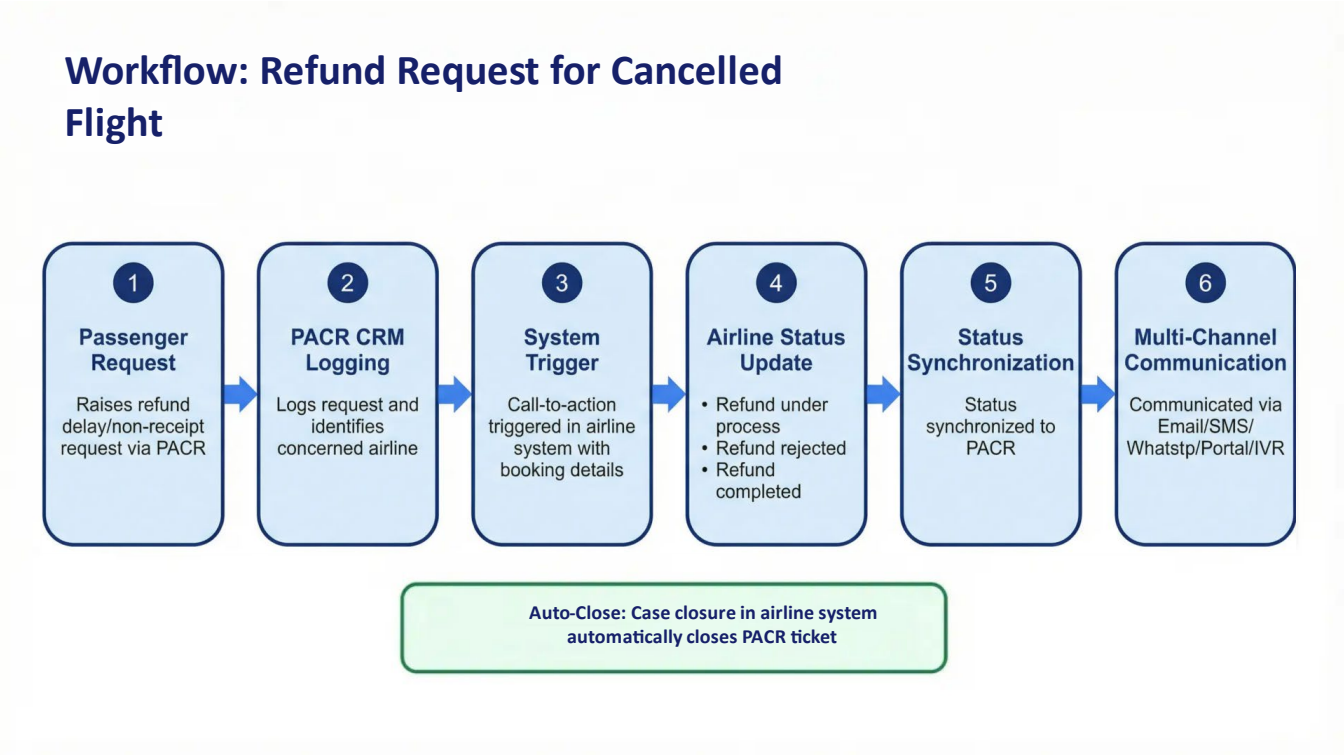
The closure will be considered complete only if the passenger confirms the same in feedback. In case the passenger is not happy with the resolution while the AI based mechanism indicates the resolution was compliant to the guidelines, a manual intervention may be required to cross check the issue with the complainant. Such cases shall go to a Case Closure Committee to take a call on the genuineness of the ask/ expectation by the Complainant. This Committee shall be chaired by the representative from MoCA/AAI/ DGCA and will decide the optimum resolution or may decide to close the case/ ticket if it is found to be ingenuine.

2.5.25. Indicative Illustration of Use Cases PACR–Stakeholder Integration

These illustrative use cases demonstrate the intended operating model of PACR as a centralized coordination and monitoring platform, enabling passengers to receive consistent updates and enabling AAI to oversee stakeholder responsiveness through an integrated omni-channel CRM, Chatbot and Helpdesk system.

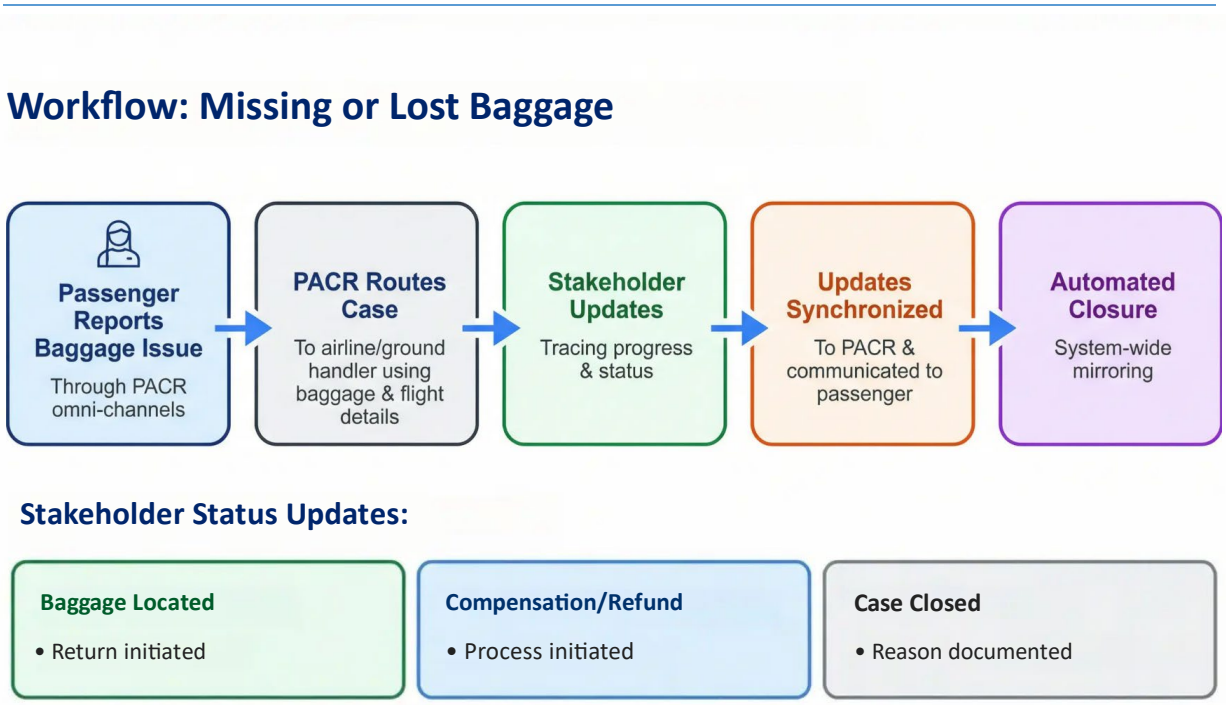
2.5.26. Refund Request for Cancelled Flight Ticket / Delay in Refund

1. A passenger raises a request through PACR regarding non-receipt or delay in refund following cancellation of a flight ticket.
2. The PACR CRM registers the request and identifies the concerned airline based on flight and booking details.
3. The PACR system triggers a call to action in the airline's CRM/service management system, transmitting passenger details, booking reference, refund category, and timelines.
4. The airline validates the request and updates its system with one of the following actions:
   - Refund under process with tentative amount and expected date
   - Refund rejected with reason
   - Refund completed with transaction reference and disbursement date
5. Each status update is synchronized back to the PACR system and automatically communicated to the passenger through configured channels (email/ SMS/ WhatsApp/portal/IVR).
6. On completion of refund or final rejection, the airline closes the case in its system, which triggers corresponding closure in the PACR system.



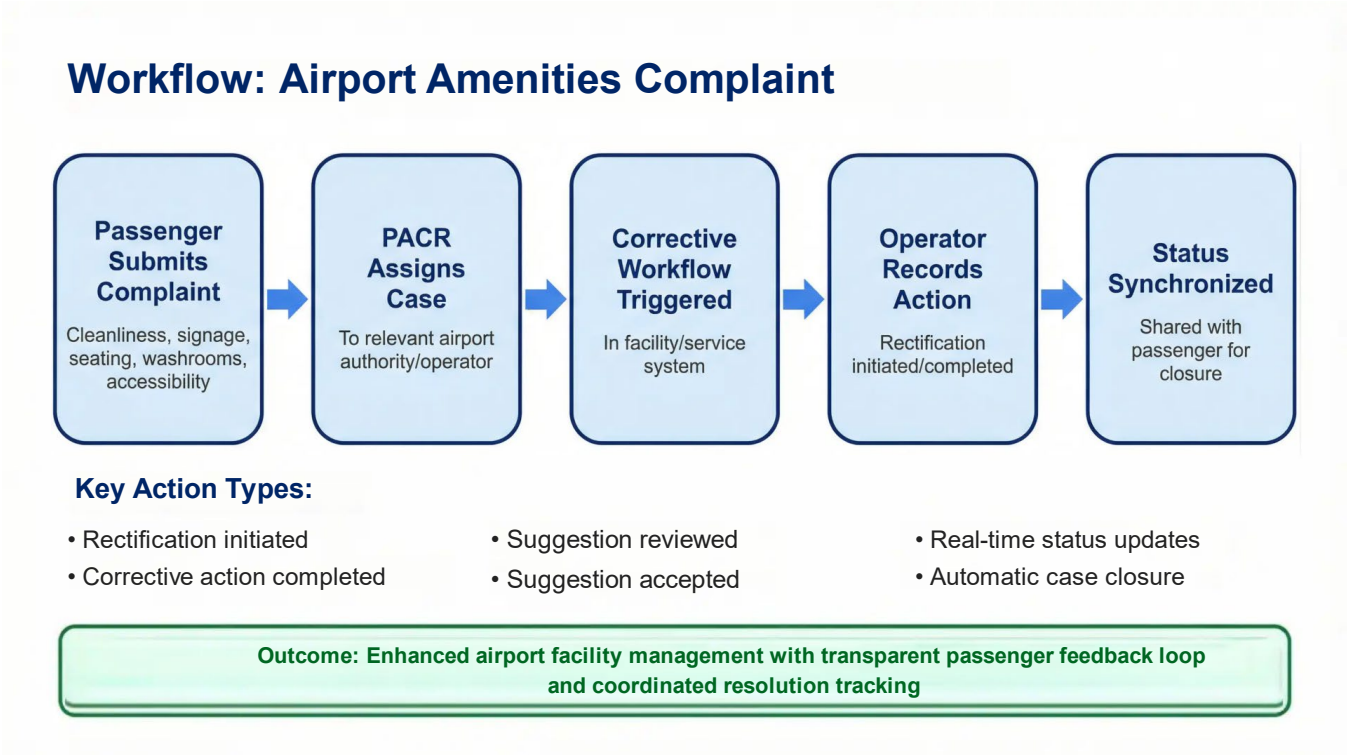
2.5.27. Missing / Lost / Left-Behind Baggage – Return or Refund

1. A passenger approaches PACR regarding missing, lost, or left-behind baggage, seeking tracking, return, or compensation/refund.
2. The PACR CRM logs the case and routes it to the responsible airline or ground handling agency based on baggage tag and flight details.
3. The PACR system triggers a call to action in the stakeholder's system with baggage reference details and passenger contact information.
4. The stakeholder updates its system as baggage tracing progresses, including:
   - Baggage located and return initiated
   - Compensation/refund process initiated
   - Request closed with reason (e.g., baggage recovered and delivered)
5. Status updates, including tentative timelines and final outcomes, are synchronized to the PACR system and communicated to the passenger.
6. Closure or reopening of the case in the stakeholder's system is mirrored automatically in the PACR system.



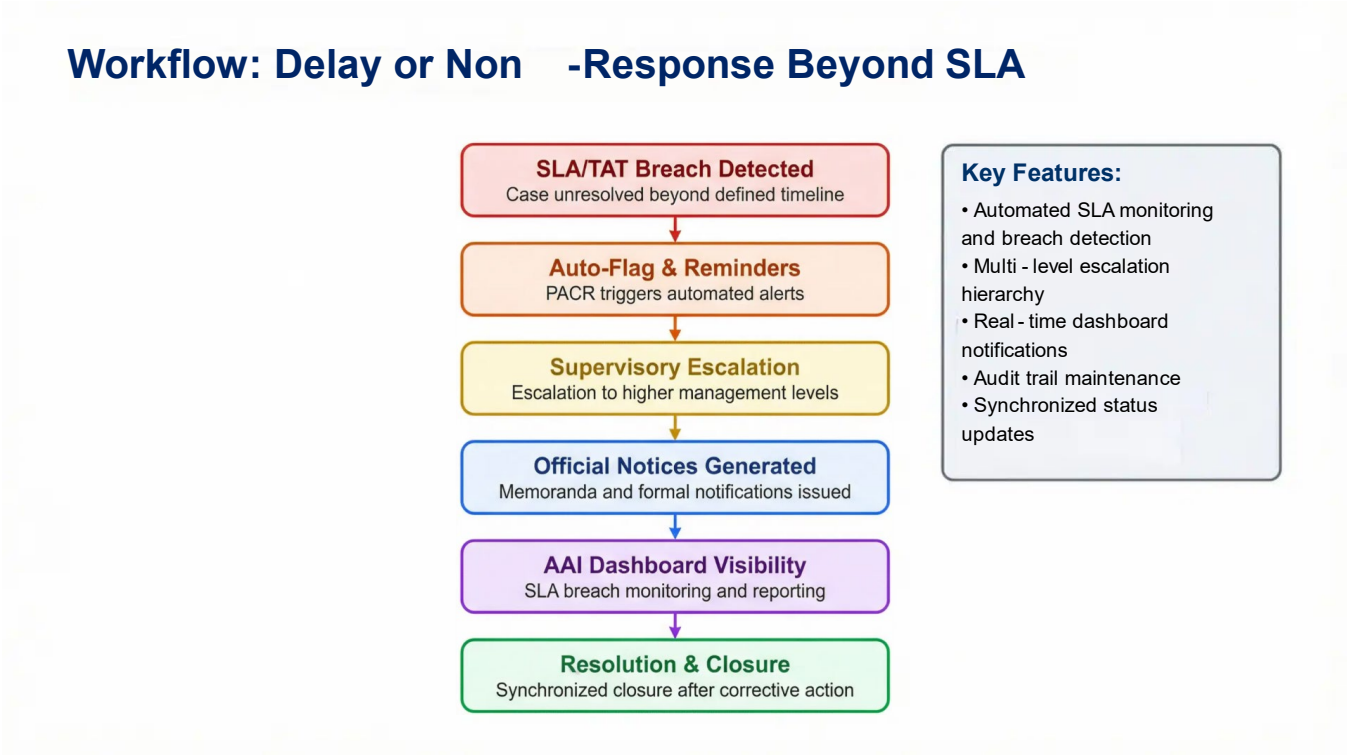
2.5.28. Airport Amenities Issue or Complaint against Airport Operators

1. A passenger submits a complaint, issue, or suggestion through PACR regarding airport amenities such as cleanliness, signage, seating, washrooms, accessibility, or facilities for senior citizens and persons with reduced mobility.
2. The PACR CRM registers the case and assigns it to the concerned airport authority or airport operator based on location and issue category.
3. A call to action is triggered in the airport authority's/operator service or facility management system.
4. The airport authority/operator records actions taken, such as:
   - Issue acknowledged and rectification initiated
   - Corrective action completed
   - Suggestion reviewed and accepted/noted
5. The status of corrective action and closure remarks are synchronized back to the PACR system.
6. The passenger receives updates and closure confirmation through PACR channels.



2.5.29. Delay or Non-Response by Stakeholder Beyond SLA

1. A passenger's case assigned through PACR remains unresolved beyond the defined TAT/SLA.
2. The PACR system automatically flags the breach and triggers:
   - Automated reminders to the concerned stakeholder
   - Escalation notifications to designated supervisory levels
3. Continued non-compliance results in:
   - Mass or case-specific notices/memoranda generated through PACR
   - Visibility of SLA breaches on AAI dashboards
4. Resolution and subsequent closure updates are synchronized Across systems once action is taken.



2.5.30. System Expectations Across All Use Cases

1. Across all scenarios, the PACR system shall ensure:
   - Bi-directional, real-time or near real-time integration with stakeholder systems
   - Unified ticketing with synchronized status, closure, and reopening
   - Omni-channel passenger communication independent of ticket origination channel
   - Complete audit trail of all actions, communications, and timelines
   - Consolidated dashboards for AAI to monitor stakeholder performance and passenger experience
2. These illustrative use cases demonstrate the intended operating model of PACR as a centralized coordination and monitoring platform, enabling passengers to receive consistent updates and enabling AAI to oversee stakeholder responsiveness through an integrated omni-channel CRM and Helpdesk system.

2.5.31. PACR Integration for any stakeholder without Native CRM Systems (e.g. Airport Operators, CISF, BCAS etc)

1. Background: Certain airport operators manage multiple airports but do not maintain a centralized digital CRM or grievance management system, relying instead on landline-based helpdesks and manual registers. This annexure illustrates how such operators shall be enabled to participate in the Passenger Assistance and Control Room (ACR) ecosystem.

3. Illustrative Stakeholder: *"AAI"* – an airport operator managing **multiple airports**, without an existing CRM system.

4. Objective: To enable airport operators without native CRM systems to digitally receive, manage, update, and resolve passenger grievances through the PACR CRM platform, Use Case Description

   - **Role-Based Access**: PACR shall provide role-based login access to the airport operator for each airport managed by it. Designated Helpdesk Head(s) / authorized team members shall be mapped to their respective airports, with access restricted to relevant tickets only.

   - **Passenger-Initiated Grievance via PACR**: Passengers may register grievances related to the operator's airports through any PACR omni-channel (call centre, AirSewa, email, social media, etc.). PACR shall register the ticket, categorize the issue, identify the responsible airport, and generate a call-to-action visible in the respective airport's PACR login.

   - **Ticket Handling by Airport Helpdesk**: The airport helpdesk shall acknowledge, update, and resolve the ticket directly within PACR. All actions and communications shall be time-stamped and tracked against applicable SLA/TAT norms.

   - Stakeholder-**Initiated Ticket Creation**: The airport helpdesk may also create tickets directly in PACR for walk-in complaints, landline calls, or operational observations. Such tickets shall follow the same lifecycle, escalation, and audit mechanisms.

   - **Ticket Number Mapping and Visibility**: PACR shall allow tagging and mapping of ticket numbers generated by the airport operator, airlines, or other stakeholders to PACR-created tickets. PACR shall be able to view the complete ticket history and maintain synchronized references for traceability.

   - **Monitoring, Escalation, and Oversight**: PACR dashboards shall provide real-time visibility to AAI and authorized users on grievance volumes, resolution timelines, and SLA compliance Across all airports of the operator. Automated reminders, escalations, and a complete audit trail shall support regulatory and administrative oversight.

5. ensuring end-to-end visibility, SLA compliance, auditability, and effective coordination with AAI and other stakeholders.

2.5.32. Data Layer

Data Layer consist of Databases mapped to the technical requirements of the platform. An indicative set of databases is given below

i. Transactions database

ii. Master database

iii. User database

iv. Configuration database

v. Unstructured Data storage

vi. Audit Logs database

vii. Legacy data storage

Bidder shall propose optimized schemas and storage type for storing, processing & retrieving data.

Additionally, suitable Backup & Archival services should be configured to ensure data resiliency, recovery and availability to meet business requirements and SLAs.
The CSP shall ensure that all data transmitted, stored and processed is protected using industry-standard encryption protocols and access controls.

2.5.33. Messenger system (Chat window)

The portal should allow chat, e-mail and notifications, comments by stakeholders that are addressing the complaints so that they can communicate through the portal and not have to coordinate manually.

i. Simple form requiring only PNR, grievance type, description, and evidence

ii. Factor authentication

iii. Option to reopen tickets

iv. Multilingual interface for wider accessibility.

The AirSewa shall be the core of the system. It will host various passenger services as detailed in the scope below. It will also act as an integration platform for other components like analytics and reporting etc.

2.5.34. Chatbot Support

The features of the Chatbot will be as follows:

1. Responding to all the passenger queries related to registration, creation, etc. on AIRSEWA website/ mobile application. If the passenger is not satisfied, Chatbot solution should have the capability to forward the query to help desk official and update the status of query to customer after its resolution.
2. Leveraging technologies like Natural Learning Process (NLP) and Artificial Intelligence (AI) to enhance contextual understanding, generative responses and learning abilities while doing end to end conversation with passengers.
3. Providing an analytical tool which can analyse the passenger data created during conversations based on various parameters like sentimental analysis, etc.
4. Interacting with the user and responding to various level queries raised in English.
5. Understanding & executing compound query, at the same time ability to identify different intents and act upon both simultaneously.
6. Not storing any of the information entered by user anywhere locally on device and it should be stored in encrypted format on servers only.

2.5.35. AI Automation suite

SI shall be responsible for automation of the following processes by use of advance AI mechanism:

1. Auto tagging and AI driven verification to reduce manual effort:
   i. The bidder shall be responsible for creating an exhaustive list of complaints that can arise, decide relevant bucketing of those into categories. This can be modified and updated to improve the efficiency of the system.
   ii. The system should be able to pick up the key words from the complaint details given by the User, map it to correct category and tag it to the relevant stakeholder/ authority/ department.
   iii. The system should be able to verify the resolution by comparing it with the relevant guidelines mapped in the system. These guidelines and regulations like Civil Aviation Regulations (CAR) and other such guidelines should be updated in real time at backend to avoid the wrong verification and closure of reports leading to multiple re-opening and escalations of the complaints.
2. Automatic routing of grievances to the relevant team/SPOC
3. Auto triggered escalation workflow especially in priority cases

   i. The bidder shall define an exhaustive list of cases where the grievance case is urgent (like health emergency, medicines involved in the baggage, admit card of a student in a baggage, etc.) and should be put on priority. This list should be signed off by MoCA/AAI and added to the backend. This list should be updated from time to time by learning from past cases if any modifications are required.
   ii. Based on the defined cases that need to be taken on priority, the system should be able to recognise such cases based on the key words used by complainant and should be highlighted as a priority case to the stakeholder tagged to the complaint. The TAT of resolution of such cases shall be lesser as defined by bidder after discussion and sign-off from MoCA/AAI.
4. Predictive congestion reporting for early issue detection:

   The system shall be able to predict congestion reporting based on data and past trends. The probable causes of congestion could be:

i. Poor performance of the airlines like multiple flight delays and cancellations

ii. Change in some regulations by government

iii. Seasonal issues like delays/ cancellations/ diversions due to fog

iv. High volume of traffic expected during special events in a city

v. Hostile / war like situation
vi. Airspace closure

vii. Airport/ runway closure due to technical issues like failure in equipment functioning on runway, etc.

To predict the same, relevant data needs to be collected/ fetched by the system from relevant stakeholders in the aviation sector. This will require multiple integrations which have been described in subsequent sections.

5. AI assistance in summarising grievance and mapping it to relevant rules for the Redressor / MoCA/AAI, to help decide the right redressal of the grievance
6. Conversational AI to translate speech to text for multiple languages for the grievances received.

7. AI based ticket classification and allocation to the concerned redressor/ stakeholder

8. AI based identification of duplicate cases.

2.5.36. Cloud and PACR Infrastructure

2.5.36.1 Overall Cloud requirements:

i. Cloud Service Provider (CSP) must be Meghraj 2.0 as managed by NIC/NICSI, for the relevant deployment model and services with ongoing STQC audit as per the MeitY empanelment process.

ii. CSP must comply with all security-related requirements, guidelines, and regulations issued by MeitY, CERT-In, RBI, Ministry of Home Affairs (MHA) or any standards body recognized by the Government of India. This includes existing and future mandatory standards that Government of India may notify from time to time.

iii. The CSP shall provide the required cloud services from its Data Centre facilities located within India.

iv. The primary and DR Data Centre must be in separate seismic zones.

v. The CSP shall ensure that underlying Data Centre facility must be at least Tier III or higher standard (certified under TIA 942 or Uptime Institute certifications) and implement tool-based processes based on ITIL standards.

vi. CSP shall also ensure that all the purchaser data and system logs shall be stored and processed within India in the MeitY-empanelled regions/facilities.

vii. Vulnerability assessment shall be conducted each quarter and to be rectified by SI & yearly penetration testing for complete applications and platform, the compliance report needs to be submitted to MoCA/AAI.

2.5.36.2 Cloud and PACR service requirements:

1. The cloud services must provide scalable, redundant and elastic compute and storage resources.

2. The service must allow configuration of VMs with required vCPU, RAM, and disk capacities.

3. The service must allow users to procure and access compute and storage capabilities remotely over SSL with two-factor authentication.

4. The service must support VM image backups and enable taking, importing, and exporting of VM instances in a MeitY approved image format.

5. Configuration and management of virtual machines must be accessible only through SSL-based VPN clients and not via the public internet.

6. SP Shall ensure that all VMs receive operating system patching, health checks, systematic attack detection, and backup functions.

7. The service must monitor VM availability and resource utilization, including RAM, CPU, disk, IOPS, and network metrics.

8. The purchaser shall retain the complete ownership of all virtual machines, templates, clones, and scripts/applications created for the organization.

9. The purchaser shall have the right to request full copies of any virtual machine at any time.

10. The purchaser shall retain the ownership of all business applications, solutions, or bespoke software installed on VMs.

11. The service must support secure administration interfaces (e.g., SSL/TLS or SSH) for authorized personnel to remotely manage their virtual instances.

12. The service must support dynamic allocation of virtual machines based on load, without service interruption.

13. The service must provide the capability to copy or clone VMs for archiving, testing, and troubleshooting.

14. The SP shall provide secure, encrypted tools and mechanisms for the purchaser or its appointed agency to define backup requirements and policies, as well as to configure, schedule, perform, and manage backups and restores for all data types including files, folders, images, system states, databases, enterprise applications etc.

15. The SP shall support data transfer back to the purchaser's environment either on demand or upon termination of the contract.

16. The SP shall not delete any purchaser data for a period of up to 90 days after contract expiry unless expressly approved by the purchaser.
17. The cloud platform must support live migration of running VMs from one host to another.
18. The service must support all APIs provided by the SP, including REST APIs.
19. The SP must offer fine-grained access controls including role-based access control (RBAC), SSL certificates and multi-factor authentication.
20. The SP shall ensure that offered cloud platform and services must support comprehensive auditing, including details of requests made, source IP, user identity, timestamp, and related events.
21. The SP shall ensure that the purchaser shall be permitted to bring, upload and run properly licensed non-OS software required for their solutions by using it directly on VMs.
22. VM CPU and RAM must automatically scale during load spikes to ensure application availability.
23. Native cloud-based storage solutions may be used if they meet all stated requirements and are supported by appropriate specifications and customer references.

2.5.36.3 Disaster Recovery (DR), Replication & Continuity Requirements:

1. Manage the network, storage, server and virtualization layers, to include performance of internal technology refresh cycles applicable to meet the SLAs.

2. Provide a secure, dual factor method of remote access which allows the purchaser's designated personnel (privileged users) the ability to perform duties on the hosted infrastructure

3. Upgrade and periodically replace hardware without financial impact to the purchaser. All the data within it shall be immediately deleted/destroyed and certify the VM and data destruction to the purchaser as per stipulations and shall ensure that the data cannot be forensically recovered.

4. Bidder to perform patch management appropriate to the scope of their control including:

    i. Alerts on the upcoming patches via email and management portal, and ability to defer or reject patches before they are applied in the next patch cycle

    ii. Patch approved VMs on the next available patch management change window

    iii. Application of automated OS security patches, unless deferred or rejected by purchaser or designated agency

    iv. Send regular approval reminders to purchaser or authorized agency designated email address five (5) days prior to patch cut-off dates

5. Bidder should undertake OS level vulnerability management – all OS images created within the cloud platform are regularly patched with the latest security updates

6. Provide the artefacts, security policies and procedures demonstrating its compliance with the Security Assessment and Authorization requirements as described in Security Requirements in this RFP.

7. Monitor availability of the servers, CSP -supplied operating system & system software, and CSP's network

8. The bidder is fully responsible for tech refreshes, patch management and other operations of infrastructure with regards to the cloud environment (DC and DR).

9. Investigate outages, perform appropriate corrective action to restore the hardware, operating system, and related tools

10. Bidder should manage CSP provisioned infrastructure including VMs as per the ITIL or equivalent industry standards.

11. Comply with technology refresh requirements as mandated by CERT-IN and MeitY

12. Software within the CSP's scope will never be more than two versions behind unless deferred or rejected by MeitY / Purchaser / Purchaser's authorized agency.

13. Manage the network, storage, server and virtualization layers, to include performance of internal technology refresh cycles applicable to meet the SLAs

14. Provide a secure, dual factor method of remote access which allows the purchaser's designated personnel (privileged users) the ability to perform duties on the hosted infrastructure

15. Upgrade and periodically replace hardware without financial impact to the purchaser. All the data within it shall be immediately deleted/destroyed and certify the VM and data destruction to the purchaser as per stipulations and shall ensure that the data cannot be forensically recovered.

16. Bidder to perform patch management appropriate to the scope of their control including:

    i. Alerts on the upcoming patches via email and management portal, and ability to defer or reject patches before they are applied in the next patch cycle

    ii. Patch approved VMs on the next available patch management change window

    iii. Application of automated OS security patches, unless deferred or rejected by purchaser or designated agency

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19. Monitor availability of the servers, CSP -supplied operating system & system software, and CSP's network

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21. Investigate outages, perform appropriate corrective action to restore the hardware, operating system, and related tools

22. Bidder should manage CSP provisioned infrastructure including VMs as per the ITIL or equivalent industry standards.

23. Comply with technology refresh requirements as mandated by CERT-IN and MeitY

24. Software within the CSP's scope will never be more than two versions behind unless deferred or rejected by MeitY / Purchaser / Purchaser's authorized agency.

2.5.36.4 Cloud and PACR management reporting requirements:

1. Provide service level management reports (as per the service levels agreed in the Service Level Agreement between the purchaser and the CSP)
2. description of major outages (including description of root-cause and fix) resulting in greater than 1-hour of unscheduled downtime within a month
3. Helpdesk and PACR tickets reports submitted
4. Monthly and quarterly utilization reports (peak and average volumetric details)
5. Bidder should provide a portal for the purchaser (administration role) which should provide data related to:
    i. Utilization reports (with threshold limits defined by the user)
    ii. SLA reports
    iii. Cloud service usage
    iv. Helpdesk and tickets
    v. User profile management
6. The bidder should set the baseline threshold limits for cloud infra utilization.
7. In the event of cloud infra utilization breaching the baseline threshold limits, the CSP is required to notify the purchaser and MSP with adequate justifications for increasing baseline capacity.

2.5.36.5 Infrastructure provisioning for cloud and PACR requirements:

The objective of this RFP is to select a Service Provider (SP) who shall be responsible for Design, Development, Integration, Deployment and Maintenance of Grievance Management System (GMS), PACR and associated Customer Relationship Module (CRM) on Cloud Infrastructure.

2.5.36.6 Cloud Deployment Model

i. The entire GMS and CRM solution shall be deployed on a **Virtual Private Cloud (VPC)**– based infrastructure.

ii. All monitoring, provisioning, should be within India and 100% isolated from other regions outside India, if in case CSP has Global presence, on Data Centres empanelled with **MeitY**.

2.5.36.7 Primary & Disaster Recovery (DR) on cloud Environment Setup

i. The solution must be deployed on **Virtual Private Cloud (VPC)**–based infrastructure and **Disaster Recovery (DR-** Cloud Based**)** sites.

ii. The infrastructure shall support **automated DR** (cloud based), ensuring seamless continuity of business operations with no single point of failure.

2.5.36.8 Scalability & Performance

i. The offered cloud platform and PACR Infra must provide **inherent auto-scaling capabilities**, enabling vertical and horizontal scaling based on workload.

ii. The system should sustain **peak load conditions without performance degradation** or service disruption. The additional resources to be reduced when the load is reduced (off peak phases).

2.5.36.9 Infrastructure Provisioning

i. The cloud environment shall include all required:

- Compute resources (servers, VMs, containers)
- Storage components (block, object, archival)
- Networking components (router, switches,)
- Cybersecurity components (VPC, firewalls, load balancers, VPN, connectivity to cloud hosting environment)

ii. Infrastructure must be equipped with **secure, redundant, and high-throughput connectivity** for the cloud hosting environment.

iii. PACR infrastructures as per Annexure-IV PACR Compliance "Minimum specification for Desktop & with IVRS"

The architecture must be optimized to ensure high availability, reliability, and consistent performance for all critical components.

2.5.36.10 Hardware & Software Requirements

i. Service Provider (SP) must ensure that **all hardware, software, middleware, and licenses** meet the requirements specified in this RFP.

ii. All components must be **OEM-recommended, supported, and scalable** for future enhancements and upgrades.

iii. SP shall ensure compliance with:

- MeitY cloud guidelines
- Industry best practices
- Relevant security frameworks and standards

2.5.36.11 Security

The SP shall provide end-to-end security, including:

i. Identity and Access Management

ii. Network security

iii. Data encryption (at rest and in transit)

iv. Monitoring and security operations

v. Regular patching and vulnerability management

2.5.36.12 Skilled Resources

The SP shall deploy qualified technical professionals for:

i. Cloud infrastructure setup

ii. Application deployment

iii. Monitoring and administration

iv. DR (On cloud) drills and failover testing

v. Security and compliance management

vi. PACR

2.5.36.13 Sizing and Assessment

i. Initial sizing of cloud infrastructure shall be based on information provided in this RFP.

ii. A comprehensive assessment and detailed sizing exercise shall be conducted by the SP post-award to ensure **optimal resource allocation**, reflecting actual business needs.

2.5.36.14 Operations, Maintenance & Support

SP shall undertake complete responsibility for:

i. Installation and deployment

ii. Ongoing maintenance

iii. Infrastructure upkeeping

iv. Monitoring and performance management

v. Ensuring SLA adherence and timely issue resolution

2.5.36.15 Availability & Resilience

i. Production SLA ≥ **99.95%**.

ii. SLA tools shall be provisioned by SI

iii. DR Site (On cloud): DR site (On cloud) must be available to maintain above SLA.

iv. The system must be highly scalable to cater maximum pick load capacity.

v. System should support VM or container orchestration.

vi. Bidder to submit High-level design (HLD), capacity plan and sizing assumptions, availability design, DR (On cloud) strategy checklist.

vii. DR (On Cloud) site geographically distant or logically isolated setup for continuity during major failures as per MeitY latest guideline.

viii. DR (On Cloud) drill may be conducted by MoCA/AAI on every six month for 24 Hours.

ix. Scalable compute resources supporting microservices, APIs, analytics, automation engines, and data services.

x. High-performance storage for transactional workloads and scalable storage for archives, logs

xi. Supports backup, archival, and retention policies.

2.5.37. Networking and communication

2.5.37.1. Overview

i. The Service Provider (SP) shall be responsible for end-to-end provisioning, deployment, management, monitoring, and security of the network and communication infrastructure required for hosting and accessing the proposed solution Across Cloud Data Centre (DC), Disaster Recovery (DR on cloud environment), PACR and all connected environments. The SP shall ensure secure, reliable, scalable, and policy-compliant connectivity in alignment with industry standards and MeitY guidelines.

ii. For PACR SP shall do procurement (refer Annexure-IV, PACR Compliance "Minimum specification for Desktop & with IVRS") and deployment/installation/ customization of all infrastructure required for related operations, including but not limited to - Telephony infrastructure (PBX, ACD, IVR, PRI/SIP/VoIP)- Call servers, UCM, storage, firewalls, switches, routers, racks - Desktops, peripherals, LAN, headsets with noise cancellation - Third-party tools, reporting systems, and utilities. SP to ensure that ICT (Infrastructure & Communication technology) infrastructure is latest & updated with all patches applied on the Software, Hardware and OS.

iii. Any other facility required to establish Call Center (CC) not listed in this section shall be responsibility of SP as mentioned elsewhere in the document.

iv. MOCA/AAI shall provide premises, electricity with power backup, water supply, PRI/SIP subscription

2.5.37.2. Bandwidth Provisioning

1. The Service Provider shall provision **requisite network bandwidth** to ensure seamless interconnectivity between:
   i. Application components,
   ii. Cloud Data Centre,
   iii. Disaster Recovery Centre (On cloud Environment).
   iv. PACR
2. The service provider is responsible for determining the **optimal network bandwidth sizing** based on:
   i. Solution architecture,
   ii. User concurrency,

iii. Application traffic behaviour,

iv. Transaction volume, and

v. Peak load requirements.

2.5.37.3. Ongoing Review

The service provider shall continuously review and optimize bandwidth requirements based on system performance and usage trends.

2.5.37.4. Core Network Components

The service provider shall provision and manage a fully functional Virtual Private Cloud (VPC)–based network infrastructure including:

i. Virtual networks

ii. Subnets

iii. Firewalls

iv. Network security groups

v. Load balancers

vi. Gateways & VPNs

vii. Routing configurations

viii. Cloud environment interconnectivity

2.5.37.5. Essential Network Services

The cloud infrastructure shall support enterprise-grade core network services including:

i. Directory Services

ii. DNS

iii. DHCP

iv. RADIUS

v. Web Server

vi. Application Server

vii. Cluster / High Availability Services

viii. Global File System Support

ix. Virtualization Support

2.5.37.6. Network Segmentation, Isolation & Logical Security

i. The Service provider shall ensure logical segregation of all information assets including:

   a. Servers
   b. Applications
   c. Databases
   d. Communication links
   e. Backup systems
   f. Call Recordings

ii. Segregation must be implemented using:

   a. VLANs
   b. Security groups
   c. Routing rules
   d. Zero-trust principles (where feasible)

2.5.37.7. Physical & Network Security Responsibilities

The SP, in coordination with the Cloud Service Provider (CSP), shall manage:

i. Physical security of data centre facilities

ii. Access controls to network equipment

iii. Firewalls, IDS/IPS, WAF, and security appliances

iv. Network infrastructure protection

v. Compliance with security policies, standards, and guidelines

vi. Security Infrastructure

2.5.37.8. Objective and Governance

The Bidder shall design, implement, operate, and continuously improve a cybersecurity and privacy-by-design framework for AirSewa and the integrated PACR environment to prevent, detect, respond to, and recover from cyber threats and risks. The framework shall align with the overall solution landscape given in this RFP and ensure secure-by-default deployments Across applications, APIs, data stores, analytics, network, endpoints, and cloud services.

The Bidder shall provide timely and standardized sharing of threat, vulnerability, and incident information with MoCA/AAI/AirSewa and PACR and ensure continuous compliance with applicable Government of India policies, laws, and guidelines.

2.6. Roles and Responsibilities of the Bidder

The Bidder's scope includes, but is not limited to, the following. All activities shall be performed at no additional cost to MoCA/AAI/AirSewa and PACR unless explicitly approved through the contract's change procedures:

2.6.1. Cyber Threat Intelligence & Sharing

Provide trusted, secure mechanisms (e.g., automated feeds, dashboards, secure portals) and formats agreed with AirSewa to share threat intelligence, vulnerabilities, IoCs, incident status, and post-incident reports.

2.6.2. Security Posture Reviews & Risk Assessments

i. Support MoCA/AAI/AirSewa and PACR during security posture reviews and third-party audits; provide remediation plans and timelines.

ii. Perform periodic, process-based risk assessments and threat modelling (people, process, technology); update the risk register and treatment plan semi-annually and after major changes/incidents.

2.6.3. Implementation & Operations

i. Implement, integrate, harden, commission, and operate the approved security solution stack per minimum requirements and referenced standards; ensure secure-by-default configurations.

ii. Deliver security services 24×7 on all in-scope components (applications, APIs, data stores, network, endpoints, analytics, cloud).

2.6.4. Monitoring, Detection & Response

i. Detect, log, classify, and report security events/incidents; coordinate containment, eradication, recovery; assist MoCA/AAI with statutory reporting (e.g., CERT-In within stipulated timelines) and data breach notifications, as applicable.

ii. Ensure SIEM and supporting technologies (e.g., EDR, WAF/API Gateway, IAM, Vulnerability Management) are integrated with the service desk and SOAR to auto-generate and track tickets and response workflows.

2.6.5. RCA/CAPA & Reporting

i. Provide Root Cause Analysis (RCA) and Corrective/Preventive Actions (CAPA) for all P1/material incidents, including attempted intrusions.

ii. Submit monthly security reports/dashboards (incidents, vulnerabilities, patch status, threat trends, control health, actions taken) and immediate notifications for critical events.

2.6.6. Configuration & Compliance Management

i. Maintain up-to-date documentation of the security architecture, data flows, runbooks, policies/standards, configurations, and inventories with version-controlled baselines.

ii. Conduct periodic configuration reviews for all security/network/cloud components; remediate inconsistencies and deviations from policy/hardening baselines.

iii. Review security policies/standards annually and propose improvements aligned to emerging threats and regulatory changes.

2.6.7. Capacity & Resilience

Perform capacity and resilience management for all security tooling and logging pipelines to meet peak loads with agreed headroom; enable auto-scaling where supported (e.g., WAF, API Gateway, SIEM ingestion).

2.6.8. SLA Compliance

Meet incident/problem response and resolution SLAs approved by MoCA/AAI (containment, eradication, recovery, closure), with defined escalation protocols.

2.6.9. Patching & Vulnerability Management

i. Test, approve, and implement patches/updates following risk-based timelines (e.g., Critical ≤ 7 days, High ≤ 15 days, Medium ≤ 30 days) or as directed by MoCA/AAI.

ii. Perform VAPT (quarterly vulnerability assessment; annual penetration testing) and promptly mitigate risks/vulnerabilities discovered during the project lifecycle.

2.6.10. Logging & Retention

Generate, schedule, and retain daily logs and incident reports; ensure ≥ 180 days online retention and extended archival as agreed, in immutable storage (e.g., WORM/object lock).

2.6.11. Integration & Elasticity

i. Integrate all proposed solutions with the central SIEM/SOAR for unified logging, correlation, response.

ii. Ensure systems/components capable of elastic scale are configured to meet demand while maintaining security policies.

iii. API specifications, integration endpoints, escalation workflows, helpline configurations and stakeholder mappings may be modified during implementation and O&M phase as per operational requirements of MoCA/AAI

2.6.12. CERT-In Compliance

Continuously comply with CERT-In advisories, guidelines, and incident reporting; keep runbooks updated.

2.6.13. Assurance

Establish and maintain all procedures, infrastructure, technology, and qualified personnel necessary to ensure the security of the AirSewa and PACR grievance environment is not compromised.

2.6.14. Security Architecture and Controls

2.6.15. Network Security (AirSewa + PACR)

2.6.16. Zero Trust & Segmentation

- Implement layered network segmentation (VPC/VNet, subnets, VLANs, security groups/NSGs, micro-segmentation) Across application tiers, databases, analytics, and management planes.
- Enforce least-privilege east-west access; deny-by-default policies; explicit egress controls.

2.6.17. Secure Connectivity & Admin Access

- All admin access via SSL/TLS VPN or bastion hosts with MFA; no direct public administration endpoints.
- DNS security, NTP time sync (per CERT-In), and secure management channels (SSH, HTTPS) only.

2.6.18. Perimeter & Ingress Controls

- Next-Gen Firewall (NGFW) with IPS/IDS at ingress/egress; WAF in front of public-facing apps and API Gateway with OWASP Top-10 protections.
- DDoS protection, bot/abuse controls, geo/ASN filtering where applicable.

2.6.19. API Security

- API Gateway enforces authentication, authorization, schema validation, rate limiting, quotas, throttling, anomaly detection, signature/timestamp checks; mutual-TLS for trusted backends.

2.6.20. Network Detection & Response

- NIDS/NDR telemetry integrated to SIEM, alerting on lateral movement, exfiltration patterns, suspicious flows.

2.6.21. Email & Messaging Security

- SPF/DKIM/DMARC for email; secure SMTP relays; link/file sandboxing for high-risk channels (email/social/WhatsApp ingest pipelines).

2.6.22. Cybersecurity Controls (Applications, APIs, Endpoints, Containers)

2.6.23. Identity & Access Management

- SSO, MFA for privileged/sensitive operations; RBAC/ABAC, SoD, PAM for admin/service accounts; SCIM/JIT provisioning; IP allowlists for admin.
- PKI/digital signatures and non-repudiation for documents/transactions where required.

2.6.24. Authentication & Passwords

- Unique IDs, strong password/passphrase policy (breached/common password blocklists), rotation and reuse prevention per policy; device/biometric options where applicable.

2.6.25. Secure SDLC & AppSec

- Align with OWASP ASVS and OWASP Top-10; implement SAST/DAST/SCA, secrets scanning, dependency pinning, SBOMs; code signing; IaC scanning for cloud resources.
- Security gates in CI/CD; pre-production VAPT and remediation prior to go-live.

2.6.26. Endpoint & Workload Protection

- EDR/EPP on servers/endpoints/containers; allow-listing where feasible; kernel-level anti-malware; quarantine/remediation orchestration via SOAR.
- Container/Kubernetes: image signing, private registries, admission controls, runtime policy enforcement, CIS benchmarks.

2.6.27. Data Protection & Privacy

- Encryption in transit (TLS 1.2/1.3) and at rest (KMS/HSM); secrets management; field-level masking and format-preserving encryption where feasible.
- Information asset classification; purpose limitation; consent capture and data minimization; privacy-by-design for analytics/training pipelines.

2.6.28. Operational Security

- Centralized SIEM with correlation rules/use-cases; SOAR playbooks for P1–P3 scenarios; red/blue/purple exercises as scheduled.
- DLP where applicable (e.g., export/download from portals/BI tools), with watermarking and access governance.
-

2.6.29. Security for Analytics & BI Tools (Power BI, Tableau, etc.)

2.6.30. Access & Identity

- SSO (SAML/OIDC), MFA for admins/publishers; service principals for automation; disable legacy auth.

2.6.31. Content Governance

- Workspace/Site strategy with environment separation (Dev/Test/Prod); Project/Folder permissions, peer review, and content promotion workflows; ownership and stewardship defined.

2.6.32. Data Security

- Row-Level Security (RLS) and Object-Level Security (OLS) for datasets; parameterized filters; data lineage and impact analysis enabled.
- Governed gateways (on-prem/cloud) for data refresh; credential vaulting; keys managed via KMS/HSM; dataset/extract encryption at rest.

2.6.33. Publishing & Sharing Controls

- Restrict public sharing and "publish to web"; allow only named users/groups; time-boxed access; watermark sensitive visuals; export/download controls and DLP rules for CSV/PDF/images.

2.6.34. Audit & Monitoring

- Enable Power BI audit logs / Tableau Server logs and admin views; integrate with SIEM (access, dataset refresh failures, abnormal exports, mass access attempts).

2.6.35. Platform Hardening

- Harden Power BI tenants, Tableau Server/Cloud (TLS, cipher suites, admin roles, backup/restore encryption, secrets rotation); VAPT on embedded analytics flows; keep within India data residency constraints where applicable.

2.6.36. Database Security

2.6.36.1 Architecture & Access

- Dedicated subnets; database firewall/NSGs; private endpoints; no public exposure.
- RBAC/ABAC for DB roles; least privilege; PAM for DBA access; break-glass accounts with approvals and full audit.

2.6.37. Encryption & Keys

- Encryption at rest (e.g., TDE) and in transit (TLS); centralized key management (KMS/HSM), rotation policies, dual control.

2.6.38. Data-level Controls

- Column/field-level masking; row-level policies; hashing/tokenization for high-risk PII where feasible.
- Stored procedures over ad-hoc queries; input validation; SQL injection protections (WAF + parameterized queries).

2.6.39. Auditing & DAM

- Database Activity Monitoring (DAM) for privileged operations, schema changes, bulk exports; alerts to SIEM; time-synced logging.

2.6.40. Vulnerability & Patch

- Regular DB vulnerability assessments; critical fixes within SLA; version currency (N-2 or better unless MoCA/AAI defers).

2.6.41. Backup & Recovery

- Encrypted backups; immutable copies; tested restores; retention aligned with policy; secure disposal and legal holds supported.

2.6.42. Monitoring, Logging, and Audit

2.6.43. Centralized Telemetry

- Application, API, infrastructure, network, endpoint, database, and analytics platform logs to centralized SIEM with correlation and behavioural analytics.

2.6.44. Auditability

- Record at minimum: login/logout attempts, user transactions, privileged actions, admin/config changes, overrides, direct data changes, API access (including app/key IDs), and security events; time synchronization Across components.

2.6.45. Search & Retention

- Provide audit queries by access type, time range, user ID, IP, host, API/key ID, correlation IDs.
- Retention: ≥ 180 days online; longer archival per MoCA/AAI/regulatory need using
  immutable storage.

2.6.46. Governance & Evidence

- Immutable audit logs; configuration snapshots; change approvals; compliance evidence packs on request.

2.6.47. Incident Response & Compliance

2.6.48. Runbooks & SOAR

- Maintain approved runbooks (P1–P4); SOAR playbooks automate triage, containment (e.g., isolate host, block IP), eradication, and recovery actions.

2.6.49. Reporting

- Notify MoCA/AAI of P1 events immediately; statutory reporting to CERT-In within mandated timelines; provide RCA/CAPA within agreed SLAs.

2.6.50. Forensics

- Preserve chain-of-custody; snapshot evidence; cooperate with law enforcement/regulators as required.

2.6.51. Patch and Vulnerability Management

- Risk-based patching windows: Critical ≤ 7 days, High ≤ 15 days, Medium ≤ 30 days (or as directed by MoCA/AAI).
- Quarterly vulnerability assessments; annual penetration testing for applications/platform; re-tests after remediation.
- Threat-driven emergency patches with MoCA/AAI's expedited approvals; change control documented.

2.6.52. Identity, Authorization, and Approvals

- Fine-grained functional and data access rights by user/role/organization; segregation of duties; workflow-based authorization; multi-level approvals.
- Restrict access to sensitive data elements by named user/group/role; field-level masking or format-preserving encryption where feasible.

2.6.53. Business Continuity & DR (On Cloud)

- Maintain approved BCP and DR Plan prior to go-live; align RTO/RPO with MoCA/AAI-approved objectives; reference 2.6.10 for cloud DR requirements.
- Robust backup procedures with immutable copies; bi-annual DR (On Cloud) drills of at least 24 hours; DR drill shall be tested with 50% load.
- High Availability (HA) deployment; failover/failback testing; continuous improvement actions recorded.

2.6.54. Tooling Stack – Minimum Capabilities

The Bidder shall provide and operate a security stack (or equivalent) covering:

- IAM/SSO/MFA/PAM, EDR/EPP, NGFW/IPS, WAF + API Gateway (OWASP Top-10 protections), DDoS protection, HIDS/NIDS/NDR, Vulnerability Management & VAPT, SIEM + SOAR, KMS/HSM, Secrets Management, RBAC/ABAC, Immutable logging (WORM/object lock), Backup/DR with immutable copies, Compliance monitoring and attestations, and full AAA (authentication/authorization/auditing) Across components.

2.6.55. Compliance, Standards, and Hosting

- Comply with ISO/IEC 27001/27002, 27017, 27018, GIGW 3.0 or latest , CERT-In Directions 28-Apr-2022, CERT-In Comprehensive Cyber Security Audit Policy guidelines (Version 1.0, July 2025), CERT-In guidelines on Software Bill of Materials/Cyber Bill of Materials, National Security Council Secretariat (NSCS) Cyber Security Audit Baseline Requirements (NSCS-46-16 Rev 1.0), DPDP Act 2023, IT Act 2000 and allied rules.
- Adhere to all applicable MeitY and MoCA/AAI security policies and guidelines.
- Host only on NIC MeghRaj/, ensuring India data residency (see 2.6.10).

2.6.56. Broad Security Requirements

1. **Compliance:** Conform to GoI guidance (e-Governance Security Guidelines, National Cyber Security Policy, GIGW 3.0, CERT-In Directions, DPDP Act 2023, etc.). Provide configurable **RBAC/ABAC** at application/module/transaction/data levels.
2. **PKI & Digital Signatures:** Use for document/transaction integrity, non-repudiation, and sender identity as required.
3. **Strong Authentication:** Unique user IDs; **SSO/MFA** for privileged/sensitive functions; support digital signatures/biometrics/devices where applicable.
4. **Passwords:** Enforce length/complexity/passphrase policy; block breached/common passwords; expiry/reuse policies as per MoCA/AAI standards.
5. **Authorization & SoD:** Fine-grained functional/data rights; **multi-level approval**; workflows; SoD.
6. **Data Access Controls:** Restrict sensitive elements by user/group/role; field-level masking/format-preserving encryption where feasible.
7. **Auditability:** Tamper-evident logs and evidence trails; evolving requirements supported.
8. **Audit Logging:** Log login/logout, user transactions, privileged/admin changes, overrides, direct data changes, API access, and security events; time sync enforced.
9. **Audit Queries & Retention:** Filter/search by type/time/user/IP/host/API/key/correlation IDs; ≥
   180 days online retention; longer immutable archival per policy.
10. **Information Asset Classification:** Classify and apply proportionate controls; all project data is owned by MoCA/AAI/AirSewa and shall not be disclosed to unauthorized parties.
11. **Malware/EDR:** On-access and periodic scans; signature/engine updates; application allow-listing where feasible.
12. **Network & Segmentation:** Robust segmentation and **Zero Trust Across** networks, servers, databases, endpoints, APIs, cloud.
13. **BCP & DR (On Cloud):** Approved plans before go-live; **immutable backups**; periodic DR/restore drills to validate RTO/RPO; **DR drill with 50% load**.

14. **Cloud Security:** Refer DC/DR controls, India residency, certifications, and cloud posture requirements.

2.6.57. IT Governance and Monitoring

2.7. Overview of Project Governance

The Service Provider (SP) shall be responsible for the end-to-end implementation, deployment, and stabilization of all modules, components, and sub-systems covered under the Scope of Work. A structured governance mechanism shall be established to ensure timely execution, transparent communication, risk mitigation, and adherence to quality and performance standards throughout the project lifecycle. PMU/Authorized MoCA/AAI officials shall review and approve SRS, UI/UX, workflows, integrations, UAT sign-off and Go-Live readiness

2.7.1. Project Monitoring Unit (PMU)

The Agency shall establish a Project Monitoring Unit (PMU) for continuous review and monitoring of the project during the implementation phase.

The PMU shall evaluate progress, verify milestone achievements, and assess compliance with contractual obligations and SLAs.

The PMU shall coordinate with the Service Provider on timelines, deliverables, acceptance testing, and risk/escalation handling.

2.7.2. Agency's Single Point of Contact (SPOC)

The Agency shall appoint a dedicated Single Point of Contact (SPOC) for all IT administration-related activities. All project-related escalations from the SP shall be routed through this SPOC. The SPOC will be responsible for:

- Reviewing SLA adherence
- Overseeing issue resolution
- Coordinating with the PMU and other stakeholders
- Facilitating approvals, clarifications, and dependencies

The Service Provider must prepare and submit a comprehensive Customization, Implementation, and Rollout Plan, outlining:

- Functional customization approach
- Deployment methodology
- Integration timelines
- Resource allocation
- Training and UAT plans
- Rollout schedule Across all modules

2.7.3. Governance Meetings

The following meetings shall be conducted regularly:

- Weekly/Monthly Progress Review Meetings between SP and stakeholders (PMU, SPOC, MoCA/AAI etc).
- Milestone Review Meetings for major deliverable approvals
- Risk & Issue Review Meetings as required

To ensures compliance, security, operational efficiency, and continuous monitoring Across the PACR and AirSewa ecosystem.

i. Alignment with national aviation guidelines, security policies, IT Act, and CERT-In directives.

ii. Periodic audits, vulnerability assessment & penetration testing as per MeitY latest guidelines.

iii. Enforcement of access controls, security baselines, and change management.

iv. Enforcement of DPDP Guidelines.

v. Proactive alerting for performance anomalies, security threats, and outages.

2.7.4. Audit and compliance

2.7.4.1. Compliance, Audits & Reporting

The CSP shall ensure comprehensive compliance with AirSewa and Government of India guidelines on security, data privacy, statutory requirements, and audit readiness throughout the lifecycle of AirSewa. The following provisions shall apply:

2.7.5. Reporting Obligations

2.7.6. Audit Facilitation

2.7.7. Mandatory Security Assessments

a) **Annual Security Audit –** The MSP shall engage a **CERT-In empanelled third-party auditor** to conduct a comprehensive annual security audit of all deployed components, in line with the **Comprehensive Cyber Security Audit Policy Guidelines** (https://www.certin.org.in/PDF/Comprehensive_Cyber_Security_Audit_Policy_Guideli nes.pdf). All associated costs shall be borne by the MSP.

b) **Forensic Investigations** – MoCA/AAI reserves the right to initiate **forensic investigations** through a third-party auditor based on credible inputs from investigating agencies. If the investigation report identifies lapses attributable to the MSP, the costs shall be borne by the CSP. In case of repeated lapses (two or more), AirSewa reserves the right to invoke termination, forfeit the security deposit, and initiate blacklisting proceedings.

c) **Internal Limited Audits** – The MSP shall conduct **half-yearly internal audits** at its own cost and shall ensure closure of all audit findings within mutually agreed timelines.

2.7.8. Right of AirSewa to Independent Audit

1. AirSewa reserves the right to engage a **CERT-In empanelled third-party auditor** to conduct comprehensive audits at any time post Go-Live.
2. The MSP shall extend full cooperation and ensure audit closure within prescribed timelines.

2.7.9. Regulatory Adherence

3. The MSP shall ensure strict compliance with **AirSewa (MoCA/AAI) and GoI guidelines on data retention, privacy, and security Across** all environments (production and non-production).
4. Any deviations identified shall be rectified by the MSP immediately at no additional cost to MoCA/AAI.
5. Failure to comply with the closure of audit findings shall result in penalties given in the RFP

- The CSP shall prepare and submit **monthly and quarterly performance reports** covering incidents, resolution timelines, system uptime, SLA compliance, and other key operational metrics as prescribed by MoCA/AAI.
- Reports shall include root-cause analysis of critical incidents, corrective and preventive measures undertaken, and recommendations for process and system improvements.
- The CSP shall support and facilitate **internal and external audits**, including IT security, statutory, compliance, and financial audits, as and when required by AirSewa.
- The CSP shall ensure full cooperation and timely access to systems, records, configurations, and documentation during audits.

2.7.10. Security

All security-related requirements outlined in this RFP, including those under Operations & Maintenance, Compliance, Audit, and Cybersecurity, shall form an integral part of the MSP's responsibilities. The CSP shall ensure that the solution is resilient, auditable, and adaptable to evolving security threats throughout the lifecycle of AirSewa 3.0.Version Upgrades

The CSP shall ensure that the AirSewa solution is designed and operated in line with the highest standards of information security, data privacy, and compliance with applicable guidelines. MoCA/AAI reserves the right to continuously monitor and inspect all inbound and outbound traffic Across hardware, software, applications, and network layers to identify, prevent, and mitigate any potential leakage of sensitive or configuration data.

In case of identification of vulnerabilities or performance gaps in hardware/software provided by the CSP, MoCA/AAI may direct the CSP to replace the affected components with equivalent or superior alternatives at no additional cost to MoCA/AAI, at any point during the contract period.

The envisaged solution shall mandatorily include:

a) **Data Masking and Sub-setting Capabilities** – The ability to obfuscate and subset sensitive information while enabling secure usage Across testing, acceptance, development, and training environments.

b) **Automated Discovery of Sensitive Data** – Identification of sensitive fields (columns, attributes) and maintenance of parent-child data relationships to ensure compliance with data protection requirements.

c) **Secure Data Sub-setting** – Selective deletion or exclusion of non-relevant or redundant records from databases to minimize exposure of sensitive information in non-production environments.

d) **Proactive Security Controls** – Implementation of monitoring, auditing, logging, and incident response mechanisms to ensure end-to-end protection of CSP's information assets.

Any version upgrade/ changes in the solution shall be implemented in development / test environment. The solution shall be migrated to the production environment only after adequate testing, submission of test completion certificate from MSP and approval by the designated MoCA/AAI officials. A quality assurance test of the changes to be implemented will be performed in a test environment prior to implementation in the production environment.

e) **Others**

The envisaged solution should also ensure maintenance of data integrity, consistency, management of Data Lifecycle, compliance to industry standards. Audit trail and logging should be enabled in the envisaged solution.

2.7.11. Capacity Building and Training

SI shall be responsible for training and capacity building of the stakeholders and shall have the following responsibilities:

1. SI will have to bear all the cost associated with the conducting such training programs. Training sessions should be conducted on a requisite mix of theory & practical operations.
2. The trainings should be conducted in Hindi and English and if any other language required by MoCA/AAI.

3. Preparation of training and support content shall be for types of complainants and all stakeholder authorities/ organisations.
4. The training and support contents may be uploaded on AirSewa platform.
5. Training to be provided multiple times, upon request from MoCA/AAI

The bidder shall be responsible for capacity building and training activities regarding the AirSewa Platform at following levels:

Table 4 Capacity building and training activities

S.No.	Stakeholder	Mode of Training	Type of Training
1.	MoCA/AAI and Government stakeholder Officials and PACR	1. Online 2. Offline	Functional
2.	Passengers	Training artifacts on website and app (text/ pdf/ images/ video, etc formats)	Functional
3.	Airport Operators	1.Online 2.Offline	Functional
4.	Airlines and private airport operators	Online	Functional
5.	Third Party Stakeholders	Online	Functional
6.	PACR stakeholders	Online	Functional

i. The bidder shall provide necessary documents and training aids required for successful and effective delivery of the training programs.

ii. The number of trainings provided here are indicative. The bidder may be required to conduct additional trainings free of cost in case more training sessions are required.

iii. Types of training the bidder shall provide:

2.8. Documentation:

The bidder shall provide following training documents:

2.8.1. System Manual:

a. This manual is intended for all the concerned stakeholders

b. Details out the system architecture including all the modules of the AirSewa Application/ web portal, Analytics, Passenger Services, Value Added Services, and Access Management, and API manager.

c. The manual shall also provide details of Data layer, System and software layer, IT infrastructure, IT Security, Control and monitoring system and support services like SLA Monitoring, Audit mechanism and log management.

2.8.2. Passenger User Manual:

a. This manual is intended for Passengers awareness

b. Details out all the Passenger Services of the AirSewa portal for instance, Passenger Registration, ID Generation, AirSewa ID Activation, AirSewa ID Update and Restore, Consent Management, Alerts and Notifications.

2.8.3. Administrator Manual:

a. This manual is intended for Administrator Users

b. Apart from passenger services, details out all the Administrator functionalities of the system covering administration of AirSewa ID Lifecycle and Access Management, Value Added Services, Biometric Services, API manager, Communication Channel, IT infrastructure and security and support services.

2.8.4. Other Training aids:

In addition to the documents stated above, the bidder shall develop and provide following training aids:

c. Interactive digital training content to be made available online on the AirSewa Web Portal and Mobile Application

d. Computer based training modules

e. Video for AirSewa functionalities, back-end modules, analytics, reporting etc.

f. Module specific presentations

g. Operations and maintenance manuals

h. Regular updates to the training content as and when system undergoes changes.

Note:

i. MoCA/AAI shall have all the rights regarding the usage, distribution and storage of the digital content.

ii. The bidder shall be responsible for timely updating the training documents, digital content as and when the system undergoes a change during the operation.

2.8.5. Training Effectiveness:

The bidder must ensure the effectiveness of training sessions, for this:

i. The bidder shall prepare a comprehensive feedback form that will capture the necessary parameters for measuring the effectiveness of the training. This format shall be finalized with MoCA/AAI.

ii. The bidder shall provide printed copy and online link of the feedback form.

iii. The bidder shall comply with the defined training SLAs.

2.8.6. Project execution requirements

Bidder shall follow the schedule as mentioned below. Each of the deliverables should be accompanied with a presentation on the deliverable by the bidder. The submission of deliverable will be deemed complete after the submission of the hard / soft copy of the deliverable and the presentation by the bidder.
The implementation plan is divided into following tracks:

i. Design Phase- **One month**

ii. Development and Implementation- **Three Months**

iii. User Acceptance and Go Live- **One Month**

iv. Operation and Maintenance- **3 Years, extendable by 2 years**

The following services at the minimum shall be provided by Bidder as part of implementation and rollout:

2.9. Design Phase

2.9.1. Project Planning and Monitoring

i. Bidder shall adopt a comprehensive and efficient Project Management methodology to ensure that project milestones are tracked and met. Bidder will be required to finalize the Project plan to determine and agree on the project expectations, ground rules, work plan, communication matrix, timelines, Quality Plan, Configuration Management Plan etc within 15 days of effective date of signing of the contract with MoCA/AAI. Following activities would be covered under Project Planning and Monitoring:

   a. Finalize a set of activities for the project with identification of resource assignments, roles and responsibilities against each activity

   b. Prepare a project plan including milestones

   c. Communicate the project plan to stakeholders

   d. Measure project deadlines and budget utilization figures

   e. Project Quality Plan shall document specific process elements and the quality actions that the project intends to implement. This shall include the derivation of quality goals, standards followed, schedule of quality assurance activities in the project, defect control, correction and preventive methodology, handling process deviations.

   f. Configuration Management Plan shall contain procedures to be implemented for managing the configuration of the software solution to be produced by the project. In this plan, bidder shall identify configuration items, responsibilities of configuration controller, access restrictions, directory structure needed for configuration management, procedure for change control, method of tracking the status of configuration items, backup procedure, configuration audits, release management, archival procedure, procedure for version / revision numbering.

The detailed project plan shall clearly specify the various project milestones and project deliverable schedules. It shall also include the following:

   a. Project Organization and Management plan

   b. Software Design and Development plan

   c. Wireframes of the User Interfaces for mobile and website (Passenger & Admin)

   d. Implementation plan

   e. Pre-commissioning, Operational and User Acceptance Testing Plan

   f. Design, Delivery and Installation Plan for Hardware and Network

   g. Training Plan

   h. Data digitization, migration, enrichment plan

i. Support Service Plan

j. Task, Time, and Resource Schedules (List of tasks, the dependency among the tasks, the duration to perform the tasks, the resources allocated to perform the tasks, the scheduled start and finish dates for the task)

k. Post-support Service Plan

l. Technical Support Plan

m. Quality Assurance and Control Process details which must include (but not limited to)

n. Detailing on Metrics, Reviews, Problem Reporting and Corrective action etc.

o. Technical and Operational Process which must include (but not limited to) detailing on Methods, Tools, Techniques etc.

ii. The detailed project plan shall be internally reviewed for completeness and correctness by Bidder and subsequently delivered to the MoCA/AAI for its review and acceptance. The mutually agreed Project Plan will form the basis for regular project monitoring.

iii. Bidder shall hold fortnightly review meetings with the MoCA/AAI and other stakeholders for providing detailed report on the progress of the project (Project Progress Report) clearly highlighting the activities completed in the reporting period, activities planned for the next reporting period, deviations from the planned dates, issues / concerns affecting the project progress, impact on the overall project timelines, project related risks with their mitigation plans.

iv. Bidder shall ensure proper configuration management functions are being performed as per the configuration management plan. Bidder's Project Manager shall review the activities periodically.

v. Bidder's independent quality assurance team shall conduct regular reviews / audits of the

project and ensure that the project adheres to the project plan documents.

2.9.2. Requirement Gathering

i. The bidder shall study the functional requirements specifications and process flow as listed in the RFP. The bidder is required to validate and further elaborate the same based on discussion with MoCA/AAI and/or their designated agencies. Functional requirements covered under various sections have been identified from business process and other documents designed during the initial phases of the project for the bidder to refer.

ii. Bidder shall understand the business processes and other related documents and seek clarifications from MoCA/AAI, if any.

iii. If found necessary to modify the designed processes and other documents for successful implementation, the same shall be discussed and the relevant documents shall be modified as and when required during the project implementation

iv. Bidder shall interact with the target end users as well as MoCA/AAI project team to gather requirements. It is expected that Bidder gathers requirements through structured questionnaires, focused interviews with user groups.

v. On gathering the requirements, Bidder shall analyse these requirements to ensure the requirements are complete, accurate, consistent and unambiguous.

vi. After the analysis, Bidder shall prepare software requirement specification (SRS) document. SRS shall contain the objectives and scope of the system, the various levels of requirements, the process model, data model, data dictionary, etc. Unresolved issues shall also be included in this document. User Role wise mapping to the various business functions with details regarding their access rights (insert / update / delete / view etc.) shall also be included in this document. Acceptance Criteria shall also be included explicitly promoting clear understanding with the end user about what the end user considers acceptable.

vii. The Bidder shall, as part of development of SRS (Software Requirement Specification), develop a system prototype to capture and demonstrate the end user requirements in the form of screens and outputs. The prototype shall be enhanced continuously during the development of SRS and SDD specifications.

viii. The SRS document shall be reviewed and approved by MoCA/AAI.

ix. Bidder shall develop a logical view of the solution to meet the user requirements. This logical view shall consist - the functional architecture of the application and the changes/new database design. Bidder shall also define standards for coding, documentation, user interfaces, etc., if the same is not already defined.

x. Bidder shall document the high-level design as System Design Document (SDD) consisting of project standards, the functional design and the database design. The SDD document shall be reviewed and approved by MoCA/AAI.

2.10. Implementation

2.10.1. Planning

Bidder, in coordination with MoCA/AAI and AAI, shall prepare a detailed Implementation Plan that shall include activities such as User Training, Data Migration / Population, and Application Rollout. Bidder shall also carry out activities to prepare documentations.

2.10.2. Data Migration and Existing vendor support

Data Migration: The existing AirSewa accounts, which might be at least 1.5 lakh or more, will have to be migrated to the new system. The bidder shall assess the requirements for the same.
Existing service providers/vendors shall extend reasonable technical and migration support during transition and stabilization phase, wherever required by MoCA/AAI

2.10.3. Software Development, Customization, Rollout for Applications

Guidelines for Software Development and Testing

i. Bidder shall carry out development or customization of all the solutions using its own infrastructure including but not limited to servers, storage, software licenses, automated testing tools etc.

ii. https://Openforge.gov.in to be used for code writing, review, merger and CI/CD for all software development and deployment activities and Bug tracking and task management.

iii. Bidder shall ensure that the COTS solution, if any selected are such that they are configured / customized with minimal modifications to the source code. Perpetual license for COTS / Proprietary products along with source code to be shared in a digital media storage device at the completion of the platform development phase of 5 months.

iv. Bidder shall implement quality standards like CMM/ CMMi for the entire life cycle of the project. The quality process shall include adequate processes for coding, change management, defect tracking, testing, review and other software development life-cycle processes.

v. Bidder shall develop and maintain the Test Environment including required licenses and hardware using its own infrastructure for the complete contract period.

The following sections describe the development activities based on traditional development methodology to be performed by the Bidder. As part of this development activity, bidder shall carry out the following tasks:

2.10.4. Software Requirement Analysis and Specification

i. Software Requirement Analysis and specification is a key stage in the project and recognizing its pivotal role in the subsequent phases, sufficient time will be provided to the bidder to capture the requirements accurately. Functional requirements covered under various solution areas have been identified from business process and other documents designed during the initial phases of the project for the bidder to refer.

ii. Bidder shall understand the business processes and other related documents and seek clarifications from MoCA/AAI, if any. Bidder shall then take a hand over these documents to MoCA/AAI.

iii. If it is necessary to modify the designed processes and other documents for successful implementation, the same shall be discussed and the relevant documents shall be modified as and when required during the project implementation

iv. Bidder shall interact with the target end users as well as MoCA/AAI project team to gather requirements. It is expected that Bidder gathers requirements through structured questionnaires, focused interviews with user groups.

v. On gathering the requirements, Bidder shall analyze these requirements to ensure the requirements are complete, accurate, consistent and unambiguous.

vi. After the analysis, Bidder shall prepare software requirement specification (SRS) document. SRS shall contain the objectives and scope of the system, the various levels of requirements, the process model, data model, data dictionary, etc. Unresolved issues should also be included in this document. User Role wise mapping to the various business functions with details regarding their access rights (insert / update / delete / view etc.) shall also be included in this document. Acceptance Criteria shall also be included explicitly promoting clear understanding with the end user about what the end user considers acceptable.

vii. The SRS document shall be reviewed and approved by MoCA/AAI.

2.10.5. Software Design specifications

i. In this phase, Bidder shall develop a logical view of the solution to meet the user requirements. This logical view shall consist - the functional architecture of the application and the changes/new database design. Bidder shall also define standards for coding, documentation, user interfaces, etc., if the same is not already defined.

ii. Bidder shall document the high-level design as System Design Document (SDD) consisting of project standards, the functional design and the database design. The SDD document shall be reviewed and approved by MoCA/AAI.

2.10.6. Preparation for Core Database design

i. Bidder shall design the Logical Data Model of the Core Database based on the understanding of the information needs of MoCA/AAI and existing information items. The Logical Data Model shall include the definition of data structure or schema diagram and data dictionaries and metadata developed explaining specific entities, relationships and dimensions identified as a part of the Logical Data Model.

2.10.7. Build Stage – Coding and Unit Testing

i. Bidder shall carry out detailed design (Low Level Design), Coding and Unit Testing phases.

ii. Bidder shall break down the high-level design into modules and programs. For every program, a unit test plan shall be created. Bidder shall also identify common routines, programs, utilities and tools for productivity improvement.

iii. In this phase, high level design document is the main input whereas the program specifications and unit test plans are the key outputs.

iv. Bidder shall identify data migration programs that need to be developed and tested. Since without them, programs for the current application cannot be tested properly, it is important that Bidder develop and test them early.

2.10.8. Documentation

a) Bidder shall develop manuals such as User Manuals, Training Manuals, FAQ & Trouble Shooting Guide, etc. as agreed upon with MoCA/AAI for different kinds of users such as System Administrator, Super user, Normal user.

b) Bidder shall ensure that the documentation is in line with the defined documentation standards.

c) Bidder shall use tools for creation of Help documents, User manuals etc. that minimize the documentation and change efforts.

d) Bidder should share cloud utilization report monthly during the development and O&M stage. A separate cloud account should be created for the AirSewa project with access to the cloud account to MoCA/AAI for visibility.

e) System Integrator (SI) shall provide a monthly Cloud Utilization Report, including detailed consumption metrics specific to the AirSewa project. The bidder must ensure that the AirSewa workload is maintained under a separate cloud account or subscription to enable clear segregation of usage and cost. Access to the AirSewa cloud account shall be provided to the Ministry of Civil Aviation (MoCA/AAI) for visibility, monitoring, and audit purposes.

f) The System Integrator (SI) shall provide logs, alerts, and corresponding reports from the deployed SIEM solution to confirm that no cybersecurity breaches, intrusions, or security incidents have occurred within the environment. These logs must be complete, unaltered, time-stamped, and submitted as per the agreed reporting frequency or upon request by the customer's security team.

g) Any cybersecurity breaches, intrusions, application defects or security incidents should be notified as per SLAs and as part of the documentation.

h) The System Integrator shall provide PACR and Cloud Downtime logs generated from the available monitoring and management tools integrated with the cloud environment. These logs must clearly indicate any service outages, performance degradation events, maintenance windows, and uptime statistics for the specified reporting period.

2.10.9. Testing Stage

Under the Testing stage, Bidder shall carryout Integration Planning & Testing as well as System Test Planning & Testing phases. The following sections explain the scope of each phase Bidder shall perform.

i. The bidder shall demonstrate the following mentioned acceptance criteria prior to acceptance of the solution as well as during project operations phase, in respect of scalability and performance etc.

ii. The bidder may propose further detailed Acceptance criteria which the MoCA/AAI will review. Once MoCA/AAI provides its approval, the Acceptance criteria can be finalized.

iii. In case required, parameters might be revised by MoCA/AAI in mutual agreement with bidder and the revised parameters shall be considered for acceptance criteria.

iv. The bidder would set up a testing environment in its own facility that is separate from the development, staging and production environment.

v. A comprehensive system should be set up that would have the capability to log & track the testing results, upload & maintain the test cases and log & track issues/bugs identified.

vi. The following table depicts the details for the various kinds of testing envisaged for the project.

Table 5 Types of envisaged testing for AIRSEWA and PACR

Type of Testing	Responsibility	Scope of Work
System Testing	The bidder	1. The bidder shall perform System testing in its own premises where the development work is being executed 2. The bidder shall prepare test plan and test cases and maintain it. MoCA/AAI may request the bidder to share the test cases and results 3. Should be performed through manual as well as automated methods 4. Automation testing tools to be provided by the bidder. MoCA/AAI doesn't intend to own these tools
Integration Testing	The bidder	1. The bidder shall perform Integration testing in its own setup 2. The bidder shall prepare and share with MoCA/AAI the Integration test plans and test cases 3. The bidder to perform Integration testing as per the approved plan 4. Integration testing to be performed through manual as well as automated methods 5. Automation testing tools to be provided by the bidder. MoCA/AAI doesn't intend to own these tools
Performance and load Testing	The bidder/ Third Party Auditor	1. The bidder shall do performance and load testing in production setup. Thereafter, the testing to be performed in a simulated environment arranged by the bidder at its own cost. 2. Various performance parameters such as transaction response time, throughput, hits per second and transactions per second etc. should be considered. 3. Load and stress testing of the AirSewa portal and PACR Setup to be performed on business transaction volume 4. Test cases and test results to be shared with MoCA/AAI. 5. Performance testing to be carried out in the exact same architecture that would be set up for production.

Type of Testing	Responsibility	Scope of Work
		6. architecture that would be set up for production. 7. The bidder needs to use performance and load testing tool for testing. MoCA/AAI doesn't intend to own these tools. MoCA/AAI if required, could involve third party auditors to monitor/validate the performance testing. Note: Before Go-Live of the project, the bidder needs to do load testing in the production environment to showcase the capability of the system to handle 50,000 concurrent user connections of the first year as per the SLAs specific to performance of the system. In case the system fails to handle the expected loads, the bidder will have to take corrective action (e.g. hardware ramp up, tuning of application / middleware / hardware, etc.) at no extra cost to MoCA/AAI. This exercise would continue yearly (at beginning of every year of O&M) basis wherein the bidder will certify MoCA/AAI that the system can handle 2x of the envisaged concurrent peak load of that year (but not less than 50,000 concurrent users). This load testing will not be allowed to be done in production environment and the bidder need to get this done at its own cost. MoCA/AAI / MoCA/AAI appointed agency will validate the test results.
Security Testing (including Penetration and Vulnerability Testing)	The bidder/ Third Party Auditor	1. The solution should demonstrate the compliance with security requirements as mentioned in the RFP including but not limited to security controls in the application, at the network layer, network, data centre(s), security monitoring system deployed by the bidder

Type of Testing	Responsibility	Scope of Work
		2. The solution shall pass vulnerability and penetration testing for rollout of each module. The solution should pass web application security testing for the portal and security configuration review of the baseline infrastructure. 3. The bidder should carry out security and vulnerability testing on the developed solution. 4. Security testing to be carried out in the exact same environment/architecture that would be set up for production. 5. Security test report and test cases should be shared with MoCA/AAI 6. Testing tools if required, to be provided by the bidder. MoCA/AAI doesn't intend to own these tools 7. During O&M phase, penetration testing to be conducted on yearly basis and vulnerability assessment to be conducted on half-yearly basis. MoCA/AAI will also involve third party auditors to perform the audit/review/monitor the security testing carried out by the bidder.
User Acceptance Testing o f AIRSEWA	MoCA/AAI or MoCA/AAI appointed Third Party Auditor	1. MoCA/AAI/ MoCA/AAI appointed third party auditor to perform User Acceptance Testing 2. The bidder to prepare User Acceptance Testing test cases 3. UAT to be carried out in the exact same environment/architecture that would be set up for production. 4. The bidder should fix bugs and issues raised during UAT and get approval on the fixes from MoCA/AAI / third party auditor before production deployment 5. Changes in the application as an outcome of UAT shall not be considered as Change Request. The bidder must rectify the observations.
Application Benchmarking	The bidder	The bidder needs to do benchmarking of the application software including APIs to meet the scalability requirements provided below:

Type of Testing	Responsibility	Scope of Work
		1. Before Go-Live (a) the bidder needs to benchmark the application software for 2x of the envisaged concurrent peak load (minimum 50,000 concurrent users). (b) AirSewa system APIs should also be benchmarked for 10X of first year projected API transaction volumes. 2. One year after Go-Live (a) the bidder needs to benchmark the application software for the peak load 2x of the concurrent peak load of that year (but not less than 50,000 concurrent users). (b) The bidder also needs to benchmark the AirSewa System API's for 100X of the first-year transaction volumes. 3. The benchmarking of the application software shall be done by a third party engaged by the MoCA/AAI. The bidder will bear the costs of such benchmarking exercises. The third party will certify the system's ability to handle the requisite load. In case the system fails to handle the expected loads, the bidder will have to take corrective action. MoCA/AAI doesn't intend to own the benchmarking tools and/or the IT infrastructure. 4. The bidder, in its bid and during project initiation needs to share the details of the benchmarking plan including the parameters to be tested with MoCA/AAI. Inputs from MoCA/AAI will be incorporated by the bidder before performing the benchmarking exercise.

Note:

a. Bidder needs to provide the details of the testing strategy and approach including details of intended tools/environment to be used by the bidder for testing in its technical proposal. MoCA/AAI does not intend to own the tools.

b. The bidder shall work in a manner to satisfy all the testing requirements and adhere to the testing strategy outlined. The bidder must ensure deployment of necessary resources and tools during the testing phases. The bidder shall perform the testing of the solution based on the approved test plan, document the results and shall fix the bugs found during the testing. It is the responsibility of the bidder to ensure that the end product delivered by the bidder meets all the requirements specified in the RFP. The bidder shall take remedial action based on outcome of the tests.

c. The bidder shall arrange for environments and tools for testing and for training as envisaged. Post Go-Live; the production environment should not be used for testing and training purpose. If any production data is used for testing, it should be masked and it should be protected. Detailed process in this regard including security requirement should be provided by the bidder in its technical proposal. The process will be finalized with the selected bidder.

d. All the Third-Party Auditors (TPA) as mentioned above will be appointed and paid by MoCA/AAI directly. The bidder can also suggest the auditors who may be appointed by MoCA/AAI at its sole discretion. All tools/environment required for testing shall be provided by the bidder.

e. STQC/Other agencies appointed by MoCA/AAI shall perform the role of TPA. The bidder needs to engage with the TPA at the requirement formulation stage itself. This is important so that unnecessary re-work is avoided and the audit is completed in time. The audit needs to be completed before Go-Live of different modules. The bidder needs to prepare and provide all requisite information/documents to third party auditor and ensure that there is no delay in overall schedule.

f. The cost of rectification of non-compliances shall be borne by the bidder.

2.11. Ongoing Administration and Maintenance requirements Operational Support

This section describes the operational requirements of the project including project management requirements, acceptance testing & certification and solution maintenance & support. The Bidder shall provide complete support for the entire IT infrastructure of the system including all the items supplied as part of the contract, for the contract period. The terms of warranty shall include that the solution supplied under this Contract shall have no defect arising from design or workmanship or from any act of omission or commission of the bidder or that may develop under normal use of the supplied solution Operational support shall ensure that the system is functioning as intended and meeting the service levels. Operational support will include:

2.11.1. Project status reporting:

Bidder shall submit progress reports monthly. The progress report shall include the following:

i. Tasks completed / Results achieved during the period (fortnight)

ii. Tasks / Results to be completed in the subsequent period.

iii. Cumulative deviations to date from schedule of progress on milestones

iv. Corrective actions to be taken to return to planned schedule of progress; Revision to planned schedule provided.

v. Other issues and outstanding problems, and actions to be taken from each stakeholder.

2.11.2. Maintenance of user documents

Bidder shall maintain the end user documents up to date as mentioned in the section

i. Software solution maintenance

The objective of application maintenance is to provide application maintenance and support services, including request-based services (problem requests/defect fixes), enhancements, configuration management and post release support. As part of these services, Bidder shall provide support for bug fixes, enhancements, operational support, and assistance to MoCA/AAI. The services include:

a. Bug-fixes and end-user problem resolution

- The end user support shall include all activities related to resolving the bugs/ defects reported by application users. Every bug / defect shall be logged. Every bug / defect shall be categorized on the severity levels.
- Bidder shall identify the solution and get the necessary approvals from MoCA/AAI and release the patch for UAT after fixing the defects.
- Bidder shall document defects / bugs encountered as well as document the resolution of the same.
- Ensure re-installations, in the event of system crash/failures.

b. New development and enhancements

- The application may require modifications or enhancements in the functionality. The enhancements or new development may also be required to fix some complex problem requests or defect fixes.
- Bidder shall ensure that correct versions of the application / program units are being considered to carry out application enhancements/ new development through configuration management plan for configuration management and version control using the version control software.

- Bidder shall support MoCA/AAI in carrying out the UAT for the modifications / enhancements.

2.12. Operations and Maintenance

The bidder will operate and maintain all the components of the AirSewa portal for a period of three (3) years from the date of Go-Live which may be extended by two (2) years on mutually agreeable terms. The Operation and Maintenance part is divided into the following parts:

2.12.1. Application Support and Maintenance

Application support includes, but not limited to, production monitoring, troubleshooting and addressing functionality/availability and performance issues and implementing system change requests etc. The bidder shall maintain the application software in good working conditions; and perform changes and upgrades to the applications as requested by the authority. All tickets related to any issue/complaint/observation about the system shall be maintained in an ITIL compliant comprehensive ticketing solution. Key activities to be performed by the bidder during the application support phase are as follows:

i. Compliance to SLA (in vol 3 of the document)

The bidder shall ensure compliance with SLAs as indicated in this section of this RFP and any upgrades/major changes to the software shall be planned by the bidder while ensuring that the SLA requirements are met at no additional cost to the authority/ MoCA/AAI.

ii. Annual Technology Support

The bidder shall be responsible for arranging annual technology support to the AirSewa portal for the OEM products provided by respective OEMs during the entire O&M phase. It is mandatory for the bidder to take enterprise level annual support over the entire contract duration at minimum for the software(s) mentioned below:

a. Operating System

b. Virtualization layers

c. Database

d. Analytics tool

e. All third-party products/ engines deployed in the AIRSEWA

iii. Application Software Maintenance

The bidder shall provide continuous and indefinite support through on-site team/telephone/E-mail/Video Conferencing/installation visits as required. The bidder shall address all the errors/bugs/gaps in the functionalities of the solution (vis-a-vis the FRS, BRD and SRS signed off) at no additional cost during the O&M phase.

All patches and upgrades from OEMs shall be implemented by the bidder. Technical upgrades of installation to the new version, as and when required, shall be done by the bidder. Any version upgrades of the software/tool/application will be done by the bidder after seeking prior approval from the MoCA/AAI and submitting the impact assessment of an upgrade as such.

Any changes/upgrades to the software performed during the support phase shall be subject to comprehensive and integrated testing by the bidder to ensure that the changes implemented in the system meet the specified requirements and do not impact any other existing functions of the system. A detailed process in this regard will be finalised by the bidder in consultation with the MoCA/AAI.

An Issue log shall be maintained by the bidder for the errors and bugs identified in the solution as well as any changes implemented in the solution and shall be periodically submitted to the MoCA/AAI.

The bidder will inform the MoCA/AAI (at least monthly) about any new updates/upgrades available for all software components of the solution along with a detailed action report. In case of critical security patches/alerts, the bidder shall inform the MoCA/AAI immediately along with any relevant

recommendations. The report shall also contain the bidder's recommendations on update/upgrade, benefits, impact analysis etc. The bidder needs to execute updates/upgrades though a formal change management process and subsequently update all documentations and Knowledge databases etc. The bidder will carry out all required updates/upgrades by following defined processes at no additional cost.

iv. Database Administration, Maintenance and Management Services

a. Management of database environment to maintain performance at optimum levels

b. End-to-end management of database on an ongoing basis to ensure smooth functioning of the same.

c. Tasks including, but not limited to, managing changes to database schema, disk space, storage, user roles etc.

d. Conduct code and configuration reviews to provide inputs to MoCA/AAI to improve the performance or resolve bottlenecks if any.

e. Performance monitoring and tuning of the databases on a regular basis including, preventive maintenance of the database as required.

f. Report backup status on a regular basis and ensure prompt problem resolution in case of failures in the backup processes.

g. Manage database upgrade or patch upgrade as and when required with minimal downtime.

h. Regular backups for all databases in accordance with the backup and archive policies and recovery whenever required with appropriate permissions. Testing of backup media should be undertaken from time to time to ensure availability of data. Bidder should coordinate with MoCA/AAI for back-up activities

v. Problem identification and Resolution:

a. Errors and bugs that persist for a long time, impact a wider range of users and are difficult to resolve in turn lead to application hindrances. The bidder shall resolve all the application problems through implementation of the identified solution (e.g. system malfunctions, performance problems and data corruption etc.)

b. Monthly reports on problems identified and resolved would be submitted to the MoCA/AAI team along with recommended solutions.

vi. Change and Version Control

All planned or emergency changes to any component of the system shall be carried out through the approved Change Management process. The bidder needs to follow all such processes (always based on industry ITSM framework). For any change, bidder shall ensure:

a. Detailed impact analysis is conducted
b. All Change plans are backed by Roll back plans
c. Appropriate communication on change required has taken place
d. Requisite approvals have been received
e. Schedules have been adjusted to minimize impact on the Production environment
f. All associated documentation is updated post stabilization of the implemented change
g. Version control is maintained for all software changes

The bidder shall define the Software Change Management and Version Control Process. For any changes to the solution, the bidder must prepare detailed documentation including proposed changes and impact to the system in terms of functional outcomes/additional features added to the system etc. The bidder shall ensure that software and hardware version control is carried out for the entire contract duration.

vii. Maintain configuration information

The bidder shall maintain version control and configuration information for application software and any relevant system documentation.

viii. Maintain System documentation

The bidder shall design, update and maintain all the System documentations of the AirSewa portal. These documents shall be up to date and shall be shared MoCA/AAI on the regular basis. The bidder should follow latest industry standards to design and maintain the documentations.

The bidder shall maintain at least the following minimum documentation with respect to the AirSewa portal:

a. High level design of whole system

b. Low level design for whole system/module design level

c. System Requirements Specifications (SRS)

d. Any other explanatory notes about system

e. Traceability matrix

f. Compilation environment

The bidder shall also ensure that any software system documentation is updated regarding the following:

a. Source code is documented

b. Functional specifications are documented

c. Application documentation is updated to reflect on-going maintenance and enhancements including FRS and SRS in accordance with the defined standards

d. User manuals and training manuals are updated to reflect on-going changes/enhancements

e. Standard practices of version control and management are adopted and followed

ix. All project documents need to follow a proper version control mechanism. The bidder will be required to keep all project documentation updated and ensure the updated project documents are submitted to the MoCA/AAI by the end of next quarter in case of any change.

x. For application support, the bidder shall assign a dedicated software support team to be based at the bidder's location as a single point of contact for resolution of all application related issues. This team will receive and resolve all the application related tickets/incidents. In the technical proposal, the bidder needs to provide the proposed application support team structure including the number of team members proposed to be deployed along with roles and skills of each. The Application support team shall comprise of the bidder's employees.

xi. Any software changes required due to problems/bugs in the developed software/application will not be considered as part of change control and will have to be completed by the bidder at no additional cost (even if it requires any enhancements/customizations).

xii. Bidder should share cloud utilization report monthly during the development and O&M stage. A separate cloud account should be created for the AirSewa project with access to the cloud account to MoCA/AAI for visibility.

xiii. The System Integrator (SI) shall provide logs, alerts, and corresponding reports from the deployed SIEM solution to confirm that no cybersecurity breaches, intrusions, or security incidents have occurred within the environment. These logs must be complete, unaltered, timestamped, and submitted as per the agreed reporting frequency or upon request by the customer's security team.

xiv. The System Integrator shall provide Cloud Downtime logs generated from the available monitoring and management tools integrated with the cloud environment. These logs must clearly indicate any service outages, performance degradation events, maintenance windows, and uptime statistics for the specified reporting period.

xv. Any cybersecurity breaches, intrusions, application defects or security incidents should be notified as per SLAs and as part of the documentation in monthly reports

2.12.2. Web based SLA Monitoring Tool

Bidder should deploy Web based SLA Monitoring Tool for monitoring all the SLAs as specified in Volume 3 of the RFP. The indicative list of features to be provided under the SLA Monitoring is provided below. Bidder must provide monitoring mechanism for all the SLAs mentioned in Volume 3 of the RFP.

a. The SLA Monitoring tool shall be tested, audited and certified for its accuracy, reliability and completeness before it is implemented by the bidder.

b. The selected tool must be from latest Gartner's magic quadrant leader's list/equivalent position

in Forrester wave/ leader in IDC marketspace (As a supporting document

c. the bidder must provide for relevant report of latest Gartner's magic quadrant leader's list/

equivalent position in Forrester wave/report of IDC).

d. The MoCA/AAI will periodically conduct quality reviews (preferably quarterly) to ensure that the tool is monitoring SLAs properly.

e. The bidder will work with MoCA/AAI officials to design these reports as per specific requirements. The solution provides both real time dashboards and historical view reporting of the SLA's at to the relevant stakeholders of MoCA/AAI.

f. The SLA monitoring tool should be able to monitor all the service levels defined in the service level agreement.

g. The SLA Monitoring tool should be web based

h. The tool should be also accessed by smart devices over internet/local airport LAN

i. The SLA monitoring tool will be tested in the Acceptance Phase for acceptance

j. The proposed tool should use smart visualization and graphics

Some of the minimum indicative features of the SLA Monitoring tool includes

a. Integrated performance view of all the SLAs along with various threshold violations for each service level.

b. Indicate the baseline, breach, lower and high-performance scores against each of the service level metric in a graphical form.

c. Dashboard must show bidder action taken on each of the SLA breach/ timelines for resolution etc.

d. The proposed tool should enable users to generate Pre-defined, Standard and Ad-hoc reports

e. Allow users to generate report with pre-defined fields to get a real-time view of the system from the portal.

f. Pre-defined Reports will be generated at pre-defined frequencies and will be available for to the users

g. Standard reports would include reports where report generation criteria are fixed and options for the same are available to the User. The User can select a combination of options and submits a request for generation of a report. The system generated the report and the same would be made available on the User dashboard.

h. Ad hoc Reports: These reports can be generated by the User by raising ad hoc queries. These would provide the ability to generate ad hoc reports as and when required by selecting any field, table or column (as per Database design) by drag and drop feature from the table.

2.12.3. Application Monitoring Services

The bidder should ensure exceptional end-user experience and consistently high service levels that meet business objectives by monitoring all end-user transactions 24x7 with low overhead. The application performance monitoring solution should accurately measure end-user transaction performance to prove IT is delivering against SLAs, business objectives and vendor commitments. APM should be deployed with the objective of improving IT productivity and control costs by quickly and accurately diagnosing problems occurring deep within the application and infrastructure, pinpoint failures and speed problem resolution.

The services to be provided by the bidder for Application Monitoring which includes following but not limited to:

a. Web/ API services

b. Application server

c. Database server

d. Middleware

e. Value Added Services

f. Other components as proposed by bidder

Application monitoring should map all transactions to the dependent infrastructure in real time for a single view of application health, business process flow and the entire transaction path to quickly triage issues

2.12.4. Storage Administration & Management Services (by Cloud Service Provider (CSP))

a. Management of storage environment to maintain performance at optimum levels.

b. Management of the storage solution including, but not limited to, storage management policy, configuration and management of disk array, SAN, tape library, etc.

c. Storage management, including but not limited to management of space, volume, Redundant Array of Inexpensive Disks configuration, LUN, zone, security, business continuity volumes, performance, etc

2.12.5. Workflow Management:

a. Workflow management module shall enable the workflows for various business processes. This shall include process tasks and routing. Workflow shall allow configuring various users such as Airport operators, Airlines, etc. into the business processes.

2.12.6. Backup and Archival Services

The services to be provided by bidder shall include:

a. Backup of storage as per the defined policies.

b. Monitoring and enhancing the performance of scheduled backups, schedule regular testing of backups and ensuring adherence to related retention policies as defined by AAI, DGCA, MoCA/AAI, and other relevant authorities.

c. Prompt execution of on-demand backups of volumes and files whenever required or in case of upgrades and configuration changes to the system.

d. Real-time monitoring, log maintenance and reporting of backup status on a regular basis.

e. Media management tasks, including, but not limited to, tagging, cross-referencing, storing, logging, testing, and vaulting in fireproof cabinets (onsite and offsite).

f. 365x24x7 support for file and volume restoration requests at the cloud (VPC)

2.12.7. Technical Support Chatbot

The technical support track includes running centralised helpdesk setup and operations for AIRSEWA project for a period of 36 months from the date of Go Live.

1. The help desk will primarily handle technical queries of internal stakeholders which includes Airport Operators, CISF Staff, Airlines, OTAs, etc. relating to AIRSEWA.

2. The bidder shall follow the implementation plan for the helpdesk launch as specified in the project timelines.

3. Helpdesk is required to ensure that the above-mentioned stakeholders can log calls and complaints for any technical issues they face while using the AIRSEWA. The following is included in the scope of work of bidder:

4. Landline/mobile number will be taken by bidder in MoCA/AAI name. This number will be published, and internal stakeholders can call this number for their issues.

5. Help Desk to have Interactive Voice Response (IVR) system for first level of call segregation.

6. In addition to the telephone call, the bidder shall also provide other channels for call logging like email and web interface.

7. Language Capabilities: Hindi and English

8. The bidder shall deploy helpdesk application accessible to all users through the AIRSEWA for logging issues.

9. The service window for Help Desk is 365X24X7 (Monday to Sunday).

10. The helpdesk is also expected to handle queries after their go live. The help desk will serve as single point of contact for reporting/resolution of all tickets (queries, errors, incidents, issues either application or infrastructure or operations related).

11. Considering the nature of the services, the bidder should decide for imparting proper training in soft skills; call handling, exposure to related application to prepare the CSEs to answer different types of queries, and on other aspects of Help Desk.

12. The bidder should ensure that all the staffs are put on actual duty only after providing them proper training

13. Before deployment of staff, he/she need to undergo training for at least a week time covering above areas. Certificate of such training need to be submitted on quarterly basis to MoCA/AAI.

The helpdesk would have following major activities and tasks:

1. Issues logged by users through Helpdesk should be accessed and serviced by the helpdesk personnel.

2. Track and route requests for service and to assist end users in answering questions and resolving problems. Assign severity level to each ticket as per the SOPs.

3. Acknowledgement should be sent to user along with service ticket number through an email immediately on call logging.

4. Routing the query received to the concerned team of the bidder for resolution of tickets

5. Notifying users the problem status and resolution through the tickets over email.

2.12.8. Proposed Project Team

1. Bidder shall provide adequate number of personnel each responsible for a specific role within the system. Bidder must provide clear definition of the role and responsibility of each individual personnel.

2. Bidder shall have a defined hierarchy and reporting structure for various teams that shall be part of the project.

3. Bidder shall ensure that all the personnel identified for this project have high level of integrity. Bidder shall obtain an undertaking from each of the personnel assigned and the same shall be submitted to MoCA/AAI as and when demanded.

4. The following table provides an indicative list of resource categories. However, bidder shall independently estimate the teams size required to meet the requirements of Service Levels as specified as part of this tender. Bidder shall propose qualified technical personnel with adequate skills levels to manage the infrastructure. Refer following table for Qualification & experience for website, web portal, application (mobile & web based.

Sr. No.	Key Personnel	Qualification
1	Project Manager-During project execution and O&M phase	· B Tech / B.E. / MCA (preferably in systems/IT)
		· Minimum 10 years' experience in IT/software development / IT System projects
		· Minimum of 2 years' experience as a Project Manager for IT Implementation Projects.
		· Exposure to Government Projects preferred.
		· Certified PMP, PRINCE 2 or equivalent.
2	Solution Architect	· B Tech / B.E. / MCA (preferably in systems/IT)
		· Minimum 7 years' experience of applied experience in web, mobile app, cloud hosting support.
		· Broad knowledge and understanding of IT concepts and architectures, coupled with practical knowledge of problem management and the principles and processes of implementing and delivering IT services
3	AI/ML Expert	BE (Computer Science or equivalent) qualified **AI/ML Expert** with **2 years of hands-on experience** in developing and deploying Artificial Intelligence, Machine Learning, and Generative AI solutions.
4	Cloud Expert	B Tech / B.E. / MCA (preferably in systems/IT) Minimum 7 years' experience of applied
5	UI/UX Designer	· Minimum B Tech / B.E. / MCA /BCA / MFA / B.Arch. / B. Design / M. Design /PG diploma in Web Design
		· Minimum 5-7 years' experience in designing for IT/ software development / IT System projects/Website development / mobile Application Development.

Sr. No.	Key Personnel	Qualification
		· Minimum of 2 years' relevant experience Web and Mobile UI-UX design and development.
		· Exposure to Government Projects preferred.
6	Developer	· Minimum B Tech / B.E. / MCA (Computer Science or IT) for all developers.
		· 5+ years' experience: Minimum 5 years' experience in IT/software development / IT System projects / Website Development/ Mobile Application Development with Minimum of 3 years' experience in website and Mobile Application development for IT projects.
		· 3-5 years' experience: Minimum 3 years' experience in IT/ Software development / website Development / Mobile Application Development with minimum of 2 years' experience in Website and Mobile Application development for IT Projects.
7	Documentation Expert	**Expert 1: Content Writer**
		· Minimum Graduation / Post Graduation in English/ Hindi PG Diploma in Mass Communication from Government recognized universities / institutes.
		· Minimum of 3 years' experience in content writing for Websites & / or Web Portals / Mobile Apps.
		Expert 2: Technical Writer
		· Minimum B.E. / B. Tech (IT / Computer Science) with excellent technical writing skills.
		· Minimum of 3 years' experience in same role (i.e. Technical Writer)
8	Mobile Application Developer	· Minimum B.E / B. Tech / MCA (IT / computer science)

Sr. No.	Key Personnel	Qualification
		· 5+ years' experience: Minimum 5 years' experience in IT/software development / IT System projects / Website Development/Mobile Application Development with minimum of 2 years' experience in same role (i.e. Mobile Application Development).
		· 3-5 years' experience: Minimum 3 years' experience in IT / software development / IT System Projects/ Website Development / Mobile Application Development with minimum of 2 years' experience in same role (i.e. Mobile Application Development)
9	Database Admin	· Minimum B Tech / B.E. / MCA (in Computer Science or Engineering) for all database admin
		· 5+ years' experience: Minimum 5 years' experience in IT/software development / IT System projects / Website Development/Mobile Application Development with Minimum of 2 years' experience in same role (i.e. Database Admin).
		· 3-5 years' experience: Minimum 3 years' experience in IT/software development / IT System projects / Website Development/Mobile Application Development with Minimum of 2 years' experience in same role (i.e. Database Admin).
10	Security Expert	· Minimum B.E / B. Tech / MCA (preferably in CS/ Electronics and Communication)
		· Minimum 7 years' experience in Cyber security (CISM/CISSP preferred)
		· Exposure to BS15000 process/ISO27001 /BS 7799/ISO 20000/ ITIL or ITIL certified. Experience working with industry-wide frameworks and standards like MITRE ATT&CK, STIX, TAXII, and SCAP

Sr. No.	Key Personnel	Qualification
11	Systems Admin	· B Tech / B.E. / MCA (preferably in Computer Science)
		· Systems Administration/System Engineer certification in Unix and Microsoft.
		· Minimum of 2 years of systems administration experience.
12	Tester (Test Engineer)	· Minimum B Tech / B.E. / MCA (preferably in Computer Science / IT) for all testers
		· 5+ years' experience: Minimum 5 years' experience in IT/software testing / IT System projects / Website testing /Mobile Application testing with Minimum of 3 years' experience in website, mobile app testing role (i.e. Tester).
		· 3-5 years' experience: Minimum 3 years' experience in IT/software development / IT System projects / Website Development/Mobile Application Development with Minimum of 2 years' experience in website, mobile app testing role (i.e. Tester).
13	Training and Change Management Expert	· Minimum B.E. / B. Tech /MCA
		· Minimum 5 years' experience in training / change management
		· Minimum of 2 years' experience in delivery of User Training Programs through Training and
		Change Management.
14	Helpdesk – During O&M phase (on site)	Minimum 3 people per shift to manage the helpdesk with minimum 1-2 years of experience.
15	PACR manpower	PACR's Call Center manager, TL, IT expert, CRO, MTS as per "Manpower Deployment and Operation of PACR"

2.13. Manpower Deployment and Operation of PACR

a) The PACR shall operate on a per-seat-per-shift model, initially comprising upto 15 seat-shifts, scalable as per MOCA/AAI requirements.

b) The agency shall deploy and manage the following strength on a 24/7 basis in three shifts:

Role	Count	Minimum Qualification & Experience
Control Room Managers	1 seat in total Across all shifts per day	Post-Graduate (Management/Aviation); 8-10 yrs exp.
Team Leads/ Trainers	3 seats in total Across all shifts per day	Graduate; 5+ yrs exp. in help desk/supervisory roles (in the field of aviation, banking)
Control Room Operators	15 seats in total Across all shifts per day	Graduate; Knowledge of English and Hindi (spoken and written), multilingual skills; Basic proficiency in computer operations, MS Office, and data entry; prior experience in multi-channel support/coordination preferably in the field of aviation operations, customer service, or other public-facing roles (desirable)
IT Support/ Security Expert	3 seat in total Across all shifts per day	Graduate or equivalent with a valid industry recognized certification in Cisco Certified Network Professional (CCNP) Security. Experience of at least 2 years in IT and telecom support role which includes Hardware, /cloud, Server, PRI, Internet Leased Line, software, application support etc. IT Experience in any call center environment/setup will be preferred.
Multi-Tasking Staff (MTS)	4 seats in total Across all shifts per day	10th/12th Pass; 1 yr or more exp.

1. The PACR will have CRO serving initially in Hindi and English. In later stages more languages may be required i.e. Tamil, Telugu, Malayalam, Kannada, Bengali, Odia, Punjabi, Gujarati, Marathi, and Assamese etc. MOCA/AAI from time-to-time may request change in composition of the agents catering to different languages as per the requirements and queue times of calls.

2. During the tenure of the contract, MOCA/AAI may ask to replace/increase with any language specified by MOCA/AAI as per project requirements. The quantity shall be provided by the Bidder as per the price quoted in this bid and on the same terms and conditions.

3. All manpower shall be on the Bidder's payroll and compliant with labour laws, EPF, ESIC, and minimum wage regulations etc.

2.14. The summary of tickets raised can be seen at Annexure-II. The ticket counts is indicative only.

2.15 Role and Responsibilities of MoCA/AAI

1. MoCA/AAI shall facilitate/ assist in providing necessary approvals for Bidder from concerned departments with respect to this project

2. MoCA/AAI shall facilitate for clearing the invoices and payments of the Bidder for smooth running of the project as per rules.

3. MoCA/AAI shall provide access to bidder to MoCA/AAI/ AAI premises for activities related to smooth running of the project.

4. MoCA/AAI shall facilitate timely approvals to the requests/ applications/ documents, etc submitted by the bidder in writing in timely manner to ensure smooth running of the project

5. MoCA/AAI shall help in knowledge transfer, providing required data related to the existing procedures, equipment, etc to the Bidder.

6. MoCA/AAI shall facilitate/ assist in providing necessary data/ support for integration for Bidder from concerned stakeholders with respect to this project

7. MOCA/AAI shall facilitate PRI/SIP subscription, premises, Table and Chairs, electricity with power backup, water supply for PACR Setup and Operations.

2.16 Warranty & Maintenance

2.16.1 Standard:

The bidder warrants that the Project, including all the system(s), materials and goods supplied pursuant to the Agreement, shall be free from any defect or deficiency in the material, design, engineering, and workmanship that prevent the system and/or any of its systems(s) from fulfilling the technical requirements or that limit in a material fashion the performance, reliability, or extensibility of the system and/or any of its sub-system(s). Commercial warranty provisions of products supplied under the Agreement shall apply to the extent they do not conflict with the provisions of this Agreement.

The bidder also warrants that the products, materials and other goods supplied under the Agreement are new, unused and incorporate all recent improvements in design that materially affect the system's or subsystem's ability to fulfill the technical requirements specified in the RFP.

2.16.2 In addition, the bidder warrants that: (i) all Goods components to be incorporated into the System form part of the Bidder/OEM's and/or JV/consortium member's current product lines.

2.16.3 The warranty period shall commence from the date of Successful implementation/ Go Live of the project or of any major component or subsystem for which phased Go-Live is provided for in the Agreement and shall extend for as follow:

Component	Period
<COTS Software>	<Till the end of the agreement >
<Bespoke Software>	<Till the end of the agreement>
PACR Licenses and Hardware	<Till the end of the agreement> All IT equipment's should have Comprehensive onsite warranty

2.16.4 AAI/Government department should approve signoff within 15 days from the submission of deliverables for Go-Live/Phased Go-live (as relevant, depending on project requirement) by the implementing agency. In case AAI/Government department fails to respond and provide feedback on the above stated submission, the deliverables will be deemed accepted for the commencement of warranty for the project. However, in case AAI confirms to bidder an alternative date, then the date would stand revised for deemed acceptance. Such revisions may be limited to 2 (two).

2.16.5 If during the warranty period any defect or deficiency is found in the material, design and performance/workmanship of the Project and other Services provided by the System Integrator, the bidder shall promptly, in consultation and agreement with AAI, and at the System Integrator's sole cost repair, replace, or otherwise make good (as the bidder shall, at its discretion, determine) such default, defect or deficiency as well as any damage to the system caused by such default, defect or deficiency.

2.16.6 The bidder may, with the consent of AAI, remove from the site any product and other goods that are defective, if the nature of the defect, and/or any damage to the System caused by the defect, is such that repairs cannot be expeditiously carried out at the site.

2.16.7 If the repair, replacement, or making good is of such a character that it may affect the efficiency of the System, AAI may give the bidder notice requiring that tests of the defective part be made by the bidder immediately upon completion of such remedial work, whereupon the bidder shall carry out such tests. If such part fails the tests, the bidder shall carry out further repair, replacement, or making good (as the case may be) until that part of the System passes such tests. The tests shall be agreed upon by AAI and the Supplier

2.16.8 If the bidder fails to commence the work necessary to remedy such defect or any damage to the System caused by such defect within a reasonable time period, AAI may, following notice to the bidder, proceed to do such work or contract a third party (or parties) to do such work, and the reasonable costs incurred by AAI in connection with such work shall be paid to AAI by the bidder or may be deducted by AAI from any amount due to the SI

2.16.9 If the System or any of its sub-systems cannot be used by reason of such default, defect or deficiency and/or making good of such default, defect or deficiency, attributable to Bidder, the warranty period for the Project shall be extended by a period equal to the period during which the Project or any of its system could not be used by AAI because of such defect and/or making good of such default, defect or deficiency.

2.16.10 Items substituted for defective parts of the System during the Warranty Period shall be covered by the Warranty for the remainder of the Warranty Period applicable for the part replaced or three (3) months, whichever is greater

2.16.11 Implied Warranty: The warranties provided herein are in lieu of all other warranties, both express and implied, and all other warranties, including without limitation that of merchantability or fitness for intended purpose is specifically disclaimed.

2.17 Insolvency of bidder

i. In the event of the bidder being adjudged insolvent or going voluntarily into liquidation or having received order or other order under Insolvency act made against Company or, the passing of any resolution, or making of any order for winding up whether voluntarily or otherwise, or in the event of the bidder failing to comply with any of the conditions herein specified, AAI shall have the power to terminate the contract without any prior notice.

ii. The Lead Bidder shall not appoint any delegate/sub-contractor without the consent in writing of AAI for the performance of Services under this contract.

iii. All the personnel proposed by the bidder except JV/Consortium services should necessarily be on the payroll of the Bidder.

iv. The bidder shall be responsible and shall ensure the proper performance of the JV/Consortium services and shall be liable for any non-performance or breach by such JV/Consortium member. The Lead Bidder indemnifies and shall keep indemnified AAI against any losses, damages, claims or such other implications arising from or out of the acts and omissions of such providers. The bidder shall be responsible for making all payments to the JV/Consortium member as may be necessary, in respect of any services performed or task executed, and AAI shall not be responsible for any part or full payment which is too due to such JV/Consortium member.

v. All rights of use of any process, product, service or facility developed or any other task performed by the JV/Consortium member for the Bidder, under this contract would lie exclusively with the AAI in perpetuity free from all liens, encumbrances and other third-party rights and the bidder shall, wherever required, take all steps that may be necessary to ensure the transfer of such ownership in favor of the AAI.

vi. Nothing in this Contract or any agreement hereunder shall relieve the bidder from its liabilities or obligations under this Contract to provide the services in accordance with this Contract. AAI reserves the right to hold the bidder liable for any act/omission of JV/Consortium member.

vii. The bidder shall ensure that the JV/Consortium member appointed is competent, professional and possesses the requisite qualifications and experience appropriate to the tasks they will perform.

2.18 System Integrator's Obligations

i. General

The bidder shall always act, in respect of any matter relating to the Contract or to the activities, as faithful to AAI, and shall at all times support and safeguard AAI's legitimate interests in any dealings with any third-parties.

ii. Standard of Performance

a. The Lead Bidder shall perform the activities and carry out their obligations hereunder with all due diligence, efficiency, in accordance with generally accepted professional standards and practices, and shall observe sound management practices, and employ appropriate technology, equipment, machinery, materials and methods.

b. The Bidder's obligations shall include provision of all the services and deliverables specified by AAI in the Scope of Work and other sections of the RFP, Contract and changes thereof to enable AAI to meet its objectives and operational requirements. It shall be the bidder's responsibility to ensure the proper and successful execution, performance and continued operation of the proposed services in accordance with and in strict adherence to the terms of their Bid, the RFP and Contract.

c. The bidder shall ensure that its resources are competent, professional and possess the requisite qualifications and experience appropriate to the task it is required to perform under this Contract.

d. All the product's licenses (subscription/ perpetual) that bidder procures should be in the name of AAI for Contract period.

e. The Bidder's representative shall have all the powers requisite for the performance of services under the Contract. The Bidder's Representative shall liaise with for the proper coordination and timely completion of the works and on any other matters pertaining to the works. He/she shall extend full co-operation to AAI in the manner required by them for supervision/ inspection/ observation of the procedures, performance, reports and records pertaining to the works. He/ she shall also have complete charge of the bidder personnel engaged in the performance of the works and to ensure internal discipline, compliance of rules, regulations and safety practice. The Bidder's Project Manager is expected to act as single point of contact for any matter related to this project till completion of all the milestones. During the Operation and Maintenance stage, Project Manager is expected to attend all the performance review meetings and shall be single point of contact for the AAI for any matter related to contract or the services.

iii. Reporting Progress

a. The bidder shall monitor progress of all the activities related to the execution of this Contract and shall submit report to the AAI, at no extra cost. These progress reports with reference to all related work and their progress to be submitted at the end of each month or before the expiry of the last day.

b. Periodic meetings shall be conducted on an ongoing basis between the representatives of AAI and the Bidder.

c. AAI reserves the right to inspect and monitor/ assess the progress/ performance of the work/ services at any time during the Contract. AAI may demand information related to project and the bidder shall provide documents, data, material or any other information which AAI may require, to enable it to assess the progress/ performance of the Work/ Service.

d. At any time during the course of the Contract, AAI shall also have the right to conduct an audit to monitor the performance by the bidder of its obligations/ functions in accordance with the standards committed to or required by AAI, the said audit can be conducted either itself or through an independent audit firm appointed by AAI as it may deem fit. The bidder undertakes to cooperate for audit and provide to AAI/ any other firm appointed by AAI, all documents and other details as may be required by them for this purpose.

iv. Conflict of Interests

The bidder shall hold the Department's interest's paramount, without any consideration for future work, and strictly avoid conflict with other assignments or their own corporate interests.

v. Prohibition of Conflicting Activities

The bidder shall not engage, and shall cause their Personnel not to engage, either directly or indirectly, in any business or professional activities which would conflict with the activities assigned to them under this Contract.

vi. Confidentiality

Except with the prior written consent of the Department, the Lead Bidder and the Personnel shall not at any time communicate to any person or entity any confidential information, maps, images, reports, etc. acquired in the course of the activities/services, nor shall the bidder and the Personnel make public the recommendations formulated in the course of, or as a result of, the activities/ services. This clause shall survive even after expiry of the Contract.

vii. Accounting, Inspection and Auditing

The bidder shall keep accurate and systematic accounts and records in respect of the activities hereunder, in accordance with internationally accepted accounting principles and in such form and detail as will clearly identify all relevant time changes and costs, and the basis thereof.

viii. Responsibilities of the bidder in respect of local laws, employment of works etc.

a. The Bidder, at all times during the continuance of this contract shall, in all its dealings with local labour for the time being employed on or in connection with the work, have due regard to all local festivals and religious and other customs.

b. The bidder shall comply with all applicable State and Central Laws, Statutory Rules, Regulations etc. such as Payment of Wages Act, Minimum Wages Act, and Workmen Compensation Act, Employer's Liability Act, Industrial Dispute Act, Employers Provident Act, Employees State Insurance Scheme, Contract Labour (Regulation and Abolition) Act 1970, Payment of Bonus & Gratuity Act and other Acts, Rules and Regulations for labour as may be enacted by the Government during the tenure of the Contract and having force or jurisdiction at the project Site. The bidder shall also give to the local Governing Body, Police and other relevant Authorities all such notices as may be required by the Law.

c. The bidder shall pay all taxes, fees, license charges, deposits, tolls, royalties, commission or other charges which may be liable on account of his operations in executing the contract.

d. The bidder shall be responsible for provision of Health and Sanitary arrangements (more particularly described in Contract Labour Regulation & Abolition Act), Safety precautions etc. as may be required for safe and satisfactory execution of contract.

e. The bidder shall arrange, coordinate his work in such a manner as to cause no hindrance to other agencies working in the same premises.

2.19 AAI's obligations

i. AAI or its nominated representative shall act as the nodal point for implementation of the contract and for issuing necessary instructions, approvals, commissioning, acceptance certificates, payments etc. to Bidder.

ii. AAI shall ensure that timely approval is provided to bidder as and when required, which may include approval of project plans, implementation methodology, design documents, specifications, or any other document necessary in fulfilment of this contract.

iii. AAI representative shall interface with Bidder, to provide the required information, clarifications, and to resolve any issues as may arise during the execution of the Contract. AAI shall provide adequate cooperation in providing details, coordinating and obtaining of approvals from various governmental agencies, in cases, where the intervention of the AAI is proper and necessary.

iv. AAI shall ensure the proper coordination between the Successful Lead Bidder and the existing bidder for smooth project transition by assigning an officer as a single point of contact (SPOC).

v. AAI may provide on Bidder's request, particulars/ information/ or documentation that may be required by bidder for proper planning and execution of work and for providing services covered under this contract and for which bidder may have to coordinate with respective vendors.

vi. AAI shall provide to bidder only sitting space and basic infrastructure (including sitting place, drinking water facility, AC, washrooms etc.) not including stationery and other consumables at the AAI office.

vii. AAI hereby agrees to provide access to the project sites within the agreed timelines. AAI agrees that the bidder shall not be in any manner liable for any delay arising out of the failure of AAI to provide access of the sites.

2.20 Submission of Project Plan

i. The Successful Lead Bidder shall submit a detailed project plan as per defined timeline.

ii. The project plan shall include details of the project showing the sequence, procedure and method in which bidder proposes to carry out the works.

iii. The Plan so submitted by bidder shall conform to the requirements and timelines specified in the Contract.

iv. AAI and bidder shall discuss and agree upon the work procedures to be followed for an effective execution of the works, which bidder intends to deploy and shall be clearly specified.

v. The detailed project plan shall clearly specify the various project milestones and project deliverable schedules. It shall also include the following:

a. Project governance and management plan,

b. Communication structure,

c. Proposed staffing,

d. Roles and responsibilities,

e. Processes and tool set to be used for quality assurance,

f. Security and confidentiality practices in accordance with industry best practices,

g. Timelines for completion of activity,

h. Delivery schedule in accordance with the Contract,

i. Data Migration Plan

vi. Approval by AAI of the Project Plan shall not relieve bidder of any of his duties or responsibilities under the Contract.

vii. If Bidder's work plans necessitate a disruption/ shutdown in the operations of AAI, the plan shall be mutually discussed and developed so as to keep such disruption/ shutdown to the barest unavoidable minimum. The bidder shall be held responsible for any time and cost arising due to failure of bidder to develop/adhere such a work plan.

2.21 Good Faith

The Parties undertake to act in good faith with respect to each other's rights under the Contract and to adopt all reasonable measures to ensure the realization of the objectives of the Contract.

The Successful Lead Bidder shall ensure and procure compliance of each of the Key Performance Indicators as indicated in Service Level Agreement and shall be penalized on failure to comply.

2.22 Compliance with Statutory and Regulatory Provisions

i. It shall be the sole responsibility of the Lead Bidder to comply with all statutory and regulatory provisions while delivering the services mentioned in this RFP, during the course of the contract. Also, the bidder needs to ensure compliance of the project with Government of India IT security guidelines including provisions of:

   a. The Information Technology Act, 2000 and amendments thereof and Guidelines /Advisories/ Orders / Notifications, DPDP act/rule published by CERT-IN/MeitY/DPIIT /DOT (Government of India) and Department/ Government of Delhi issued till the date of publishing of tender notice. Periodic/ Latest amendments in these Guidelines /Advisories/ Orders / Notifications during project duration need to be complied with.

ii. To encourage "Make in India" and promote manufacturing and production of goods and services in India as per the revised "Public Procurement (Preference to Make in India), Order 2017", circulated by the Department of Promotion of Industry and Internal trade, Ministry of Commerce & Industry, Govt. Of India vide Order No. P-45021/2/2017-PP (BE-II) dated 04.06.2020 will be applicable for this tender.

2.23 Disputes and arbitration

Amicable Settlement: The Parties shall use their best efforts to settle amicably all disputes arising out of or in connection with the Agreement or the interpretation thereof.

Conciliation: In the event of any dispute, difference of opinion or dispute or claim arising out of the Agreement or breach, termination, shall firstly be attempted to be settled by conciliator appointed/ nominated by AAI on receipt of such requests from either party. The conciliator shall make the settlement agreement. The settlement agreement shall be final and binding on the parties. The settlement agreement shall have the same status and effect of an Arbitration Award under the Arbitration and Conciliation Act 1996. If the conciliation failed, the party may refer the matter to Arbitration to resolve the disputes.

- If during the subsistence of this Contract or thereafter, any dispute between the Parties hereto arising out of or in connection with the validity, interpretation, implementation, material breach or any alleged material breach of any provision of this Contract or regarding any question, including as to whether the termination of this Contract by one Party hereto has been legitimate, the Parties hereto shall endeavour to settle such dispute amicably and/or by Conciliation to be governed by the Arbitration and Conciliation Act, 1996 or as may be agreed to between the Parties. The attempt to bring about an amicable settlement is considered to have failed as soon as one of the Parties hereto, after reasonable attempts, which attempt shall continue for not less than 21 (Twenty-One) days, gives 30 (thirty) day notice to refer the dispute to arbitration to the other Party in writing.

- The Arbitration proceedings shall be governed by the Arbitration and Conciliation Act, 1996. The Arbitration proceedings shall be held in Delhi, India.

- The Arbitration proceeding shall be governed by the substantive laws of India.

The proceedings of Arbitration shall be in Hindi/ English language.

- Except as otherwise provided elsewhere in the Contract if any dispute, difference, question or disagreement arises between the parties hereto or their respective representatives or assignees, at any time in connection with construction, meaning, operation, effect, interpretation or out of the Contract or breach thereof the same shall be referred to a Tribunal of three (3) Arbitrators, constituted as per the terms of and under the (Indian) Arbitration and Conciliation Act, 1996. Each party to the Contract shall appoint or nominate one Arbitrator each, the two Arbitrators so appointed/ nominated by the Parties herein shall together choose the third Arbitrator, who will be the Presiding Arbitrator of the Tribunal.

- Both the Parties will make attempts to appoint arbitrators with background of Information Technology related activities, Information Technology related acts, understanding of the processes and activities of emergency response organizations (preferably ambulatory services) and an understanding of the software applications.

- In case, a party fails to appoint an arbitrator within 30 days from the receipt of the request to do so by the other party or the two Arbitrators so appointed fail to agree on the appointment of third Arbitrator within 30 days from the date of their appointment upon request of a party, the Justice of the Delhi High Court or any person or institution designated by him shall appoint the Arbitrator or Presiding Arbitrator upon request of one of the parties.

- Any letter, notice or other communications dispatched to the bidder relating to either arbitration proceeding or otherwise whether through the post or through a representative on the address last notified to the AAI by the bidder shall be deemed to have been received by the bidder although returned with the remarks, refused 'undelivered' where about not known or words to that effect or for any other reasons whatsoever.

- If the Arbitrator so appointed dies, resigns, incapacitated or withdraws for any reason from the proceedings, it shall be lawful for the bidder to appoint another person in his/ her place in the same manner as aforesaid. Such person shall proceed with the reference from the stage where his/ her predecessor had left if both parties consent for the same.

- It is a term of the Contract that the party invoking arbitration shall specify all disputes to be referred to arbitration at the time of invocation of arbitration and not thereafter
- It is also a term of the Contract that neither party to the Contract shall be entitled for any interest on the amount of the award
- The Arbitrator shall give reasoned award and the same shall be final, conclusive and binding on the parties
- The fees of the arbitrator, costs and other expenses incidental to the arbitration proceedings shall be borne equally by the parties
- Each Party shall bear the cost of preparing and presenting its case, and the cost of arbitration, including fees and expenses of the arbitrators, shall be shared equally by the Parties unless the award otherwise provides.
- With respect to any dispute arising out of or related to this Contract, the parties' consent to the exclusive
- jurisdiction of, and venue in, the District Court located in Delhi, India.

2.24 Service Level Agreement (SLA)

2.24.1. SLA for AirSewa

For purposes of the SLA, the definitions and terms as specified in the document along with the following terms shall have the meanings set forth below:

I. "Downtime" – Time period for which the specified services/ components/ outcomes are not available in the concerned period which would exclude downtime owing to Force Majeure, scheduled maintenance time (approved by AAI).

II. "Incident" – Any event/ abnormalities in the service being rendered, that may lead to disruption in normal operations and services to the end user.

III. "Ticket Resolution time" - means time taken by the Helpdesk to resolve (diagnosing, troubleshooting and fixing) a service request for any incident.

IV. "Ticket Closure time"- This includes the ticket resolution time and approval time taken by department post resolution of ticket. If it is identified that issue is not resolved then "Ticket resolution Time" will be addition of Ticket closure time and actual ticket resolution time for SLA computation.

V. "Scheduled Maintenance Time" – Time period for which the specified services/ Application is not available due to scheduled maintenance activity pre-approved by AAI.

VI. "Service Request" - Request from a user for reporting incident or to seek support, delivery, information, advice or documentation.

VII. "Total Time" – Total number of hours in the quarter (or the concerned period) being considered for evaluation of SLA performance.

VIII. "Uptime" – Time period for which the specified services/ outcomes are available in the period being considered for evaluation of SLA. Formulae for calculation of Uptime:

$$uptime\ (\%) = \{1 - (Downtime/(\ Total\ time - scheduled\ maintenance\ time)\ \}\ x\ 100\%$$

The prime objective of service levels is to ensure quality of services from the Bidder, in an efficient manner as defined in RFP. To meet this objective, the bidder will provide the Service Levels in accordance with the performance levels as set out in detail in this Section i.e., Service Level Agreement.

The service level targets define the levels of service to be provided by the bidder to AAI for the duration of this contract or until the stated SLA targets are amended.

2.24.2. General Principles of SLA:

I. This Agreement shall govern the provision of the Contracted professional services of the bidder to the AAI.

II. The bidder shall comply to service level requirements for availability and quality of services, throughout the period of the Contract or extensions, if any.

III. 24x7 shall mean hours between 12:00 AM -11.59 PM midnight on all days of the week.

IV. The bidder needs to perform periodic maintenance/upgradation on hosting environment for the purposes of system upgrades, maintenance, and backup procedures. The maintenance can be done only on Scheduled Maintenance Time with at least 3 days prior approval from AAI. However, no downtime will be allowed for the critical modules. For those services, the bidder needs to devise a mechanism for the uninterrupted services even in case of any maintenance and updates through DR sites.

V. The purpose of imposing penalty on account of SLA is to ensure the performance of the services as per defined parameters. In view of this, if the performance of the services is not improved, it shall constitute enough grounds for the annulment of the Contract.

2.25 Liquidated Damages

Time is the essence of the project and the delivery dates are binding on the Bidder. In the event of delay or any gross negligence in implementation of the project for causes solely attributable to the Bidder, in achieving the milestones, the AAI shall be entitled to recover from the bidder as liquidated damages. Delays that are not attributed to bidder shall not be considered for the penalty.

The liquidated damages subject to a maximum of 10% of the Contract Value.

If the cumulative liquidated damages go beyond 10% of the Contract value, AAI reserves the right to take appropriate action including termination.

This right to claim any liquidated damages shall be without prejudice to other rights and remedies available to AAI under the Contract and law.

The applicable Liquidated damages may be recovered by deduction of the same value from first or any invoice of CAPEX or OPEX as defined in payment terms. Any such recovery or liquidated damages shall not in any way relieve the bidder from any of its obligations to complete the Work or from any other obligations and liabilities under the Contract.

The SLA parameters shall be monitored on a quarterly basis as per the individual SLA parameter requirements while reporting shall be monthly or instance/incident/event bases. However, if the performance of the system/services is degraded significantly at any given point in time during the contract and if the immediate measures are not implemented and issues are not rectified to the complete satisfaction of AAI or an agency designated by them, then AAI will have the right to take appropriate disciplinary actions including termination of the contract.

Phase	Deliverable	Timelines	Measurement Interval	Penalty
Resource Deployment	Deployment of all Key resources whose resume has been submitted in the proposal	T+ 15 Days (T1)	Weekly after target date of completion of activity	0.5% of the R1 for delay of each week beyond the targeted timeline
Design Phase	Project Plan	T+ 1 (Month (T2)	Weekly after target date of completion of activity	1% of the R1 value for delay of each week beyond the targeted timeline
	Resource Deployment			
	Project Charter			
	Software Requirement Specification Document			
	System Design Document			
	Solution Architecture			
	Integration plan			
	API Architecture			
	Cloud Considerations & incremental strategy			
	Cloud Security architecture			
Development and Implementation Phase	Development of website & Application on Cloud	T+3 (Months) (T3)	Weekly after target date of completion of activity	1% of the R1 value for the delay of each week beyond the targeted timeline
	Analytics and Reporting			
	Functional Testing			
	Unit Testing			
	Performance Testing			
	Vulnerability Assessment			
	User Acceptance Testing Report			
Integration	Complete integration with all systems			

Phase	Deliverable	Timelines	Measurement Interval	Penalty
Final Acceptance	Acceptance of complete application by AAI	T+ 4.5 (Months) (T4)	Weekly after target date of completion of activity	1% of the R1 value for the delay of each week beyond the targeted
Go Live	Go Live			
Final SRS Sign-off	Final SRS Sign-off	T+ 5 (Months) (T5)	Weekly after target date of completion of activity	1% of the R1 value for the delay of each week beyond the targeted timeline

R1: Price quoted in financial bid for Design, Implementation & Go-Live phase.

2.26 Performance Service Levels

Sr.No.	Activity	Target	Measurement Interval	Target	Penalty
1	Availability / Uptime	Measured as uptime or downtime calculated from time ticket was raised to the time of ticket closure	Per quarter	Baseline: >= 99.95% uptime ofthe Software and services in a quarter as per availability SLA measurement	NIL
				Less than 99.95% uptime, upto 95%	((Baseline% - Achieved Service level%)/ Baseline%) x Quarterly O&M Payment) Baseline = 99.95%
				Less than 95% uptime	Department may invoke termination and exit clauses part from the penalty clause outlined for availability in

Sr.No.	Activity	Target	Measurement Interval	Target	Penalty
					the previous sub clause
2.	Cyber security clause	Cybersecurity Audit and safe to host certificate issued by Cert-In empaneled 3rd party agency on hosting platform	Per Quarter basis	No Breach	3% of Quarterly O&M Cost in case of any breach.
3.	Vulnerability Assessment	Authenticated Mode Assessment should be done for hosting platform and application on a quarterly basis. The report to be Submitted to AAI.	Quarterly	100% coverage of assets as per respective VA report for that Quarter	Any non compliance to the target will lead to a penalty applicable 0.5% of Quarterly O&M Cost.
4.	Penetration testing	Penetration Testing (external) will be conducted once every year. All detected vulnerabilities to be closed within one month from the date of identified vulnerabilities, start date will be Go-live date. The report to be submitted to AAI.	Yearly	100% coverage of assets as per respective PT report for that Year.	Any non compliance to the target will lead to a penalty of 0.5% of Yearly O&M Cost.

5.	Non-Compliance to DPDP IT Act 2000 and its amendments or any other guidelines or laws issued by AAI, MeitY or Govt. of India	Strictly to be followed	Appropriate certificates/ proof to produced as per law / guideline	No Breach	3% of Quarterly O&M payment in case of any breach.
6.	Report on attempted attacks/unauthorized access to database.	Any attacks / unauthorized access on DB including any unlawful attempt of data extraction via any medium	Weekly	Baseline: 100% of such attempt to be reported within 15 Minutes of occurrence. Breach: < 100%	1% of Quarterly O&M Cost for each security incident not reported within the designated time
7.	Patch upgrade	Time to time patch upgrade in case of new release of any underlying library / software	6 Month	No Breach	0.5% of Two Quarter O&M Cost on each such cases o breach of SLA.
8.	Response Time for all Transactions like log-in, payment gateway	Response time of All grievance transactions	98% of business transactions within the limit of 2 seconds measured quarterly basis.	Breach: < 99%	.001 % for each transaction would be deducted from the quarterly O&M payment.

Note:- If any penalty provision is defined by Government for violation of any act/rule in above table or during execution of the works of the bidder, the penalty provision will supersede the penalty defined in the table. i.e. penalty provision of Government will be applicable.

2.27 Incident Management SLAs

Severity of Outages / Application defects is defined as below –

- **Severity 1: Critical**

Showstoppers involving major functional failure in the application. There are no usable workarounds available to resolve the problem. Affects majority of the users and airports (more than 25%).

- **Severity 2: High**

Users face severe functional defects in the application irrespective of the cause. Workarounds is available. Affects majority of the users (more than 25%).

- **Severity 3: Moderate**

Moderate functional defects in the application irrespective of the cause. Has a convenient and readily available workaround. Affects a few users (less than 25%)

- **Severity 4: Low**

Requiring cosmetic functional changes. Does not require any workaround. It may include user query / suggestions but has no business impact.

Penalty applicable on bidder as per the breach of SLA for Incident Management

Sr.No.	Activity	Benchmark - Time to respond and resolve the issue	Measurement Interval	Target	Applicable Penalty
1	Application Outage / Defect with critical severity (Severity 1)	• Response Time: 15 minutes • Resolution Time: 4 hours	Per Quarter basis	Response and Resolution to the issues as per benchmark outlined	5% of Quarterly O&M receivable in case of any breach.
2	Application Outage / Defect with High Severity	• Response Time: 30 minutes • Resolution Time: 8 hours	Per Quarter basis		3% of Quarterly O&M receivable in case of any breach.
3	Application Outage / Defect with Moderate Severity	• Response Time: 3 hours • Resolution Time: 24 hours	Per Quarter basis		2% of Quarterly O&M receivable in case of any breach.
4	Application Outage / Defect with low Severity	• Response Time: 24 hours • Resolution Time: 3 days	Per Quarter basis		1%of Quarterly O&M receivable in case of any breach.

- **Response Time** is the amount of time between when an incident is identified / reported and when the bidder **acknowledges** it and begins working on it.
- **Resolution Time** is the total amount of time from when an incident is logged until the service is restored to normal by the Bidder.

2.28 SLA for PACR

Sl. N.	Activity	Target/Service level	Penalty for delays beyond
PACR Setup			
1	Deployment of Manpower	100% deployment at T + 4.5 months	Rs. 1000 per day per 'SEAT BASED ROLES" for every day of delay.
2	Go-Live & Stabilization	T + 5 Months	Penalty of Rs 2,000 per day
PACR Operations			
1	Maintenance of Call Center – IT infrastructure provided by selected Bidder desktops, Server, headsets, UCM, Mouse, firewall etc or any Software, application provided by Selected Bidder not working.	1 day	If any of the system is down beyond 1 day, penalty will be levied per day per item/system at the rate of 1,000.
2	Manpower deployed must be on the payroll of Lead Bidder	100% compliance at any audit instance	Resource not on Bidder payroll: penalty of Rs. 5,000 per instance per resource (not seat)
3	Manpower Replacement on request	Within 30 days of intimation	Rs 1,000 per day per 'SEAT BASED ROLES"
4	Manpower shortage due to relieving, absent and holiday	100% Physical 'SEAT BASED ROLES" head Count/Presence may be verified by client on daily basis. Bidder needs to manage week-offs, leave, shrinkage, relieving and absenteeism without any manpower deviation on daily basic.	Penalty of Rs 500 per day of absence per 'SEAT BASED ROLES"
	Unified Operator Productivity (Inbound Calls / Outbound Calls / Emails / Chats)	Net productive time: 7 hours per shift (excluding breaks/training) Target: 60 Productivity Units (PU) per operator per shift PU Normalization: – Inbound Call: 1 PU per call – Outbound Call: 1 PU per connected & logged call – Email/Ticket (included not limited to Social media ticket, Online Ticket): 0.5 PU per response of email – Chat Session: 0.75 PU per completed chat • Target may be met through any combination of channels	Penalty based on shortfall from 60 PU (per operator per shift): Shortfall up to **7 PU**: No penalty (grace margin) Shortfall of **8–20 PU**: ₹500 Shortfall of **21–40 PU**: ₹1,000 Shortfall of **more than 40 PU**: ₹1,500

6	Increase of 'SEAT BASED ROLES"	Increase of seats within 15 days of intimation.	Rs. 2,000 per 'SEAT BASED ROLES" per day beyond the service level
7	Telephony/CTI server	Availability 99 %	Rs. 5000 per drop below 99%

Note: For any breach of SLA which is beyond the purview of the Bidder will be mutually discussed and suitable action can be proposed. SLA penalties for PACR Operations will not be applicable till 60-days post deployment of manpower. Verifiable dashboard should be available for all above SLA data in AirSewa Dashboard

SLA penalties shall be computed monthly and capped at 10% of monthly invoice of PACR. All payments are subject to statutory deductions. Payment does not imply acceptance. AAI's interpretation shall prevail. Provisions relating to payments, penalties, recovery, audits, verification, interpretation, dispute resolution, and statutory compliances shall survive expiry or termination.

2.29 Financial Bid Formats

i. The Bidder shall submit the financial bid in the format provided in this section

ii. The bid value shall include all taxes and shall be in Indian Rupees (₹).

iii. The bid value shall be mentioned in words and figures. If there is a discrepancy between words and figures, the amount in words will prevail.

iv. The bidder should provide all prices, and quantities as per the prescribed formats provided below. Bidder should not leave any field blank. In case the field is not applicable, Bidder must indicate "0" (Zero) in all such fields.

v. MoCA or AAI reserves the right to ask the Agency to submit proof of payment against any of the taxes, duties, or levies indicated.

vi. MoCA or AAI shall take into account all Taxes, Duties, and Levies for the purpose of Evaluation.

vii. The bidder shall provide a breakup of all Taxes, Duties and Levies wherever applicable and or payable.

viii. OPEX shall be distributed for 3 Years wherever applicable.

ix. No OPEX as AMC charges or recurring charges can be left blank for any component. In case the AMC charges or recurring charges are 0, then it will be the responsibility of the Successful Bidder to provide the services for the complete tenure of the project.

x. All the rates shall be applicable throughout the life cycle of the project and the Successful Bidder has to provide the same on the same rates in case of any supply/ demand by the department.

xi. CAPEX to be considered as new procurement in the project lifecycle w.e.f. signing of contract.

xii. No separate payment will be considered for the deployed human resources during the Implementation stage.

xiii. Year wise human resource cost for O&M Stage shall be provided by the Successful Bidder

xiv. Indicative number of units are given in the financial table, successful Bidder may modify as per solution requirements

xv. All payments to be paid to the successful bidder shall be inclusive of all statutory levies, duties, taxes, and other charges whenever levied/applicable.

xvi. MoCA reserves the right to reject any proposals, whole or in part, to waive any or all informalities and disregard non-conforming, non-responsive or conditional proposals.

xvii. Successful Bidder must submit the itemized summary of BoQ and other item with all relevant details including description, quantities and rates to the department as prescribed in the RFP.

xviii. Successful Bidder to provide cost of any missing item in BOQ under "Others", breakup of the other Items needs to be provided by Successful Bidder before issuance of LOI.

xix. The Bidder can quote only one OEM against any product, quoting multiple OEM may lead to rejection of bid.

xx. The Bidder has to ensure compliances of the objectives and functional requirements of the project. In order to ensure this, bidder has to provide required items (if any) at no additional cost.

xxi. No payments of AMC of the product within the warranty periods shall be made on any component.

xxii. The Bidder can refer to Annexure 1 of this Scope of Work. for the indicative cost component of the under each cost heads

2.30 Financial Bid Format

Sr. No.	Scope Component		Amount (INR)
1.Design, Implementation & Go-Live (R1)- M1 to M5 and PACR as R1pacr			
1.1	AirSewa Grievance Redressal Platform (A)	R1	In word and figure
1.2	Integrations (B)	R1	In word and figure
1.3	Human Resource Cost (C)	R1	In word and figure
1.4	Capacity Building & Training (D)	R1	In word and figure
1.5	Any other cost (E)	R1	In word and figure
1.6	PACR Setup (In F)	R1pacr Setup	In word and figure
		Sub Total	
2.O&M for 36 Months (R2)			
2.1	Operation & Maintenance – 36 Months, equal (12) quarter instalments		In word and figure
2.2	R2Pacr operation (In G)	R2Pacr operation	In word and figure
Sub Total			
Total (1+2)			
Tax (GST@18%)			
Total (In Numbers)			
Total (In words)			

Reference:

- M: Month starting from signing of contract

2.31 Detailed breakup of AirSewa Grievance Redressal Platform Development cost (A)

S.No.	Module	Amount (INR)	Estimated Man-Month efforts	Estimated Man-Month blended Rates (INR)
1	Requirement Gathering			
2	Design & SRS			
3	Environment setup			
4	Development of Air Sewa			
a.	User Registration Module			
b.	Grievance Registration Module			
c.	Ticket Allocation Engine			
d.	Communication Module			
e.	AirSewa Agent Module			
f.	Redressor Module			
g.	Administrator Module			
h.	SLA & Escalation Module			
i.	MoCA Dashboard Module			
j.	Flight Status and Tracking Module			
5	UAT testing & incorporating feedback			
6	VAPT & Go Live			
7	AI Services - Verification of closure			
8	AI Services – Chatbot			
9	AI Services - Classification of issues			
10	AI Service - Tagging of duplicate requests			
11	UAT of AI services & Go Live			
12	Any other cost			

2.32 Detailed breakup of Human Resource cost (C)

Sr. No.	Title of resource	Person-month Rates	(R1) Design, Implementation & Go-Live		
			# of resources	Person-month Effort	Total Amount
1	Project Manager				
2	Cloud Expert				
3	Security Expert				
4	UI Designer				
5	Developer				
6	Quality Assurance Expert				
7	Documentation Expert				
8	Mobile Application Developer				
9	Database Admin				
10	Systems Admin				
11	Software Tester				
12	Business Analyst				
13	Training and Change Management Expert				
14	Technical Helpdesk support (24x7)				
15	Any other resource				

2.33 Detailed breakup of cost associated with capacity building and training (D)

Sr. No.	Scope Component	Design, Implementation & Go-Live
		M1 to M5
		Unit Cost (INR)
1	Offline training on functions after go-live	
2	Online training on functions after go-live	
3	Development of material (videos/ pdf) for website tech help uploaded before go-live	
4	Refresher courses (online)	
5	Additional on demand training (offline)	
6	Any other cost	

2.34 Detailed breakup of cost associated with PACR set up and Operation (F)

R1pacr Setup: Setup, Implementation & Go-Live (Lump-Sum excluding Manpower)

Cost shall cover all activities up to Go-Live & Stabilization as defined in the RFP. SIP/PRI subscription shall be procured by MOCA and shall not be included in this lump-sum cost.

A	**B**
Description	Cost (INR, Excl. GST)
Total Lump-Sum Cost	**Insert cost here**

Note: Lump-sum shall be released milestone-wise as per Payment Terms (Infrastructure Delivery, Installation, UAT, Training, CERT-In Audit etc.).

R2Pacr operation: Operator Seat-Shift Cost (Base Operations including Manpower)

Operator refers to call-handling resources. Total approved operator seats = 15 seats per day (all shifts combined).

CRO per-Seat-per-Shift Cost for Year 1 **(C)**	**Seats cumulative of all shifts in a day** **(D)**	**Total CRO Cost for 3 Years** **(E)**
Insert Rate here	**15**	**E = 365 x 3.4725xCxD**

Note: The selected Bidder has to maintain sufficient no. of resources as specified at the section "Manpower Deployment and Operation of PACR". Costs associated with all types of resources specified under "Manpower Deployment and Operation of PACR" should be adjusted/included within the rate of above CRO seats (C of Table II). No separate billing/amount will be paid for any other type of resources.

Per-seat-per-shift Cost (as quoted in "C") will be increased by 15% per year (after completion of 1st year and subsequently after 2nd year) for all manpower provided for calculating total manpower cost for 3 years. Therefore, CRO Cost for three years will be calculated as **3.4725.**

Important Notes:

1. Seat-shift model applies; manpower headcount is not the billing unit.
2. Annual escalation on Operator seat cost shall be 15% year-on-year.

2.35 Deliverables & Payment Milestones for AirSewa

A. Design and Development Phase

Phase	Deliverable	Timelines	Milestones	Payment Schedule
Resource Deployment	Deployment of all Key resources whose resume has been submitted in the proposal	T+ 15 (Days) (T1)	Deployment of Key Resources	10% of R1
Design Approval	Project Plan	T+1 (Month) (T2)	Submission and acceptance of final design deliverables	10% of R1
	Resource Deployment Plan			
	Project Charter			
	FRS & BRD (Business Requirements Document)			
	Software Requirement Specification Document			
	System Design Document			
	Solution Architecture			
	Integration plan			
	API Architecture			
	Cloud Considerations & incremental strategy			
	Cloud Security architecture			
Development and Implementation	Development of website & Application on Cloud	T+ 4 (Months) (T3)	Development of AirSewa & Soft launch & Integration with all system	30% of R1
	Analytics and Reporting			
	Functional Testing			
	Unit Testing			
	Performance Testing			
	Vulnerability Assessment			
	User Acceptance Testing Report			
Integration	Complete integration with all systems			
Final Acceptance	Acceptance of complete application by customer	T+ 5 (Months) (T4)	Final Acceptance & Go Live	35% of R1
Go Lives	Go Live			
Final SRS Sign-off & Source code	Final SRS Sign-off, Source Code submission and cloud utilization reports for development period submitted monthly	T+ 5 (Months) (T5)	Final SRS Sign-off	15% of R1

R1: Price quoted in financial bid for Design, Implementation & Go-Live phase

B. Operation and Maintenance Phase

Operation & Maintenance for 3 years, expandable by 2 years.

Table: Payment Milestone & Deliverables - Operation and Maintenance Phase

Phase	Deliverable	Timelines	Milestones	Payment Schedule
Operation and Maintenance	SLA Compliance report Project Status Report Change Control Note and Report Issue/ Incident reports Process audit reports Critical reports like call analytics, problem management, capacity management, availability management, etc. Cloud utilization reports	T5+36 Months	Project Status Reports	Quarterly Payments of the O&M Fees quoted by the Lead Bidder
Help Desk Support	Monthly log of service calls and problem resolution reports for Helpdesk	T5+36 Month		
Capacity Building and Training	Training to personnel, manuals, material and training completion report	T5+36 Months		

The payment will be made by AAI to the bidder on quarterly basis. The quarterly bill will be submitted by the Bidder to the AAI who will in turn release the payment as depicted in the table above, if there is no dispute and after verification/audit of the bills. SLA amount will be deducted in receptive bills in case of any breach in SLA.

i. Adherence to SLA & Milestone plan

a. The payment to the Lead Bidder shall be governed by the SLAs defined in this RFP.
b. The payment to the Lead Bidder shall also depend on the adherence of the Lead Bidder to the timelines specified for implementation as specified in the RFP.

ii. Third Party Audits and Reviews

Cost of any third-party audit – initial and annual audits as mentioned in RFP - shall be borne by the Lead Bidder. Agency for the same will be selected by AAI; however, payment to the agency will be made by the Lead Bidder.

iii. Taxes and Statutory Payments

a. All payments agreed to be made by AAI to the Lead Bidder in accordance with the Bid shall be inclusive of all statutory levies, duties, taxes and other charges whenever levied/applicable.

b. The Bidder shall bear all personal/income taxes levied or imposed on it and its personnel, etc. because payment received under this Contract. The Bidder shall bear all income/corporate taxes, levied or imposed on the Bidder on account of payments received by it from AAI for the work done under this Contract. The Bidder shall bear all other taxes such as GST, custom duty, etc. if there is change in tax rate or introduction of any new tax, same shall be passed to each party i.e. if tax rate reduces, Bidder will charge lower rate and if tax rate increases or new tax is introduced, bidder will be compensated for additional percentage or new tax.

2.36 Payment Terms for PACR services

These Payment Terms shall form an integral and inseparable part of the RFP, Agreement, Service Level Agreement (SLA), and the Work Order issued by the AAI.

The bidder shall submit a Quarterly invoice for PACR Manpower services within seven (7) calendar days from the close of the relevant billing month, supported by the Monthly SLA Compliance and Performance Report.

Table 1: Payment milestone of PACR

Sl. No.	Milestone / Item	Payment & Conditions
1	IT Infrastructure Delivery subject to PACR Compliance "Minimum specification for Desktop & with IVRS	40% of "B of R1pacr Setup table from financial bid/sheet"; payable on AAI-certified delivery
2	Installation & Commissioning subject to PACR Compliance "Minimum specification for Desktop & with IVRS	20% of "B of R1pacr Setup table from financial bid/sheet"; payable on AAI acceptance
3	User Acceptance Testing (UAT) and testing -Testing Stage and submission of Cert-In, VA-PT, Regulatory Compliance (TRAI/OSP/TCCCPR) Telephony Security (SIP/VoIP Controls)	30% of B of R1pacr Setup table from financial bid/sheet"; payable on successful execution of AAI-approved test cases, resolution of all Critical/High defects, and AAI's written sign-off
4	Training Completion	10% of B of R1pacr Setup table from financial bid/sheet"; payable on completion of AAI-approved training for all deployed roles, handover of training material/SOPs, and AAI's written certification
6	Base Seat-Shifts with all manpower required	Per-seat-per-shift rate; billed monthly; seat must be logged-in & operational
7	Additional manpower resources and Infra	As per the table given below on -*Additional Manpower & Infrastructure Payments*
10	SLA Penalties of PACR	As per SLA; capped at 10% of monthly invoice value

Go-Live shall not be permitted unless UAT, Training, and CERT-In (Pre Go-Live) milestones are completed and accepted by the AAI.

2.36.1. Additional Manpower & Infrastructure Payments

Deployment under this section shall be only upon prior written approval of the AAI and payable in addition to Phase-wise payments.

Table 2: Additional Manpower & Infrastructure

Sl. No.	Resource / Item	Payment Formula & Conditions
1	Control Room Manager Seat per shift	Per month cost (3 × 30 × P); beyond baseline manpower
2	Team Leads / Trainers Seat per shift	Per month cost (2 × 30 × P); combined role
3	Control Room Operator Seat per shift	Per month cost (1 × 30 × P); beyond approved seats
4	MIS / Data Analytics Expert	Per month cost (2 × 30 × P)
5	IT Support / Security Expert Seat per shift	Per month cost (2 × 30 × P)
6	Full-stack Developer	Per month cost (2.5 × 30 × P)
7	Multi-Tasking Staff (MTS) seat per shift	Per month cost (1 × 30 × P)
8	Additional Seat Infrastructure	P × 100 (one-time); includes desktop, peripherals, headset, CTI/CRM & all required licenses
Where **P** = **0.9** x **Q**; Additional manpower and infrastructure shall not dilute seat-shift obligations under the SLA.		

2.36.2. Calculation for a month

a.

Amount for one month = (N1 × Q)
+ (N2 × P)
+ Cost of additional manpower resources as per above table other than additional CRO per-seat-per shift (as per above table in para 30.3)
+ Approved Additional Infrastructure Charges (if any)
+ IT Infrastructure Delivery (as applicable)
+ Installation & Commissioning (as applicable)
+ User Acceptance Testing (as applicable)
+ Training Completion (as applicable)
+ CERT-In Audit (as applicable)
+ ISO 27001:2022 and VAPT for PACR (as applicable)
− SLA Penalties (as applicable)
− Statutory Deductions (if any).

Where is:

N1 = Number of seat-shift done by CRO upto 15 shift per day in the relevant month.
(Note: Any shifts exceeding the required number specified in the work order on a given day will not be counted.)
N2= Number of seat-shift done by CRO above 15 shift per day in the relevant month.
(Note: Any shifts exceeding the required number specified in the work order on a given day will not be counted.)

Q = **Approved Per-Seat-Per-Shift Rate quoted in the Financial Bid/Sheet** (i.e. Year 1 is C, Year 2 is 1.15xC and Year 3 is 1.15x1.15xC.)

Note: This calculation is for one month hence quarterly bill may be prepared accordingly.

GOVERNANCE SCHEDULE

1. PURPOSE

The purpose of this Schedule is to:

i. establish and maintain the formal and informal processes for managing the relationship between AAI and the bidder (including the outputs from other Schedules to this Agreement;

ii. define the principles that both Parties wish to follow to ensure the delivery of the Services;

iii. ensure the continued alignment of the interests of the Parties;

iv. ensure that the relationship is maintained at the correct level within each Party;

v. create the flexibility to revise and maintain the relationship and this Agreement during the Term;

vi. set out the procedure for escalating disagreements; and

vii. enable contract administration and performance management.

2. GOVERNANCE STRUCTURE

i. Project Managers: The relationship under this Agreement will be managed by the Project Managers appointed by each Party, who will provide the interface between the executive management of the respective Parties.

ii. Project Implementation Unit (PIU): Within 7 days following the Effective Date, AAI or its nominated agencies and the bidder shall each appoint a Project Manager. In the event that either Party wishes to substitute its Project Manager it will do so in manner in which the original appointment is made and notify the other Party of such substitution as soon as reasonably practicable but at the latest within 7 days of the substitution.

iii. The Project Managers shall have responsibility for maintaining the interface and communication between the Parties.

iv. The PIU will meet formally on a fortnightly / monthly / quarterly, as required, basis at a time and location to be agreed between them. These meetings will cover, as a minimum, the following agenda items: (i) consideration of Quarterly Performance Reports; (ii) consideration of matters arising out of the Change Control Schedule; (iii) issues escalated in accordance with the escalation procedure as set out in the Governance Schedule; (iv) matters to be brought before the PIU in accordance with the MSA and the Schedules; (v) any matter brought before the PIU by the bidder under this Article; and (vi) any other issue which either Party wishes to add to the agenda.

v. In the event that there is any material factor which affects the delivery of the Services or the terms of payment as stated in the Terms of Payment Schedule, the Parties agree to discuss in the PIU any appropriate amendment to the Agreement or any Service Level Agreements or Statement of Works including any variation to the terms of payment as stated in the Terms of Payment Schedule. Any variation so agreed shall be implemented through the change control procedure as set out in the Change Control Schedule.

vi. The other Party ("Respondent") shall have the right to respond to the Dispute Notice within 7 days after receipt of the Dispute Notice. In the event that the parties are unable to resolve the Disputed Matter within a further period of 7 days, it shall refer the Disputed Matter to next level of the dispute resolution for action as per the process mentioned.

vii. All negotiations, statements and / or documentation pursuant to these Articles shall be without prejudice and confidential (unless mutually agreed otherwise).

viii. If the Disputed Matter is having a material effect on the operation of the Services (or any of them or part of them) the Parties will use all their respective reasonable endeavours to reduce the elapsed time in reaching a resolution of the Disputed Matter.

AUDIT, ACCESS AND REPORTING

1. PURPOSE

This Schedule details the audit, access and reporting rights and obligations of AAI or its nominated agency and the System Integrator.

2. AUDIT NOTICE AND TIMING

2.1 As soon as reasonably practicable after the Effective Date, the Parties shall use their best endeavours to agree to a timetable for routine audits during the Project Implementation Phase and the Operation and Management Phase. Such timetable during the Implementation Phase, AAI or its nominated agency and thereafter during the operation Phase, AAI or its nominated agency shall conduct routine audits in accordance with such agreed timetable and shall not be required to give the bidder any further notice of carrying out such audits. Exit Management plan shall be furnished in writing to AAI or its nominated agencies within 90 days from the Effective Date of this Agreement.

2.2 AAI or its nominated agency may conduct non-timetabled audits at his/ her own discretion if it reasonably believes that such non-timetabled audits are necessary as a result of an act of fraud by the System Integrator, a security violation, or breach of confidentiality obligations by the System Integrator, provided that the requirement for such an audit is notified in writing to the bidder a reasonable period time prior to the audit (taking into account the circumstances giving rise to the reasonable belief) stating in a reasonable level of detail the reasons for the requirement and the alleged facts on which the requirement is based. If the bidder considers that the non-timetabled audit was not appropriate, the matter shall be referred to the escalation procedure as set out in the Governance Schedule.

2.3 The frequency of audits shall be a (maximum) half yearly, provided always that AAI or its nominated agency shall endeavour to conduct such audits with the lowest levels of inconvenience and disturbance practicable being caused to the System Integrator. Any such audit shall be conducted by with adequate notice of 2 weeks to the System Integrator.

2.4 AAI will ensure that any 3rd party agencies (except CAG) appointed to conduct the audit will not be the competitor of bidder and will be bound by confidentiality obligations.

3. ACCESS

The bidder shall provide to AAI or its nominated agency reasonable access to employees, JV/consortium members, suppliers, agents and third party facilities as detailed in the RFP, documents, records and systems reasonably required for audit and shall provide all such persons with routine assistance in connection with the audits and inspections. The Chairperson, PIU / Steering Committee shall have the right to copy and retain copies of any relevant records. The bidder shall make every reasonable effort to co-operate with them.

4. AUDIT RIGHTS

4.1 AAI or its nominated agency shall have the right to audit and inspect suppliers, agents and third party facilities (as detailed in the RFP), data centres, documents, records, procedures and systems relating to the provision of the services, but only to the extent that they relate to the provision of the services, as shall be reasonably necessary to verify:

i. The security, integrity and availability of all data processed, held or conveyed by the Partner on behalf of AAI and documentation related thereto;

ii. That the actual level of performance of the services is the same as specified in the SLA;

iii. That the bidder has complied with the relevant technical standards, and has adequate internal controls in place; and

iv. The compliance of the bidder with any other obligation under the MSA and SLA.

v. Security audit and implementation audit of the system shall be done once each year, the cost of which shall be borne by the System Integrator.

vi. For the avoidance of doubt the audit rights under this Schedule shall not include access to the System Integrator's profit margins or overheads, any confidential information relating to the System Integrator' employees, or (iii) minutes of its internal Board or

vii. Board committee meetings including internal audit, or (iv) such other information of commercial-in-confidence nature which are not relevant to the Services associated with any obligation under the MSA.

5. AUDIT RIGHTS OF JV/CONSORTIUM MEMBERS, SUPPLIERS AND AGENTS

5.1 The bidder shall use reasonable endeavours to achieve the same audit and access provisions as defined in this Schedule with JV/consortium member who supply labour, services in respect of the services. The bidder shall inform AAI or its nominated agency prior to concluding any JV/consortium or supply agreement of any failure to achieve the same rights of audit or access

5.2 REPORTING: The bidder will provide quarterly reports to the Chairperson, PIU / Steering committee regarding any specific aspects of the Project and in context of the audit and access information as required by AAI or its nominated agency.

6. ACTION AND REVIEW

6.1 Any change or amendment to the systems and procedures of the bidder, where applicable arising from the audit report shall be agreed within thirty (30) calendar days from the submission of the said report.

6.2 Any discrepancies identified by any audit pursuant to this Schedule shall be immediately notified to AAI or its nominated agency and the bidder Project Manager who shall determine what action should be taken in respect of such discrepancies in accordance with the terms of the MSA

7. TERMS OF PAYMENT

AAI shall bear the cost of any audits and inspections. The terms of payment are exclusive of any costs of the bidder and the JV/consortium, for all reasonable assistance and information provided under the MSA, the Project Implementation, Operation and Management SLA by the bidder pursuant to the Schedule. All licenses, subscriptions, warranties and assets procured under the project shall be in the name of AAI. Pending payments may be withheld in case of unresolved license/subscription transfer issues during Exit Management

8. RECORDS AND INFORMATION

For the purposes of audit in accordance with this Schedule, the bidder shall maintain true and accurate records in connection with the provision of the services and the bidder shall handover all the relevant records and documents upon the termination or expiry of the MSA

EXIT MANAGEMENT SCHEDULE

1. PURPOSE

1.1 This Schedule sets out the provisions, which will apply on expiry or termination of the contract, the Project Implementation, Operation and Management SLA.

1.2 In the case of termination of the Project Implementation and/or Operation and Management, the Parties shall agree at that time whether, and if so during what period, the provisions of this Schedule shall apply.

1.3 The Parties shall ensure that their respective associated entities carry out their respective obligations set out in this Exit Management Schedule.

2 TRANSFER OF ASSETS

2.1 AAI shall be entitled to serve notice in writing on the bidder at any time during the exit management period as detailed hereinabove requiring the bidder and/or its JV/consortium member to provide AAI with a complete and up to date list of the Assets within 30 days of such notice. AAI shall then be entitled to serve notice in writing on the bidder at any time prior to the date that is 30 days prior to the end of the exit management period requiring the bidder to sell the Assets, if any, to be transferred to AAI or its nominated agencies at book value as determined as of the date of such notice in accordance with the provisions of relevant laws.

2.2 In case of contract being terminated by AAI, AAI reserves the right to ask bidder to continue running the project operations for a period of 6 months after termination orders are issued.

2.3 Upon service of a notice under this Article the following provisions shall apply:

i. in the event, if the Assets to be transferred are mortgaged to any financial institutions by the bidder, the bidder shall ensure that all such liens and liabilities have been cleared beyond doubt, prior to such transfer. All documents regarding the discharge of such lien and liabilities shall be furnished to AAI.

ii. All risk in and title to the Assets to be transferred / to be purchased by AAI pursuant to this Article shall be transferred to AAI, on the last day of the exit management period.

iii. AAI shall pay to the bidder on the last day of the exit management period such sum representing the Net Block (procurement price less depreciation as per provisions Of Companies Act) of the Assets to be transferred as stated in the Terms of Payment Schedule.

iv. Payment to the outgoing bidder shall be made to the tune of last set of completed services / deliverables, subject to SLA requirements.

v. The outgoing bidder will pass on to AAI and/or to the Replacement bidder, the subsisting rights in any leased properties/ licensed products on terms not less favourable to AAI/ Replacement Bidder, than that enjoyed by the outgoing Bidder.

3 COOPERATION AND PROVISION OF INFORMATION

3.1 During the exit management period:

i. The bidder will allow AAI or its nominated agency access to information reasonably required to define the then current mode of operation associated with the provision of the services to enable AAI to assess the existing services being delivered;

ii. promptly on reasonable request by AAI, the bidder shall provide access to and copies of all information held or controlled by them which they have prepared or maintained in accordance with this agreement relating to any material aspect of the services (whether provided by the bidder or JV/consortium member appointed by the Bidder). AAI shall be entitled to copy of all such information. Such information shall include details pertaining to the services rendered and other performance data. The bidder shall permit AAI or its nominated agencies to have reasonable access to its employees and facilities as reasonably required by the Chairman, PIU to understand the methods of delivery of the services employed by the bidder and to assist appropriate knowledge transfer.

4 CONFIDENTIAL INFORMATION, SECURITY AND DATA

4.1 The bidder will promptly on the commencement of the exit management period supply to AAI or its nominated agency the following:

i. information relating to the current services rendered and customer and performance data relating to the performance of JV/consortium member in relation to the services;

ii. documentation relating to Computerization Project's Intellectual Property Rights;

iii. documentation relating to JV/consortium member;

iv. all current and updated data as is reasonably required for purposes of AAI or its nominated agencies transitioning the services to its Replacement bidder in a readily available format nominated by AAI, its nominated agency;

v. all other information (including but not limited to documents, records and agreements) relating to the services reasonably necessary to enable AAI or its nominated agencies, or its Replacement bidder to carry out due diligence in order to transition the provision of the Services to AAI or its nominated agencies, or its Replacement bidder (as the case may be).

4.2 Before the expiry of the exit management period, the bidder shall deliver to AAI or its nominated agency all new or up-dated materials from the categories set out in Schedule above and shall not retain any copies thereof, except that the bidder shall be permitted to retain one copy of such materials for archival purposes only.

4.3 Before the expiry of the exit management period, unless otherwise provided under the MSA, AAI or its nominated agency shall deliver to the bidder all forms of bidder confidential information, which is in the possession or control of Chairperson, PIU or its users.

5 EMPLOYEES

5.1 The Promptly on reasonable request at any time during the exit management period, the bidder shall, subject to applicable laws, restraints and regulations (including in particular those relating to privacy) provide to AAI or its nominated agency a list of all employees (with job titles) of the bidder dedicated to providing the services at the commencement of the exit management period.

5.2 The Promptly on reasonable request where any national, regional law or regulation relating to the mandatory or automatic transfer of the contracts of employment from the bidder to AAI or its nominated agency, or a Replacement bidder ("Transfer Regulation") applies to any or all of the employees of the System Integrator, then the Parties shall comply with their respective obligations under such Transfer Regulations.

6 TRANSFER OF CERTAIN AGREEMENTS

On request by AAI or its nominated agency the bidder shall effect such assignments, transfers, licences and sub-licences as the Chairperson, PIU may require in favour of the Chairperson, PIU, or its Replacement bidder in relation to any equipment lease, maintenance or service provision agreement between bidder and third party lessors, vendors, and which are related to the services and reasonably necessary for the carrying out of replacement services by AAI or its nominated agency or its Replacement System Integrator.

7 RIGHTS OF ACCESS TO PREMISES

7.1 At any time during the exit management period, where Assets are located at the System Integrator's premises, the bidder will be obliged to give reasonable rights of access to (or, in the case of Assets located on a third party's premises, procure reasonable rights of access to) AAI or its nominated agency and/or any Replacement bidder in order to make an inventory of the Assets.

7.2 At any The bidder shall also give AAI or its nominated agency or its nominated agencies, or any Replacement bidder right of reasonable access to the Implementation Partner's premises and shall procure AAI or its nominated agency or its nominated agencies and any Replacement bidder rights of access to relevant third party premises during the exit management period and for such period of time following termination or expiry of the MSA as is reasonably necessary to migrate the services to AAI or its nominated agency, or a Replacement System Integrator.

8 GENERAL OBLIGATIONS OF THE System Integrator

8.1 At any The bidder shall provide all such information as may reasonably be necessary to effect as seamless a handover as practicable in the circumstances to AAI or its nominated agency or its Replacement bidder and which the bidder has in its possession or control at any time during the exit management period.

8.2 For the purposes of this Schedule, anything in the possession or control of any System Integrator, associated entity, or JV/consortium member is deemed to be in the possession or control of the Bidder.

8.3 The bidder shall commit adequate resources to comply with its obligations under this Exit Management Schedule.

9 EXIT MANAGEMENT PLAN

9.1 The bidder shall provide AAI or its nominated agency with a recommended exit management plan ("Exit Management Plan") which shall deal with at least the following aspects of exit management in relation to the MSA as a whole and in relation to the Project Implementation, and the Operation and Management SLA.

i. A detailed program of the transfer process that could be used in conjunction with a Replacement bidder including details of the means to be used to ensure continuing provision of the services throughout the transfer process or until the cessation of the services and of the management structure to be used during the transfer;

ii. plans for the communication with such of the JV/consortium member, staff, suppliers, customers and any related third party as are necessary to avoid any material detrimental impact on AAI's operations as a result of undertaking the transfer;

iii. (if applicable) proposed arrangements for the segregation of the Bidder's networks from the networks employed by AAI and identification of specific security tasks necessary at termination;

iv. Plans for provision of contingent support to AAI, and Replacement bidder for a reasonable period after transfer.

9.2 The bidder shall re-draft the Exit Management Plan annually thereafter to ensure that it is kept relevant and up to date.

9.3 Each Exit Management Plan shall be presented by the bidder to and approved by AAI or its nominated agencies.

9.4 The terms of payment as stated in the Terms of Payment Schedule include the costs of the bidder complying with its obligations under this Schedule.

9.5 In the event of termination or expiry of MSA, and Project Implementation, each Party shall comply with the Exit Management Plan.

9.6 During the exit management period, the bidder shall use its best efforts to deliver the services.

9.7 Payments during the Exit Management period shall be made in accordance with the Terms of Payment Schedule.

9.8 The selected bidder shall provide complete transition support including source code, credentials, KT sessions, deployment support, documentation and technical handholding to AAI/replacement agency

9.9 The selected bidder shall provide minimum 2 months transition support after contract expiry/termination or till successful transition to replacement agency, whichever is later.…

This Exit Management plan shall be furnished in writing to AAI or its nominated agencies within 90 days from the Effective Date of this Agreement

CHANGE CONTROL SCHEDULE

1 Introducing a Change

1.1 AAI shall have the right to propose, and subsequently require, the Project Manager to order the Supplier from time to time during the performance of the Contract to make any change, modification, addition, or deletion to, in, or from the System (interchangeably called Change), provided that such Change falls within the general scope of the System, does not constitute unrelated work, and is technically practicable, taking into account both the state of advancement of the System and the technical compatibility of the Change envisaged with the nature of the System as originally specified in the Contract.

A Change may involve, but is not restricted to, the substitution of updated Information Technologies and related Services in accordance with system/product Upgrades.

1.2 The Supplier may from time to time during its performance of the Contract propose to AAI (with a copy to the Project Manager) any Change that the Supplier considers necessary or desirable to improve the quality or efficiency of the System. AAI may at its discretion approve or reject any Change proposed by the Supplier.

1.3 Notwithstanding Clauses 1.1.1. and 1.1.2 of Schedule I of this document, no change made necessary because of any default of the Supplier in the performance of its obligations under the Contract shall be deemed to be a Change, and such change shall not result in any extra cost or Time for Achieving Operational Acceptance.

1.4 The procedure on how to proceed with and execute Changes is specified below.

1.5 Moreover, AAI and Supplier will agree, during development of the Project Plan, to a date prior to the scheduled date for Operational Acceptance, after which the Technical Requirements for the System shall be frozen. Any major Change initiated after this time will be dealt with after Operational Acceptance.

2 Changes originating from AAI

2.1 If AAI proposes a Change pursuant to Clause mentioned, it shall send to the Supplier a request for Change Proposal,‖ requiring the Supplier to prepare and furnish to the Project Manager as soon as reasonably practicable a Change Proposal, which shall include the following:

- brief description of the Change;
- impact on the Time for Achieving Operational Acceptance;
- detailed estimated cost of the Change;
- effect on Functional Guarantees (if any);
- Effect on any other provisions of the Contract.

2.2 Prior to preparing and submitting the Change Proposal, the Supplier shall submit to the Project Manager a Change Estimate Proposal, which shall be an estimate of the cost of preparing the Change Proposal, plus a first approximation of the suggested approach and cost for implementing the changes. Upon receipt of the Supplier‘s Change Estimate Proposal, AAI shall do one of the following:

2.2.1 Accept the Supplier‘s estimate with instructions to the Supplier to proceed with the preparation of the Change Proposal;

2.2.2 Advise the Supplier of any part of its Change Estimate Proposal that is unacceptable and request the Supplier to review its estimate;

2.2.3 Advise the Supplier that AAI does not intend to proceed with the Change.

2.2.4 Upon receipt of AAI‘s instruction to proceed under Clause mentioned, the Supplier shall, with proper expedition, proceed with the preparation of the Change Proposal, in accordance with Clause mentioned .The Supplier, at its discretion, may specify a validity period for the Change

Proposal, after which if AAI and Supplier has not reached agreement in accordance with Clauses mentioned shall apply.

2.2.5 The pricing of any Change shall, as far as practicable, be calculated in accordance with the rates and prices included in the Contract. If the nature of the Change is such that the Contract rates and prices are inequitable, the parties to the Contract shall agree on other specific rates to be used for valuing the Change.

2.2.6 Upon receipt of the Change Proposal, AAI and the Supplier shall mutually agree upon all matters contained in the Change Proposal. Within fourteen (14) days after such agreement, AAI shall, if it intends to proceed with the Change, issue the Supplier a Change Order. If AAI is unable to reach a decision within fourteen (14) days, it shall notify the Supplier with details of when the Supplier can expect a decision. If AAI decides not to proceed with the Change for whatever reason, it shall, within the said period of fourteen (14) days, notify the Supplier accordingly. Under such circumstances, the Supplier shall not be entitled to reimbursement of any costs incurred by it in the preparation of the Change Proposal.

2.2.7 If AAI and the Supplier cannot reach agreement on the price for the Change, an equitable adjustment to the Time for Achieving Operational Acceptance, or any other matters identified in the Change Proposal, the Change will not be implemented.

3 Changes originating from Bidder

3.1 If the bidder proposes a Change pursuant to Clause mentioned, the bidder shall submit to the project manager a written application for Change Proposal, giving reasons for the proposed Change and including the information specified in Clause mentioned. Upon receipt of the application for Change Proposal, the Parties shall follow the procedures outlined in Clauses mentioned. However, should AAI/ AirSewa and PACR project committee (comprising relevant stakeholders as per discretion of AAI) choose not to proceed or AirSewa and PACR project committee and the bidder cannot come to Contract on the change during any validity period that the bidder may specify in its application for Change Proposal, the bidder shall not be entitled to recover the costs of preparing the application for Change Proposal, unless subject to an Contract between AAI and the bidder to the contrary.

3.2 The bidder needs to make any changes in software suggested by AAI or the bidder in accordance with Clause mentioned. It is expected that any minor changes shall be carried out by bidder without any extra cost. In case of substantial change requests, extra cost estimated for the change shall be paid by AAI.

3.3 A Change control board is to be constituted comprising of technical team members from AAI

3.4 . The change management process as per ITIL 4 will apply. The details of the change management process will be defined in the inception phase.

Change Control Notice (CCN) Format

Change Control Note	CCN Number
Part A: Initiation	
Title:	
Originator:	
Sponsor:	

Date of Initiation:	
Details of Proposed Change	
(To include reason for change and appropriate details/specifications. Identify any attachments as A1, A2, and A3 etc.)	
Authorized by :	Date:
Name:	
Signature: Received by the IP	Date:
Name:	
Signature:	
Change Control Note	CCN Number:
Part B : Evaluation	
(Identify any attachments as B1, B2, and B3 etc.) Changes to Services, charging structure, payment profile, documentation, training, service levels and component working arrangements and any other contractual issue.	
Brief Description of Solution:	
Impact:	
Deliverables:	
Timetable:	
Charges for Implementation: (including a schedule of payments)	
Other Relevant Information: (including value-added and acceptance criteria)	
Authorized by the Implementation Partner	Date:
Name:	
Signature:	

4 Acceptance Criteria

4.1 Acceptance of Software

1. General

i. The system shall support secure, role-based access for at least the following roles:

Complainants, Redressors, Redressor Administrators, Air Sewa Agents and AAI officers

ii. The system shall be usable on modern web browsers (Latest LTS version of Chrome, Edge, Firefox) and responsive on standard mobile devices Android and iOS)

iii. All acceptance criteria shall be testable with clear pass/fail outcomes.

2. Grievance Intake and Registration

i. The system shall allow users to submit grievances via web portal, mobile app, chatbot and other inbound channels as outlined in the scope of work

ii. The system shall capture at minimum: complainant identity (or anonymous flag), contact details, grievance category and subcategory, description, date/time, location (if applicable), and supporting documents.

iii. The system shall auto-generate a unique grievance ID immediately upon successful submission and display/communicate it to the complainant.

iv. The system shall validate required fields, show user-friendly error messages, and prevent submission until mandatory data is provided.

v. The system shall send an acknowledgment (on-screen and via email and SMS)

vi. The system shall allow saving a draft grievance and resuming later.

3. Workflow, Routing, and SLAs

i. The system shall automatically route new grievances to the appropriate Redressor Officer or queue based on configured business rules.

ii. The system shall support configurable workflow states (e.g., New, Under Review, Pending Clarification, Resolved, Closed, Escalated) and associated transitions.

iii. The system shall support configurable SLA definitions per category or priority (e.g., acknowledgment within X hours, first response within Y days, resolution within Z days).

iv. The system shall automatically flag and highlight SLA breaches and near-breaches in Redressor Operator and Admin dashboard and to the AAI dashboard.

v. The system shall support manual reassignment and escalation with audit logging of who changed what and when.

vi. The system shall support multi-level escalation paths (e.g., Officer → Supervisor → Nodal Officer) based on configurable rules or manual triggers.

4. Communication and Notifications

i. The system shall provide a secure portal view for complainants to track grievance status and view a status history.

ii. The system shall send configurable notifications (email/SMS/in-app) on key events: registration, assignment, request for additional information, resolution proposal, closure, escalation, request for feedback and reopening of ticket.

iii. The system shall allow officers to request clarifications from complainants and record responses within the same case thread.

iv. The system shall maintain a communication log for each grievance, including internal notes and external messages, time-stamped and user-stamped.

5. Case Handling and Resolution

i. The system shall allow officers to record investigations, actions taken, and proposed resolutions against each grievance.

ii. The system shall allow attaching and viewing supporting documents and evidence (text, images, PDFs, and other file formats as decided) with size and type restrictions configurable by AAI Admin.

iii. The system shall allow capturing resolution outcome, resolution date, and resolution category (e.g., Resolved – Action Taken; Rejected – Insufficient Grounds; Referred – Out of Jurisdiction).

iv. The system shall allow capturing complainant satisfaction/acceptance (e.g., “Accepted”, “Not Accepted”) prior to final closure where process requires it.

v. The system shall prevent closure of a grievance unless all mandatory resolution fields have been entered.

6. Search, Reporting, and Analytics

i. The system shall provide search and filtering across grievances by ID, complainant, category, date range, status, location, officer, and escalation level.

ii. The system shall provide standard reports, including at least:

- Volume of grievances by category, geography, channel, and time period.
- SLA compliance (on-time vs delayed) by category and responsible unit.
- Ageing analysis of open grievances.

iii. The system shall allow export of report results to CSV/XLSX, subject to role permissions.

iv. The system shall support configurable dashboards for different roles (e.g., summary tiles, trend charts, SLA alerts).

7. Security, Privacy, and Compliance

i. The system shall implement role-based access control so that users see only data they are authorized to access.

ii. The system shall support strong authentication mechanisms (e.g., SSO, 2FA) as per organization policy.

iii. The system shall ensure data in transit, rest and under processing is encrypted using HTTPS with current best-practice protocols.

iv. The system shall support encryption at rest for sensitive fields as per applicable regulations.

v. The system shall support masking or restricted access to personally identifiable information (PII) for specified roles.

vi. The system shall maintain an audit trail of all create/update/delete actions on grievances and configuration, with user, timestamp, and action details.

8. Configuration and Administration

i. The system shall allow authorized administrators to configure grievance categories, subcategories, priority rules, routing rules, and SLAs without code changes.

ii. The system shall allow configuration of notification templates and channels for each event type.

iii. The system shall support configuration of working hours, holidays, and calendar rules for SLA calculations.

iv. The system shall provide user and role management (create, deactivate, assign roles, reset passwords where not delegated to SSO).

v. The system shall allow configuration of multilingual labels and content

9. Performance, Availability, and Scalability

i. The system shall support concurrent usage by at least 100 internal users and 500 external users concurrently, without degradation beyond SLA thresholds (2 seconds per transaction maximum).

ii. The system shall load core screens (dashboard, grievance view, grievance search) within 2 seconds under normal load.

iii. The system shall meet a minimum uptime as defined in SLA, excluding scheduled maintenance windows.

iv. The system shall support horizontal or vertical scaling to handle increased grievance volumes without major re-architecture.

10. Usability and Accessibility

i. The system shall follow the MeitY's GIGW and AAI's directions for UX and provide consistent navigation, labelling, and error handling.

ii. The system shall be accessible and adhere to the chosen accessibility standard as per MeitY GIGW standards

iii. The system shall support at least 12 major Indian languages in addition to English and Hindi where in scope, with clear switching options.

2.18 Acceptance Criteria for PACR

"Acceptance Criteria for Software" is fulfilled along with the following:

S.N	Category	Key Activities / Deliverables	Tentative Timeline T	Responsibility
1	Requirements Finalization & Solution Design for PACR	Requirement validation, solution architecture, integration design, security design, final sign-off	T +1 Month	Bidder (with AAI inputs)
2	Documentation for PACR	Functional, technical, integration, SRS, Design for Prototype in Figma / adobe etc. security, operational documents, SOPs and user manuals and update from time	T +2 Month	Bidder
3	Helpdesk, Network & Setup	Infra's Infra provisioning, setup, sizing, commissioning and scalability planning (Desktop, headset, patch cord UCM, Storage, router, lan switches, Servers, Server Racks, firewall, load balancer, sufficient power points and plugs electrical works from the power source, adequate UPS (main and fail-over) or any other devices not mentioned but required for making it functional) procurement, Setup and commissioning of Contact Centre and Technical Helpdesk infrastructure. Establishment of secure network connectivity and interfaces	T + 4 Month	Bidder
4	CTI, IVR, UCM API Integrations with AirSewa	Integration of voice channel and any other component of PACR with AirSewa, Power BI etc.	T + 4 Month	Bidder
5	Manpower	Deployment, training and management of manpower as required in the document	T + 4.5 Month	Bidder

6	Security, Testing & Audit	Security controls, logging, access management, unit/ system/ integration/ performance testing, third-party audit & certification. Submission of Third-party audit and certification (Regulatory Compliance (TRAI/ OSP/ TCCCPR), Telephony Security (SIP/ VoIP Controls), CERT-In / STQC / empaneled agencies) prior to Go-Live	T + 5 Month	Bidder
7	Go-Live & Stabilization	Go-Live after stabilization, handholding, issue resolution and transition to steady-state operations and completion of all the above activities	T + 5 Month	Bidder

Annexure-II

PACR Tickets Summary 01st - 31st July, 2026

Platform	Total Ticket Generated
Twitter	16223
Instagram	669
Facebook	483
Outbound	176
Inbound	36
YouTube	3
Total	17590

Annexure-III

3 Compliance to Functional Requirement specifications

The below FRS is indicative only.

3.1. User Registration Module Requirements Compliance

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
1	User	User Type	The system should allow User to identify themselves from the predefined list of stakeholder categories eligible to register a grievance. The stakeholder categories shall include: 1. Passenger 2. AAI 3. Airline 4. Airport Operator 5. BCAS 6. DGCA 7. Airport Security Group (ASG) 8. Flying Training Organization (FTO) 9. Others	
2	Complainant	Complainant Registration	The system should allow Complainant to register using email	
3	Complainant	Complainant Registration	The system should allow Complainant to register using mobile	
4	Complainant	Complainant Registration	The system should allow Complainant to authenticate using OTP on email/mobile	
5	Complainant	Complainant Registration	The system should allow Complainant to authenticate using multi factor authentication	
6	Complainant	Complainant Login	The system should allow Complainant to login with registered credentials	
7	Complainant	Complainant Login	The system should allow Complainant to login as guest with mobile number	
8	Complainant	Complainant Login	The system should allow Complainant to login as guest with email ID	
9	Complainant	Complainant Login	The system should allow Complainant the option to give consent to save their data and create an account when they submit grievance through guest login	
10	Complainant	Complainant Registration	The system should allow Complainant to set password with standard password policy with length and complexity validation	

11	Complainant	Complainant Login	The system should ensure Complainant session expires after predefined minutes of inactivity	
12	Complainant	Complainant Login	The system should allow Complainant to reset login password using factor authentication	
13	Complainant	Complainant Login	The system should allow Complainant to keep username as registered mobile number or email id	
14	Complainant	Complainant Login	The system should ensure the username field should automaticaly detect the username is mobile or email ID	
15	Complainant	Complainant Login	The system should ensure the username field has 10 digit validation for mobile number	
16	Complainant	Complainant Login	The system should ensure the username field has @xyz.com validation for email ID	
17	Complainant	Complainant Registration	The system should allow Complainant to register using international mobile number	
18	Complainant	Complainant Registration	The system should ensure the country code is captured while registering using mobile	
19	Administrator	Redressor Nodal Officer Registration	The system should allow administrator to create redressor account for Redressor Nodal Officer	
20	Administrator	Redressor Nodal Officer Registration	The system should ensure administrator uploads authorisation letter from redressor organisation authorising the nodal officer	
21	Administrator	Redressor Nodal Officer Registration	The system should ensure administrator to fill up details of nodal officer - email ID, phone number, organisation name, designation, backup delegate email ID, backup delegate mobile number	
22	Administrator	Redressor Nodal Officer Registration	The system should allow administrator to reset the password of Redressor Nodal Officer on request of redressor	

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
23	Redressor Nodal Officer	Redressor Nodal Officer Registration	The system should allow Redressor Nodal Officer to reset default password using mul factor authentication	
24	Redressor Nodal Officer	Redressor Staf Registration	The system should allow Redressor Nodal Officer to create redressor account fo redressor staff	
25	Redressor Nodal Officer	Redressor Staff Registration	The system should allow Redressor Nodal Officer to reset the password on request o redressor	
26	Redressor Nodal Officer	Redressor Staff Registration	The system should ensure Redressor Nodal Officer to fill up details of staff - email ID phone number, designation, access level (L1/L2/L3), access type (read/write)	
27	Redressor Nodal Officer	Redressor Staff Registration	The system should ensure Redressor Nodal Officer gets error pop-up in case the staf is already registered using validation check of phone number and/or email ID	
28	Redressor Nodal Officer	Redressor Staff Registration	The system should ensure Organisation name is auto filled for Redressor Staff based o organisation name of Redressor Nodal Officer	
29	Redressor Nodal Officer	Redressor Staff Registration	The system should allow Redressor Nodal Officer to disable Redressor Staff credentials in case of exit from organisation	

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
30	Redressor Nodal Officer	Redressor Staff Registration	The system should allow Redressor Nodal Officer to suspend Redressor Staff credentials in case of any investigation	
31	Redressor Nodal Officer	Redressor Staff Registration	The system should allow Redressor Nodal Officer to release Redressor Staff credentials that have been previously suspended	
32	Redressor Staff	Redressor Staff Registration	The system should allow Redressor Staff to reset default password using factor authentication	
33	Redressor Nodal Officer	Redressor Staff Registration	The system should allow Redressor Nodal Officer to change access level of Redressor Staff	
34	User	Flight Tracking	The system should allow User to fetch current status and estimated time of arrival and landing at the correct airport by entering flight number without login	
35	User	Flight Tracking	The system should allow User to receive regular updates through notifications, SMS, messenger app (Arratai/ WhatsApp/ Telegram, etc) message, email, etc to give real-time status of the flight	
36	User	Flight Tracking	The system should allow User to opt for receiving regular updates for a particular flight number	

3.2. Grievance Registration Module Requirements Compliance

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
1	Complainant	Grievance Category	The system should allow Complainant to select the type of grievance from a dropdown list of categories. The list of categories shall include: 1. Flight related grievance (flight delay/cancellation/modification and/or baggage related) 2. Airport related grievance (Hygiene, charging points, facilities, security) 3. On-ground grievance (wheelchair assistance, airport stores, Taxi Aggregators, Meet & Greet service providers) 4. Government department related grievances (Customs, immigration, DGCA, etc.) 5. Others	
2	Complainant	Grievance Form - Flight related	The system should allow Complainant to submit grievance on provision of mandatory field - Contact Details	
3	Complainant	Grievance Form - Flight related	The system should allow Complainant to submit grievance on provision of mandatory field - Description of grievance	
4	Complainant	Grievance Form - Flight related	The system should allow Complainant to submit grievance on provision of mandatory field - Date & Time of event	
5	Complainant	Grievance Form - Flight related	The system should allow Complainant to submit grievance on provision of mandatory field - Airline	

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
6	Complainant	Grievance Form - Flight related	The system should allow Complainant to submit grievance on provision of mandatory field - Service Request Number	
7	Complainant	Grievance Form - Flight related	The system should allow Complainant to submit grievance on provision of mandatory field - PNR	
8	Complainant	Grievance Form - Flight related	The system should allow Complainant to submit grievance on provision of mandatory field - Flight number	
9	Complainant	Grievance Form - Flight related	The system should allow the information related to flight number and PNR be auto fetched once Complainant enters the Airline and Service Request Number	
10	Complainant	Grievance Form - Flight related	The system should ensure no duplicate grievance is registered against the given Service Request Number	
11	Complainant	Grievance Form - Flight related	The system should ensure Complainant fills all mandatory fields before allowing submission of the grievance	
12	Complainant	Grievance Form - Flight related	The system should allow Complainant to submit any supporting documents in PDF/JPG/PNG format	

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
13	Complainant	Grievance Form - Flight related	The system should ensure File size for supporting documents does not exceed 10MB	
14	Complainant	Grievance Form - Flight related	The system should ensure the standard TAT (based on agreed SLAs) is displayed on the screen once the Complainant selects the grievance category	
15	Complainant	Grievance Form - Flight related	The system should ensure validation of TAT (based on agreed SLAs) with respect to the Service Request logged date and time	
16	Complainant	Grievance Form - Flight related	The system should ensure Complainant can submit grievance only after the TAT exceeds the agreed SLAs / TAT SLA breach	
17	Complainant	Grievance Form - Flight related	The system should allow Grievances to be tagged "critical" in case critical category keywords are mentioned in the grievance description	
18	Complainant	Grievance Form - Flight related	The system should allow Complainant to submit the grievance before TAT SLA Breach in case grievance is tagged "critical"	
19	Complainant	Grievance Form - Flight related	The system should ensure Complainant agrees to have read privacy notice and checks the consent checkbox for storing contact information	

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
20	Complainant	Grievance Form - Flight related	The system should ensure a unique AirSewa ticket ID is generated upon successful submission of grievance form	
21	Complainant	Grievance Form - Airport related	The system should allow Complainant to submit grievance on provision of mandatory field - Contact Details	
22	Complainant	Grievance Form - Airport related	The system should allow Complainant to submit grievance on provision of mandatory field - Description of grievance	
23	Complainant	Grievance Form - Airport related	The system should allow Complainant to submit grievance on provision of mandatory field - Date & Time of event	
24	Complainant	Grievance Form - Airport related	The system should allow Complainant to submit grievance on provision of mandatory field - Airport	
25	Complainant	Grievance Form - Airport related	The system should ensure Complainant fills all mandatory fields before allowing submission of the grievance	
26	Complainant	Grievance Form - Airport related	The system should allow Complainant to submit any supporting documents in PDF/JPG/PNG format	

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
27	Complainant	Grievance Form - Airport related	The system should ensure File size for supporting documents does not exceed 10MB	
28	Complainant	Grievance Form - Airport related	The system should ensure the standard TAT (based on agreed SLAs with Airport Operators) is displayed on the screen once the Complainant selects the grievance category	
29	Complainant	Grievance Form - Airport related	The system should allow Grievances to be tagged "critical" in case critical category keywords are mentioned in the grievance description	
30	Complainant	Grievance Form - Airport related	The system should ensure Complainant agrees to have read privacy notice and checks the consent checkbox for storing contact information	
31	Complainant	Grievance Form - Airport related	The system should ensure a unique AirSewa ticket ID is generated upon successful submission of grievance form	
32	Complainant	Grievance Form - On-ground Grievances	The system should allow Complainant to submit grievance on provision of mandatory field - Contact Details	
33	Complainant	Grievance Form - On-ground grievaces	The system should allow Complainant to submit grievance on provision of mandatory field - Description of grievance	

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
34	Complainant	Grievance Form - On-ground Grievances	The system should allow Complainant to submit grievance on provision of mandatory field - Date & Time of event	
35	Complainant	Grievance Form - On-ground Grievances	The system should allow Complainant to submit grievance on provision of mandatory field - Airport	
36	Complainant	Grievance Form - On-ground Grievances	The system should ensure Complainant fills all mandatory fields before allowing submission of the grievance	
37	Complainant	Grievance Form - On-ground Grievances	The system should allow Complainant to submit any supporting documents in PDF/JPG/PNG format	
38	Complainant	Grievance Form - On-ground Grievances	The system should ensure File size for supporting documents does not exceed 10MB	

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
39	Complainant	Grievance Form - On-ground Grievances	The system should ensure the standard TAT (based on agreed SLAs with on-ground service providers) is displayed on the screen once the Complainant selects the grievance category	
40	Complainant	Grievance Form - On-ground Grievances	The system should ensure Grievances to be tagged "critical" in case critical category keywords are mentioned in the grievance description	
41	Complainant	Grievance Form - On-ground Grievances	The system should ensure Complainant agrees to have read privacy notice and checks the consent checkbox for storing contact information	
42	Complainant	Grievance Form - On-ground Grievances	The system should allow a unique AirSewa ticket ID is generated upon successful submission of grievance form	
43	Complainant	Grievance Form - Govt. Dept. related Grievances	The system should allow Complainant to submit grievance on provision of mandatory field - Contact Details	
44	Complainant	Grievance Form - Govt.	The system should allow Complainant to submit grievance on provision of mandatory field - Description of grievance	

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
		Dept. related Grievances		
45	Complainant	Grievance Form - Govt. Dept. related Grievances	The system should allow Complainant to submit grievance on provision of mandatory field - Date & Time of event	
46	Complainant	Grievance Form - Govt. Dept. related Grievances	The system should allow Complainant to submit grievance on provision of mandatory field - Which department is the grievance against?	
47	Complainant	Grievance Form - Govt. Dept. related Grievances	The system should allow Complainant to submit grievance on provision of optional field - Airport	
48	Complainant	Grievance Form - Govt. Dept. related Grievances	The system should allow Complainant to submit grievance on provision of optional field - PNR	
49	Complainant	Grievance Form - Govt. Dept. related Grievances	The system should allow Complainant to submit grievance on provision of optional field - Flight number	

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
50	Complainant	Grievance Form - Govt. Dept. related Grievances	The system should allow Complainant to submit grievance on provision of optional field - Passport Number	
51	Complainant	Grievance Form - Govt. Dept. related Grievances	The system should ensure Complainant fills all mandatory fields before allowing submission of the grievance	
52	Complainant	Grievance Form - Govt. Dept. related Grievances	The system should allow Complainant to submit any supporting documents in PDF/JPG/PNG format	
53	Complainant	Grievance Form - Govt. Dept. related Grievances	The system should ensure File size for supporting documents does not exceed 10MB	
54	Complainant	Grievance Form - Govt. Dept. related Grievances	The system should ensure the standard TAT (based on agreed SLAs with regulators) is displayed on the screen once the Complainant selects the grievance category	
55	Complainant	Grievance Form - Govt.	The system should ensure Grievances to be tagged "critical" in case critical category keywords are mentioned in the grievance description	

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
		Dept. related Grievances		
56	Complainant	Grievance Form - Govt. Dept. related Grievances	The system should ensure Complainant agrees to have read privacy notice and checks the consent checkbox for storing contact information	
57	Complainant	Grievance Form - Govt. Dept. related Grievances	The system should allow a unique AirSewa ticket ID is generated upon successful submission of grievance form	
58	Airsewa	Grievance Input - Inbound email data parsing	The system should allow Email grievances ingestion from PACR	
59	Airsewa	Grievance Input - Inbound call data parsing	The system should allow IVRS grievances ingestion from PACR	
60	Airsewa	Grievance Input - Social Media Ingestion	The system should allow Social Media grievances ingestion from PACR	

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
61	CPGRAMS	Grievance Input - CPGRAMS	The system should allow ingestion of Aviation/AAI related grievances raised on CPGRAMS from PACR	
62	Complainant	Additional Input Required	The system should allow Complainant to submit additional information in case grievance is sent back to Complainant by AirSewa Operator seeking additional information	
63	Complainant	Additional Input Required	The system should allow Complainant to submit additional information in case grievance is sent back to Complainant by Redressor Staff seeking additional information	
64	Complainant	Status	The system should allow Complainant to view status of the grievance	
65	Complainant	Status	The system should allow Complainant to view the trail of grievance resolution	
66	Complainant	Feedback	The system should allow Complainant to give feedback on the resolution after status is marked "Resolved"	
67	Complainant	Feedback	The system should allow Complainant to give feedback on the redressor staff (in case the Complainant was contacted) after status is marked "Resolved"	
68	Complainant	Feedback	The system should allow Complainant to give feedback on the AirSewa Operator (in case the Complainant was contacted) after status is marked "Resolved"	

3.3. Ticket Allocation Engine Requirements Compliance

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
1	Backend (automated)	Grievance Source	The system should ensure recording of source of the grievance which may be AirSewa portal, email, inbound call, social media or CPGRAMS	
2	Backend (automated)	Grievance Assignment	The system should allow Grievance to be assigned to "Airline" for Flight related grievance (flight delay/cancellation/modification and/or baggage related)	
3	Backend (automated)	Grievance Assignment	The system should allow Grievance to be assigned to "Airport" for Airport related grievance (Hygiene, charging points, facilities, security)	
4	Backend (automated)	Grievance Assignment	The system should allow Grievance to be assigned to "Ground Service Provider" for On-ground grievance (wheelchair assistance, airport stores, Taxi Aggregators, Meet & Greet service providers)	
5	Backend (automated)	Grievance Assignment	The system should allow Grievance to be assigned to "Regulator" for Regulatory related grievances (Customs, immigration)	
6	Backend (automated)	Grievance Labelling	The system should allow Labelling of severity levels of grievances based on keywords mentioned in the grievance description in accordance with keywords based severity matrix (annexure-##)	
7	Backend (automated)	Grievance TAT	The system should ensure Revised TAT (based on agreed SLAs for critical tag) is displayed against grievances labelled with Severity-1 and Severity-2	
8	Backend (automated)	Auto-Routing Rule	The system should ensure Automatic assignment of Severity-1 cases to redressor staff at L2 level and above	
9	Backend (automated)	Auto-Routing Rule	The system should allow Automatic assignment of Severity-2, Severity-3 and Severity-4 cases to redressor staff at L1 level	

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
10	Backend (automated)	Auto-Routing Rule	The system should allow Airlines to configure auto-routing rules for sub-categories of grievances	
11	Backend (automated)	Auto-Routing Rule	The system should allow Airlines to follow auto-routing of grievances based on sub-category SPOCs defined in Annexure-##	
12	Backend (automated)	Status	The system should allow Status to be "Under Assessment" once the grievance is submitted	
13	Backend (automated)	Status	The system should allow Status to be "Assigned" once the grievance is assigned	
14	Backend (automated)	Status	The system should allow Status to be "In-Progress" once the grievance is accessed by redressor staff	
15	Backend (automated)	Status	The system should allow Status to be "On Hold - Awaiting passenger inputs" in case the grievance is put on hold by redressor staff on account of seeking additional information from user	
16	Backend (automated)	Status	The system should allow Status to be "Provisionally Resolved" in case the grievance resolution is finalised by the redressor but the remedy process is incomplete	
17	Backend (automated)	Status	The system should allow Status to be "Resolved" in case the grievance resolution is finalised by the redressor and the remedy process is also complete	
18	Backend (automated)	Event logs	The system should ensure Preservation of event logs - trail of grievance routing, rerouting, resolution, timestamp, redressor staff details and comments (if any)	
19	Backend (automated)	Language	The system should allow to check for language of tickets and auto-translate to English in case of other language	

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
20	Backend (automated)	Deduplication	The system should allow Automatically flag duplicate grievances inbound from multiple channels	
21	Grievance fields	Viewing grievance	The system should ensure when any stakeholder searches for and opens a grievance record, the system shall display the Grievance Timestamp	
22	Grievance fields	Viewing grievance	The system should ensure when any stakeholder searches for and opens a grievance record, the system shall display the Grievance ID	
23	Grievance fields	Viewing grievance	The system should ensure when any stakeholder searches for and opens a grievance record, the system shall display the Severity Level	
24	Grievance fields	Viewing grievance	The system should ensure when any stakeholder searches for and opens a grievance record, the system shall display the Severity Level Reason (cite the keywords that led to severity categorisation)	
25	Grievance fields	Viewing grievance	The system should ensure when any stakeholder searches for and opens a grievance record, the system shall display the Subcategory	
26	Grievance fields	Viewing grievance	The system should ensure when any stakeholder searches for and opens a grievance record, the system shall display the Description of grievance	
27	Grievance fields	Viewing grievance	The system should ensure when any stakeholder searches for and opens a grievance record, the system shall display the Assignment redressor organisation	
28	Grievance fields	Viewing grievance	The system should ensure when any stakeholder searches for and opens a grievance record, the system shall display the Assignment redressor team	
29	Grievance fields	Viewing grievance	The system should ensure when any stakeholder searches for and opens a grievance record, the system shall display the Assignment redressor staff	

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
30	Grievance fields	Viewing grievance	The system should ensure when any stakeholder searches for and opens a grievance record, the system shall display the Escalation level	
31	Grievance fields	Viewing grievance	The system should ensure when any stakeholder searches for and opens a grievance record, the system shall display the SLA policy applicable	
32	Redressor Staff	Severity Level Override	The system should allow L2 level and above staff to override the automated severity level based on manual assessment of case	
33	Redressor Staff	Severity Level Override	The system should ensure severity level override allows a grievance to be upgraded to higher severity level and does not allow a grievance severity to be downgraded to a lower level	
34	Redressor Staff	Severity Level Override	The system should allow severity level downgrade in special cases, however the grievance would route to L3 level staff or nodal officer for approval, only after which the severity level be downgraded	
35	Redressor Nodal Officer	Severity Level Override	The system should allow approval of severity downgrade request processed by L1, L2, L3 level redressor staff	
36	AirSewa Operator	Severity Level Override	The system should allow AirSewa agent to override severity level of grievance allowing to upgrade severity level	
37	AirSewa Operator	Severity Level Override	The system should allow AirSewa agent to override severity level of grievance allowing to downgrade severity level	
38	Redressor Staff	Additional Redressor	The system should allow redressor staff to add additional redressors in case the ticket/ complaint required attention/ intervention from multiple redressors	

3.4. Communication Module Requirements Compliance

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
1	User	Registration	The system should allow OTP to be sent on mobile/email for factor authentication - login/guest login/password change	
2	User	Registration	The system should allow notification to be sent to the User once the account is registered	
3	User	Registration	The system should allow notification to be sent to the User once the account password is modified	
4	User	Grievance Registration	The system should allow notification to be sent on successful submission of grievance along with unique grievance ID	
5	User	Grievance Status	The system should allow notification to be sent if status changes to "On Hold - Awaiting passenger inputs" in case the grievance is put on hold by redressor staff on account of seeking additional information from user	
6	User	Grievance Status	The system should allow notification to be sent if status changes to "Provisionally Resolved" in case the grievance resolution is finalised by the redressor but the remedy process is incomplete	
7	User	Grievance Status	The system should allow notification to be sent if status changes to “Resolved" in case the grievance resolution is finalised by the redressor and the remedy process is also complete	
8	General	Message	The system should allow rich text messages with variable customisation	
9	General	Message	The system should ensure Personal Identifier Information (PII) is masked in all communications	
10	User	Notifications	The system should allow User to set notification preferences to mobile	
11	User	Notifications	The system should allow User to set notification preferences to email	

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
12	User	Notifications	The system should allow User to set notification preferences to mobile and email	
13	User	Notifications	The system should allow User to set notification preferences to none i.e. no notifications be sent on email or mobile	
14	Redressor Nodal Officer	Registration	The system should allow Notification to be sent to the Redressor Nodal Officer once the administrator creates the account	
15	Redressor Staff	Registration	The system should allow Notification to be sent to the Redressor Staff once the Redressor Nodal Officer creates the account	
16	Redressor Nodal Officer	Daily report	The system should allow Daily summary report of grievances added, resolved, escalated and on hold be sent to Redressor Nodal Officer	
17	Administrator	Daily report	The system should allow Daily summary report of Redressor wise grievances added, resolved, escalated and on hold be sent to Administrator	
18	MoCA officials	Daily report	The system should allow Daily summary report of Redressor wise grievances added, resolved, escalated and on hold be sent to listed MoCA officials' email ID	
19	Redressor Nodal Officer	TAT SLA Breach	The system should ensure Notification is sent to Redressor Nodal Officer for grievances unattended beyond agreed SLAs - need clarity on this - users would file grievance on AirSewa only in case TAT SLA is breached by Redressor, so would all grievances be assumed TAT SLA Breach	
20	Redressor Staff	Escalation	The system should ensure Notification is sent to L2 level Redressor Staff in case grievance is escalated to L2 level	
21	Redressor Staff	Escalation	The system should ensure Notification is sent to L3 level Redressor Staff in case grievance is escalated to L3 level	

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
22	Redressor Nodal Officer	Escalation	The system should ensure Notification is sent to Redressor Nodal Officer in case grievance is escalated to L3 level	
23	Redressor Nodal Officer	Grievance re-open	The system should ensure Notification is sent to Redressor Nodal Officer in case a grievance is reopened for the second time	
24	AirSewa Operator	Grievance re-open	The system should ensure Notification is sent to AirSewa Operator in case a grievance is reopened for the second time	
25	AirSewa Operator	Escalation	The system should allow AirSewa operator to trigger notification to Redressor Staff and Redressor Nodal Officer in case a grievance is to be escalated	
26	AirSewa Operator	Additional Information Required	The system should allow AirSewa operator to trigger notification to User to seek additional information in the grievance on portal	
27	Redressor Staff	Additional Information Required	The system should allow Redressor Staff to trigger notification to User to seek additional information in the grievance on portal	
28	Administrator	Notification Logs	The system should allow Administrator to view logs of all notifications sent to User	
29	Administrator	Notification Logs	The system should allow Administrator to view logs of all notifications sent to Redressor Staff	
30	Administrator	Notification Logs	The system should allow Administrator to view logs of all notifications sent to Redressor Nodal Officer	
31	Administrator	Notification Logs	The system should ensure Preservation of all notification logs	

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
32	User	Tech Support	The system should allow all redressor users to access the manuals / videos related to training content in order to effectively use the portal	
33	User	Tech Support	The system should allow all redressor users to ask queries to a chatbot/ tech help team to help resolve their issues, help them in troubleshooting the issues related to the portal	
34	User	Tech Support	The system should allow all redressor users to give feedback to improve the portal	
35	User	Tech Support	The system should allow all complainants to access manuals / videos related to effectively use the portal	

3.5. AirSewa Agent Module Requirements Compliance

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
1	AirSewa Agent	Grievance Assignment	The system should allow AirSewa Agent to conduct assessment of Grievances in bulk	
2	AirSewa Agent	Grievance Assignment	The system should allow AirSewa Agent to take bulk action of Grievances - override severity level	
3	AirSewa Agent	Grievance Assignment	The system should allow AirSewa Agent to take bulk action of Grievances - escalate Grievances from L1 to L2 Redressor Staff	
4	AirSewa Agent	Grievance Assignment	The system should allow AirSewa Agent to take bulk action of Grievances - escalate Grievances from L2 to L3 Redressor Staff	
5	AirSewa Agent	Grievance Assignment	The system should allow AirSewa Agent to take bulk action of Grievances - escalate Grievances from L3 to Redressor Nodal Officer	
6	AirSewa Agent	Grievance Assignment	The system should allow AirSewa Agent to merge duplicate grievances	
7	AirSewa Agent	Grievance Assignment	The system should allow AirSewa Agent to label a grievance as a parent while merging duplicate grievance	
8	AirSewa Agent	Grievance Assignment	The system should allow AirSewa Agent to label a grievance as a child while merging duplicate grievance	
9	AirSewa Agent	Grievance Assignment	The system should allow AirSewa Agent to view links of all related grievances under the parent labelled grievance	
10	AirSewa Agent	Grievance Assignment	The system should allow AirSewa Agent to override auto-assignment of grievance redressor	

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
11	AirSewa Agent	Grievance Assignment	The system should allow AirSewa Agent to override auto-assignment of grievance severity level	
12	AirSewa Agent	Monitoring	The system should allow AirSewa Agent to view workload/assigned grievances of each Redressor Staff	
13	AirSewa Agent	Monitoring	The system should allow AirSewa Agent to view workload/assigned grievances of each Redressor	
14	AirSewa Agent	Monitoring	The system should allow AirSewa Agent to view grievances based on type of grievance	
15	AirSewa Agent	Monitoring	The system should allow AirSewa Agent to view grievances based on sub-category of grievance	
16	AirSewa Agent	Monitoring	The system should allow AirSewa Agent to view grievances based on date/time of grievance	
17	AirSewa Agent	Monitoring	The system should allow AirSewa Agent to view grievances based on age of grievance	
18	AirSewa Agent	Grievance Assignment	The system should allow AirSewa Agent to change the status of grievance	
19	AirSewa Agent	Grievance Assignment	The system should allow AirSewa Agent to add additional information to the grievance based on communication with user	
20	AirSewa Agent	Reporting & Analytics	The system should allow AirSewa Agent to view summary of grievance data on an interactive dashboard	
21	AirSewa Agent	Reporting & Analytics	The system should allow AirSewa Agent to download entire detailed grievance history as a PDF, CSV, excel or any other required format	

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
22	AirSewa Agent	Reporting & Analytics	The system should allow AirSewa Agent to download detailed list of grievances as a PDF, CSV, excel or any other required format - based on grievance status	
23	AirSewa Agent	Reporting & Analytics	The system should allow AirSewa Agent to download detailed list of grievances as a PDF, CSV, excel or any other required format - based on grievance age	
24	AirSewa Agent	Reporting & Analytics	The system should allow AirSewa Agent to download detailed list of grievances as a PDF, CSV, excel or any other required format - based on grievance severity	
25	AirSewa Agent	Reporting & Analytics	The system should allow AirSewa Agent to download detailed list of grievances as a PDF, CSV, excel or any other required format - based on grievance type or sub-category	
26	AirSewa Agent	Reporting & Analytics	The system should allow AirSewa Agent to download detailed list of grievances as a PDF, CSV, excel or any other required format - based on grievance redressor	
27	AirSewa Agent	Escalation	The system should allow AirSewa Agent to escalate grievance from L1 to L2 or from L2 to L3 based on triage	

3.6. Redressor Module Requirements Compliance

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
1	Redressor Staff	Access to assigned grievance	The system should allow Redressor staff to access the grievances in queue assigned to redressor by the automated system based on grievance type, sub-category	
2	Redressor Staff	Access to assigned grievance	The system should ensure L1 level Redressor staff is able to access the Severity-2, Severity-3 and Severity-4 grievances in queue	
3	Redressor Staff	Access to assigned grievance	The system should ensure L2, L3 level Redressor staff and Redressor Nodal Officer are able to access the Severity-1, Severity-2, Severity-3 and Severity-4 grievances in queue	
4	Redressor Staff	Search grievance	The system should allow Redressor staff to search grievance based on Grievance ID	
5	Redressor Staff	Search grievance	The system should allow Redressor staff to search grievance based on User contact details	
6	Redressor Staff	Search grievance	The system should allow Redressor staff to search grievance based on User PNR details	
7	Redressor Staff	Search grievance	The system should allow Redressor staff to search grievance based on User Service Request Number	
8	Redressor Staff	Filter grievances	The system should allow Redressor staff to filter grievance based on Grievance Type	
9	Redressor Staff	Filter grievances	The system should allow Redressor staff to filter grievance based on Grievance Sub-Category	

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
10	Redressor Staff	Filter grievances	The system should allow Redressor staff to filter grievance based on Grievance Age	
11	Redressor Staff	Filter grievances	The system should allow Redressor staff to filter grievance based on Grievance Status	
12	Redressor Staff	Filter grievances	The system should allow Redressor staff to filter grievance based on Grievance Date/time	
13	Redressor Staff	Update Status	The system should allow Redressor staff to update status of the grievance as "Provisionally Resolved" in case the grievance resolution is finalised by the redressor but the remedy process is incomplete	
14	Redressor Staff	Update Status	The system should allow Redressor staff to update status of the grievance as "Resolved" in case the grievance resolution is finalised by the redressor and the remedy process is also complete	
15	Redressor Staff	Update Status	The system should allow Redressor staff to update status of the grievance as "On Hold - Awaiting passenger inputs" in case the grievance is put on hold on account of seeking additional information from user	
16	Redressor Staff	Trigger notification	The system should allow Redressor staff to trigger notification to grieved user to seek additional inputs for "On Hold" grievances	
17	Redressor Staff	Reopen Ticket	The system should allow Redressor staff to reopen a "resolved" grievance in case the circumstances of the grievance change - need to check if we want to allow redressor to reopen a ticket	
18	Redressor Staff	Resolution Proof	The system should ensure Redressor staff uploads proof for provisionally resolved grievance before changing status to "Provisionally Resolved"	

Sl. No.	Stakeholder	Area	Requirement	Whethercomplied or not (Yes/No)
19	Redressor Staff	Resolution Proof	The system should ensure Redressor staff uploads proof for resolved grievance before changing status to "Resolved"	
20	Redressor Staff	Grievance assignment	The system should allow L2, L3 level Redressor staff and Redressor Nodal Officer are able to override the assignment and reassign to other redressor	
21	Redressor Nodal Officer	Reporting & Analytics	The system should allow Redressor Nodal Officer to download detailed list of grievances as a PDF, CSV, excel or any other required format - based on grievance status	
22	Redressor Nodal Officer	Reporting & Analytics	The system should allow Redressor Nodal Officer to download detailed list of grievances as a PDF, CSV, excel or any other required format - based on grievance age	
23	Redressor Nodal Officer	Reporting & Analytics	The system should allow Redressor Nodal Officer to download detailed list of grievances as a PDF, CSV, excel or any other required format - based on grievance severity	
24	Redressor Nodal Officer	Reporting & Analytics	The system should allow Redressor Nodal Officer to download detailed list of grievances as a PDF, CSV, excel or any other required format - based on grievance type or sub-category	
25	Redressor Nodal Officer	Reporting & Analytics	The system should allow Redressor Nodal Officer to download detailed list of grievances as a PDF, CSV, excel or any other required format - based on grievance redressor	
26	Redressor Staff	Escalation	The system should allow Redressor staff to escalate grievance from L1 to L2 or from L2 to L3 based on assessment of criticality and severity	
27	Redressor Nodal Officer	Monitoring	The system should allow Redressor Nodal Officer to view workload/assigned grievances of each Redressor Staff	
28	Redressor Nodal Officer	Search grievance	The system should allow Redressor Nodal Officer to search grievance based on Grievance ID	

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
29	Redressor Nodal Officer	Search grievance	The system should allow Redressor Nodal Officer to search grievance based on User contact details	
30	Redressor Nodal Officer	Search grievance	The system should allow Redressor Nodal Officer to search grievance based on User PNR details	
31	Redressor Nodal Officer	Search grievance	The system should allow Redressor Nodal Officer to search grievance based on User Service Request Number	
32	Redressor Nodal Officer	Filter grievances	The system should allow Redressor Nodal Officer to filter grievance based on Grievance Type	
33	Redressor Nodal Officer	Filter grievances	The system should allow Redressor Nodal Officer to filter grievance based on Grievance Sub-Category	
34	Redressor Nodal Officer	Filter grievances	The system should allow Redressor Nodal Officer to filter grievance based on Grievance Age	
35	Redressor Nodal Officer	Filter grievances	The system should allow Redressor Nodal Officer to filter grievance based on Grievance Status	
36	Redressor Nodal Officer	Filter grievances	The system should allow Redressor Nodal Officer to filter grievance based on Grievance Date/time	
37	Redressor Nodal Officer	Filter grievances	The system should allow Redressor Nodal Officer to filter grievance based on Redressor Staff name/ID	
38	Redressor Nodal Officer	Feedback	The system should allow Redressor Nodal Officer to view feedback of each of the redressor staff against each grievance resolved	
39	Redressor Nodal Officer	Feedback	The system should allow Redressor Nodal Officer to view average feedback of each of the redressor staff	

3.7. Redressor Administrator Module Requirements Compliance

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
1	AirSewa Agent	Grievance Assignment	The system should allow AirSewa Agent to conduct assessment of Grievances in bulk	
1	Administrator	Grievance Assignment	The system should allow Administrator to conduct assessment of Grievances in bulk	
2	Administrator	Grievance Assignment	The system should allow Administrator to take bulk action of Grievances - override severity level	
3	Administrator	Grievance Assignment	The system should allow Administrator to take bulk action of Grievances - escalate Grievances from L1 to L2 Redressor Staff	
4	Administrator	Grievance Assignment	The system should allow Administrator to take bulk action of Grievances - escalate Grievances from L2 to L3 Redressor Staff	
5	Administrator	Grievance Assignment	The system should allow Administrator to take bulk action of Grievances - escalate Grievances from L3 to Redressor Nodal Officer	
6	Administrator	Grievance Assignment	The system should allow Administrator to merge duplicate grievances	
7	Administrator	Grievance Assignment	The system should allow Administrator to label a grievance as a parent while merging duplicate grievance	
8	Administrator	Grievance Assignment	The system should allow Administrator to label a grievance as a child while merging duplicate grievance	
9	Administrator	Grievance Assignment	The system should allow Administrator to view links of all related grievances under the parent labelled grievance	

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not
10	Administrator	Grievance Assignment	The system should allow Administrator to override auto-assignment of grievance redressor	
11	Administrator	Grievance Assignment	The system should allow Administrator to override auto-assignment of grievance severity level	
12	Administrator	Monitoring	The system should allow Administrator to view workload/assigned grievances of each Redressor Staff	
13	Administrator	Monitoring	The system should allow Administrator to view workload/assigned grievances of each Redressor	
14	Administrator	Monitoring	The system should allow Administrator to view grievances based on type of grievance	
15	Administrator	Monitoring	The system should allow Administrator to view grievances based on sub-category of grievance	
16	Administrator	Monitoring	The system should allow Administrator to view grievances based on date/time of grievance	
17	Administrator	Monitoring	The system should allow Administrator to view grievances based on age of grievance	
18	Administrator	Grievance Assignment	The system should allow Administrator to change the status of grievance	
19	Administrator	Grievance Assignment	The system should allow Administrator to add additional information to the grievance based on communication with user	
20	Administrator	Reporting & Analytics	The system should allow Administrator to view summary of grievance data on an interactive dashboard	
21	Administrator	Reporting & Analytics	The system should allow Administrator to download entire detailed grievance history as a PDF, CSV, excel or any other required format	

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
22	Administrator	Reporting & Analytics	The system should allow Administrator to download detailed list of grievances as a PDF, CSV, excel or any other required format - based on grievance status	
23	Administrator	Reporting & Analytics	The system should allow Administrator to download detailed list of grievances as a PDF, CSV, excel or any other required format - based on grievance age	
24	Administrator	Reporting & Analytics	The system should allow Administrator to download detailed list of grievances as a PDF, CSV, excel or any other required format - based on grievance severity	
25	Administrator	Reporting & Analytics	The system should allow Administrator to download detailed list of grievances as a PDF, CSV, excel or any other required format - based on grievance type or sub-category	
26	Administrator	Reporting & Analytics	The system should allow Administrator to download detailed list of grievances as a PDF, CSV, excel or any other required format - based on grievance redressor	
27	Administrator	Escalation	The system should allow Administrator to escalate grievance from L1 to L2 or from L2 toL3 based on triage	
28	Administrator	Feedback	The system should allow Administrator to view feedback of each of the redressor staff against each grievance resolved	

29	Administrator	Feedback	The system should allow Administrator to view feedback of each of the AirSewa Operator against each grievance resolved	
30	Administrator	Monitoring	The system should allow Administrator to monitor First Response Time of each grievance	
31	Administrator	Monitoring	The system should allow Administrator to monitor Response Time of each grievance	
32	Administrator	Monitoring	The system should allow Administrator to monitor grievance reopening statistics	
33	Administrator	API	The system should allow Administrator to integrate APIs with Grievance Portal	
34	Administrator	API	The system should ensure Administrator solely controls the API key for integration	

3.8. SLA & Escalation Module Requirements Compliance

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
1	Backend (automated)	SLA	The system should allow multiple SLA policies to run in parallel based on categories, priorities and redressor	
2	Backend (automated)	Escalation	The system should allow automatic escalation from L1 to L2 and L2 to L3 in case of TAT SLA breach	
3	Backend (automated)	SLA	The system should allow notification to be sent to L3 Level Redressor staff and Redressor Nodal Officer for unresolved grievances approaching the threshold of TAT SLA	
4	Backend (automated)	SLA	The system should allow recording of First Response Time of each grievance	
5	Backend (automated)	SLA	The system should allow recording of Response Time of each grievance	
6	Backend (automated)	SLA	The system should allow recording of grievance reopening statistics	
7	Backend	SLA	The system should allow SLAs to be reconfigured based on changing CAR rules and terms of MoU with Redressors	
8	Backend	SLA	The system should ensure Resolution provided is as per rules defined for each grievance type and sub-category	
9	Backend	SLA	The system should ensure that for "refund related" grievance type, the Refund amount is equal to or greater than the refund calculated based on the CAR rules	
10	Backend	SLA	The system should allow assessment of quality of resolution by redressor	
11	Backend	SLA	The system should allow rating of the redressor based on the quality of resolution	

3.9. MoCA Dashboard Module Requirements Compliance

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
1	AirSewa Agent	Grievance Assignment	The system should allow AirSewa Agent to conduct assessment of Grievances in bulk	
1	MoCA Stakeholders	Monitoring	The system should allow MoCA to view links of all related grievances under the parent labelled grievance	
2	MoCA Stakeholders	Monitoring	The system should allow MoCA to view workload/assigned grievances of each Redressor Staff	
3	MoCA Stakeholders	Monitoring	The system should allow MoCA to view workload/assigned grievances of each Redressor	
4	MoCA Stakeholders	Monitoring	The system should allow MoCA to view grievances based on type of grievance	
5	MoCA Stakeholders	Monitoring	The system should allow MoCA to view grievances based on sub-category of grievance	
6	MoCA Stakeholders	Monitoring	The system should allow MoCA to view grievances based on date/time of grievance	
7	MoCA Stakeholders	Monitoring	The system should allow MoCA to view grievances based on age of grievance	
8	MoCA Stakeholders	Monitoring	The system should allow MoCA to view summary of grievance data on an interactive dashboard	
9	MoCA Stakeholders	Monitoring	The system should allow MoCA to download entire detailed grievance history as a PDF, CSV, excel or any other required format	

Sl. No.	Stakeholder	Area	Requirement	Whether complied or not (Yes/No)
10	MoCA Stakeholders	Monitoring	The system should allow MoCA to download detailed list of grievances as a PDF, CSV, excel or any other required format - based on grievance status	
11	MoCA Stakeholders	Monitoring	The system should allow MoCA to download detailed list of grievances as a PDF, CSV, excel or any other required format - based on grievance age	
12	MoCA Stakeholders	Monitoring	The system should allow MoCA to download detailed list of grievances as a PDF, CSV, excel or any other required format - based on grievance severity	
13	MoCA Stakeholders	Monitoring	The system should allow MoCA to download detailed list of grievances as a PDF, CSV, excel or any other required format - based on grievance type or sub-category	
14	MoCA Stakeholders	Monitoring	The system should allow MoCA to download detailed list of grievances as a PDF, CSV, excel or any other required format - based on grievance redressor	
15	MoCA Stakeholders	Monitoring	The system should allow MoCA to view feedback of each of the redressor staff against each grievance resolved	
16	MoCA Stakeholders	Monitoring	The system should allow MoCA to view feedback of each of the AirSewa Operator against each grievance resolved	
17	MoCA Stakeholders	Monitoring	The system should allow MoCA to monitor First Response Time of each grievance	
18	MoCA Stakeholders	Monitoring	The system should allow MoCA to monitor Response Time of each grievance	
19	MoCA Stakeholders	Monitoring	The system should allow MoCA to monitor grievance reopening statistics	

Annexure-IV

3.10. PACR Compliance "Minimum specification for Desktop & with IVRS"

(All the compliance will be checked and approved by MOCA before deployment)

S.No.	Module	Remarks	Compliance (Yes/No)
1	IT Hardware (1 MFP printer) All-in-Desktop and head set (For each resources of PACR)	All-in-One-Desktop • Form factor AIO • Processor Make: Intel • Processor Generation:12 or above • Processor Description: Intel Core i7 • Operating System (Factory Pre-Loaded): Windows 11 Professional • MS office 365 or latest • RAM Size (GB): 16 • RAM Expandability up to: 32 GB • Type of Drives used to populate the Internal Bays: SSD • Total HDD Capacity (GB): 0 • Total SSD Capacity (GB): 512	

S.No.	Module	Remarks	Compliance (Yes/No)
		• Display Size (INCHES): 20 and above • Display Type: Non-Touch • Panel Technology: IPS • Display Resolution (PIXELS): 1920x1080 • Availability of Webcam integrated with Display: Yes • On Site OEM Warranty (Year): 3 Printer—1 no. • Print Technology: Laser • Type of Machine: Multifunction Machine • Type of Printing: Mono • Cartridge Technology: Composite Cartridge • Platen/Flatbed Size: A4 • Paper Size (Original/Image): A4/A4 • RAM size (MB): 512 • Hard Disk Capacity (GB):0	

S.No.	Module	Remarks	Compliance (Yes/No)
		• Minimum Speed per Minute as per ISO/IEC 24734 in A4 Size-Mono: 40 • Minimum Speed per Minute as per ISO/IEC 24734 in A4 Size-Colour: 0 • Scanning Feature Availability: Yes • Duplexing Feature Availability: Yes • Faxing Feature Availability: Yes • Networking Feature Availability: Yes • Original Document Feeder Type: SPDF • Number of Main Paper Tray: 1 • Bypass Facility: Yes • On Site OEM Warranty (Year): 3Warranty • Three-year comprehensive hardware warranty. • Head set with Microphone (with noise cancellation facility) --- 15 nos. • Headphone • Noise Cancellation • Microphone	

S.No.	Module	Remarks	Compliance (Yes/No)
		• Wired • Standard support for 3 years	
2	Infra related to Call Handling (2 nos. each (for fail over/high availability))	CTI/Telephony Server, UCM, Storage, SIP/PRI Converter, Router, Firewall, Load balancer, LAN, Cable/Network Switches, , sufficient power points and plugs electrical works (including electrical from the power source provided by MoCA, adequate UPS (main and fail-over) and any other infrastructure/ software required to make the PACR fully functional and ready for BCP	

S.No.	Requirement Specifications (Omni channel call center solution with IVRS)	Compliance (Yes/no)
1.	Provide Support in mapping new and existing short code/toll-free number/ long-code/PRI or SIP Number	
2.	Configuration of IVRS call flows as per the requirement and modify as required	
3.	Configuration of Call flow of IVRS for multiple projects, Multiple Indian language/dialect as per the requirement	
4.	Have facility to create multiple accounts with separate reports. Individual users would have the same credentials to access multiple accounts.	
5.	Provide GUI to manage the IVRS menu options, Add and delete users, Add and delete departments,	
6.	Integration and Configuration of API as per the IVRS call flow defined for all services as per the need of the projects.	
7.	Support Click to call services and can be configured as per the requirement based on the requirement of project. Should also support outgoing calling through mobile dialer app to call directly from mobile phone/Landline and build all analytics on such call to give seamless mobile experience in outgoing call and building all analytics regarding the user to assist their work during remote working scenario.	
8.	Provide Cumulative and segregated dashboard for reporting which can be configured to customize the GUI reports (Weekly, Monthly, Quarterly, and Yearly)	
9.	Time-based routing, Contact Based routing, Language / domain-basedrouting, Call Transfer, Call Conferencing	
10.	Dedicated Dashboard with analytics and call logs (if required, Power BI tool licenses will be provided by MOCA for further analytics requirement)	

S.No.	Requirement Specifications (Omni channel call center solution with IVRS)	Compliance (Yes/no)
11.	Complete contact centre solution including Agent and supervisor console Routing basis agent status availability, skills, time of day. In call abilities like call hold, call transfer, muteAuto-dialer and Progressive.	
12.	Missed call solution – Scale concurrency on demand	
13.	OBD blaster via UI, APIs and FTP. Campaign level reporting on UI	
14.	Speech to Text & Text to Speech functionality	
15.	SMS GUI for campaign creation and campaign specific reporting Detailed analytics at campaign level reporting	
16.	Role based access. To Customize new roles as per requirement on the fly Manage and create new accounts from UI. To Customize permission for role at feature level . Activity logs to check history of actions performed by each user	
17.	Backup and restoration policy and mechanism	
18.	Call Dashboard	
19.	Call Analytics	
20.	Call & SMS Reports	

S.No.	Requirement Specifications (Omni channel call center solution with IVRS)	Compliance (Yes/no)
21.	Call Recordings	
22.	Virtual Number facility	
23.	Multiple Operator Backups	
24.	Voice APIs	
25.	SMS APIs	
26.	Campaign APIs	
27.	CRM Integration APIs	
28.	Follow up module: Auto follow up task addition rules-based system to let CCE follow up with missed calls or even connected calls.	
29.	Accessibility of multiple accounts with same login credentials and switching from one account to another without logging multiple times.	
30.	Having facility to fetch the information on API from the customer data base and informing the same to the caller on a real time basis.	
31.	Mobile dialer app to support outgoing calls through a centralized number to ensure all calls are getting tracked and analytics are being built and giving employees seamless mobile experience while working from home	
32.	Live call transfer feature to transfer calls from one agent to other, one state to another and one call center to another.	
33.	Multi-tenant architecture so that it is scalable immediately to other functions as well.	

S.No.	Requirement Specifications (Omni channel call center solution with IVRS)	Compliance (Yes/no)
34.	Have facility to integrate with existing Chatbot or AI Assistant (ChatBOT Licenses/tools cost will be procured/provided by AAI. The agency has to integrate the same.)	
35.	Support for Chatbot and WhatsApp and social media integration (Licenses/tools cost will be procured/provided by AAI. The agency has to integrate the same.)	
36.	Voice with AI and ML support & Integration (Licenses/tools cost will be procured/provided by AAI. The agency has to integrate the same.)	
37.	Automated outbound call dialer on dashboard and with APIs	
38.	Facility for Reports to be downloaded from dashboard	
39.	Call handling functionality such as hold, mute, conference, transfer and hang-up, call barging facility, call queue waiting (with caller number & estimate time of connecting) should be available on CRM. CCE should also be able to forward the calls / call conferencing with other CCE using their extension numbers.	
40.	Customization of CRM popup screen, if any with integration to CRM of Developed by AAI. Facility for agent to receive, Hang-up and dispose the call with various options on MOCA CRM along with the profile of the caller (from MOCA CRM).	
41.	CRM integration with Web app and Mobile app (Using API or other techniques). APIs for the call flow would be provided by the vendor (including click to call / CRM integration of soft phone etc.).	
42.	Facility to listen ‘live calls’ without the agent or caller knowing for quality monitoring purpose.	

S.No.	Requirement Specifications (Omni channel call center solution with IVRS)	Compliance (Yes/no)
43.	Real-time reporting dashboard to monitor calls center metrics& SLA in real-time with the desired/required KPIs	
44.	Reporting Module - Historical reporting displays call center metrics from any defined time point in the past it may include various reporting features such as hourly, daily, weekly, monthly, live calls, agent who disconnect call, actual time of call and real time on IVRS, queue report, utilization of channels usage, missed call, call forwarded to another agent, call conference, call dropped and annual statistics for detailed long-term analysis, campaign wise report, agent wise report (call picked, disconnected, missed), report on listened best practice by the user, recorded best practice by the user per lab, Peak hours, etc.	
45.	Omni channel Services for channels like voice, WhatsApp, chat, ChatBOT. (Licenses/tools cost will be procured/provided by AAI. The agency has to integrate the same.)	
46.	Auto assignment of call drop to the relevant CCE.	
47.	Setup advertisement (if required) at the time of call hold by callers (voice files to be provided by AAI)	
48.	Surveys, feedback and poll using voice channel	
49.	Tele-Marketing–Outbound Calls	
50.	Alert/Message if System is down	
51.	Provision to set the ‘call receiving time’ for CCE as per their availability / Leave management of CCE	
52.	Should have open CTI protocol	

3.3. **Compliance to Technical Requirements**

Design Principle	**Description**	**Whether complied or not (Yes/No)**
Accessibility	The platform should support multiple access channels such as web, mobile app, IVRS, WhatsApp, walk-in assisted filing. Forms should be designed in easy-to-interpret language, with suitable font types and sizes to ensure readability. Design should be inclusive to ensure accessibility for users with low technical literacy. Suitable accessibility tools and mechanisms should be provided as per latest GIGW and as amended from time to time (Guidelines for Indian Government Websites) norms to ensure accessibility for Specially-abled users, including WCAG 2.2 guidelines,.	
Performance metrics	The system should support a minimum of 500 Concurrent users including both complainants, actioning departments and admin users. Transaction processing timelines and execution should be within the defined SLAs	
Security by design	• Ensure the developed software is compliance with industry cyber-security frameworks (e.g., CERT-In, ISO 27001, OWASP). • Enforce encryption in transit, at rest and during processing. • Use secure coding practices and conduct regular code scans, vulnerability assessments, and penetration testing to identify buds.	

Privacy & Data Protection	• Adhere strictly to applicable data-protection laws (DPDP, IT Act 2000 and amendments) and government policies. • Implement privacy-first data handling, with data minimization and audit controls. • Enable granular consent management and role-based access controls (RBAC).	
Modular, Open and Interoperable Architectures	Modularization, standard interfaces, and interoperability to enable flexible upgrades and horizontal and vertical scalability of the platform. Microservices approach should be followed to ensure the same.	

Design Principle	Description	Whether complied or not (Yes/No)
User friendly interfaces & inclusive UX	Web portal and mobile apps must conform to GIGW 3.0. Solution should provide simple and standardized user experience	
Multilingual	Content must support major Indian languages and ensure responsive layouts for all language scripts	
Availability, reliability and serviceability	Solutions must achieve a level of availability, reliability and serviceability in line with the business criticality and requirements.	
Scalability, flexibility	Architecture must ensure that system can deliver consistently high performance and handle a large volume of transactions as defined in the Performance requirements. The system should permit vertically and horizontal scalability to handle increate in the scales upward and must anticipate order-of-magnitude growth beyond today's state-of-the-art standard (user scale, data scale, and application scale). The architecture should allow for scalability as and when the number of registered resources have increased. Provide a flexible platform to support ongoing changes in the business and effective integration of new business initiatives. A microservices based architecture should be followed to ensure a scalable. Modular platform with circuit break mechanisms for troubleshooting and platform resilience	
Openness and Extensibility	• Adoption of open API, open standards and wherever prudent open-source products are of paramount importance for the system. This will ensure the system is lightweight, scalable and secure. Openness comes from use of open standards and creating vendor neutral APIs and interfaces for all components.	
Innovative, sustainable and modern solutions	• Technology solutions should be innovative and apply modern approaches as per current trends. It should enable more effective ways to meet airport related business demand and requirements.	

Design Principle	Description	Whether complied or not (Yes/No)
Resilience and Robustness	The platform should promote resilience between its platform components and act as a robust solution. It should have ability to provide and maintain an acceptable level of service in case of faults and challenges to the normal platform operation.	
Responsive Change Management	Changes to the platform information environment are implemented in a timely manner.	
Control Technical Diversity	Technological diversity should be controlled to minimize the non-trivial cost of maintaining expertise in and connectivity between multiple processing environments.	
Security risks are considered	Technology environment should maintain a level of security that is commensurate with the risk and magnitude of the harm that could result from the loss, misuse, disclosure or modification of information.	
Simplify and standardise	Standardise the processes and services. Introduction of new solutions should simplify the operations.	
Technology Refresh	System should have provision to adopt technology refresh in every 3-5 years or whenever there are significant updates and upgrades available	

Application Principle	Description	Whether complied or not (Yes/No)
Supreme User Interface / Experience	Solution should provide simple and standardised user experience as per GIGW norms. Applications are easy to use. The underlying technology is transparent to users.	
Technology Agnostic	The solution should be developed using open-source libraries and software that are compatible with multiple operating systems and infrastructures, ensuring that it remains vendor- and platform-agnostic and can be easily ported across different environments	

Data Principle	Description	Whether complied or not (Yes/No)
Data security and privacy	Data should be protected from unauthorised use and disclosure	
Data is accessible and available	Quality data should be available and accessible to enable users to perform their functions	
Common Vocabulary and Data Definitions	Data is defined consistently throughout, and the definitions are understandable and available to all users. Information, data and its metadata are consistently defined and structured through a Data Model. Data definitions and vocabulary are easy to comprehend and published to all users. Calculations, terms, definitions and business rules defined and approved by the business for the business and to be used by the entire business.	